

A10-P

Programmer's Manual

Personal ID Number Pad
With EMV Level 2 Transaction Capabilities
For BV2 Version
(POS-A Specification)

PM 0122-C1

Revision 8

2022-04-01

FEDERAL COMMUNICATIONS COMMISSION STATEMENT

This device complies with Part 15 of the FCC Rules. Operation is subject to the following two conditions: (1) this device may not cause harmful interference, and (2) this device must accept any interference received, including interference that may cause undesired operation.

NOTE

This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to Part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communications. Operation of this equipment in a residential area is likely to cause harmful interference in which case the user will be required to correct the interference at his own expense.

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AGENCY APPROVED

- FCC class B
- CE class B

WARRANTY

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Document History

Document Version	Apply to FW version	Change
01	A10-P	First Release
02	A10-P	Modify the description of Encrypted Session Key in Message T15 / T35 / T61 / T81.
03	A10-P	Add new T67 / T68 command
04	A10-P	Extend Z2/Z3/Z50 command function
05	B1.10-K1.11-R1.12-S1. 11-A1.10	Add US Common Debit
06	B1.10-K1.11-R1.13-S1. 12-A1.10	Add new T37 / T38 command
07	B1.10-K1.11-R1.13-S1. 12-A1.10	Modify the description of the self-define tags
08	B1.10-K1.11-R1.14-S1. 12-A1.12	Add I08 / I09 command and modify 08 command

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Section 1 Introduction

The A10-P is a countertop PIN entry device with payment card readers (MSR, ICC, and optionally RFID.)

This manual describes the A10-P system with a pre-loaded EMV payment application developed by UIC.

The A10-P POS-A EMV application is fully compliant to UIC PIN Pad APIs/Commands set to offer following major functions:

1. PIN Entry Device (PED): A10-P can encrypt ANSI X9.8 standard PIN block by DES and TDES algorithm, using master/session key or DUKPT as key management scheme. In addition, it can encrypt EMV Level 2 specified PIN block by DES or RSA algorithm.
2. EMV Lv2 terminal: A10-P can handle most of the EMV Level 2 transaction flow, especially card holder verification (CVM) process, and send transaction result to its host machine.
3. Proprietary purse or loyalty programs: Thanks to the buildin NFC reader, A10-P can support most of closed loop contactless programs and proprietary contactless cards reading.

Meanwhile, A10-P is capable of running user applications to replace the UIC pre-loaded application per customer's requests. please contact UIC Sales or technical support for the additional information to develop the customized application resided inside A10-P.

A10-P Specifications

OS	Linux
Processor	Microchip SAMA5D28
Memory	DDR2 64MB
Storage	Flash 256MB
Display	2.8" color LCD 320x240 pixels with backlight
Physical Key	18 keys; 3 programmable function keys included
Magnetic Card Reader	Bi-directional, support triple tracks (ISO7811) Speed 5 to 40 IPS, 1,000,000. swipe times
Smart Card Reader	ISO7816 / EMV Level 1 compatible Supports 5V/3V, 500,000. inserted times
RFID Reader (optional)	ISO 14443 type A & type B / Contactless EMV Level 1 compatible. Antenna around the display.
SAM	2 SAMs (optional)
Key management method	TDES: Master / session key and DUKPT RSA: Chain of trust maintained by firmware and OTP flash.
MTBF	50,000 hours

Electrical Specifications

Power Requirements	5 VDC, 500 mA
Main Connector	RS232 DB9, or USB type A ¹
Peripheral Ports	LAN: RJ45 (optional)
ICC Voltage	5V
SAM Voltage	3V

Environmental Specifications

Temperature	Operating: 32°F to 122°F (0°C to 50°C) non-condensing Storage: -4°F to 158°F (-20°C to 70°C) non-condensing
Humidity	5% to 90% non-condensing

Communication Interface

Interface Protocols:

USB Protocol	USB 2.0 full speed (11Mbps) with CDC-ACM protocol (virtual COM) VID = 0x6352, PID = 0x294A
RS232 Protocol	9600bps, no parity, 8 data bits, 1 stop bit.

A10-P:

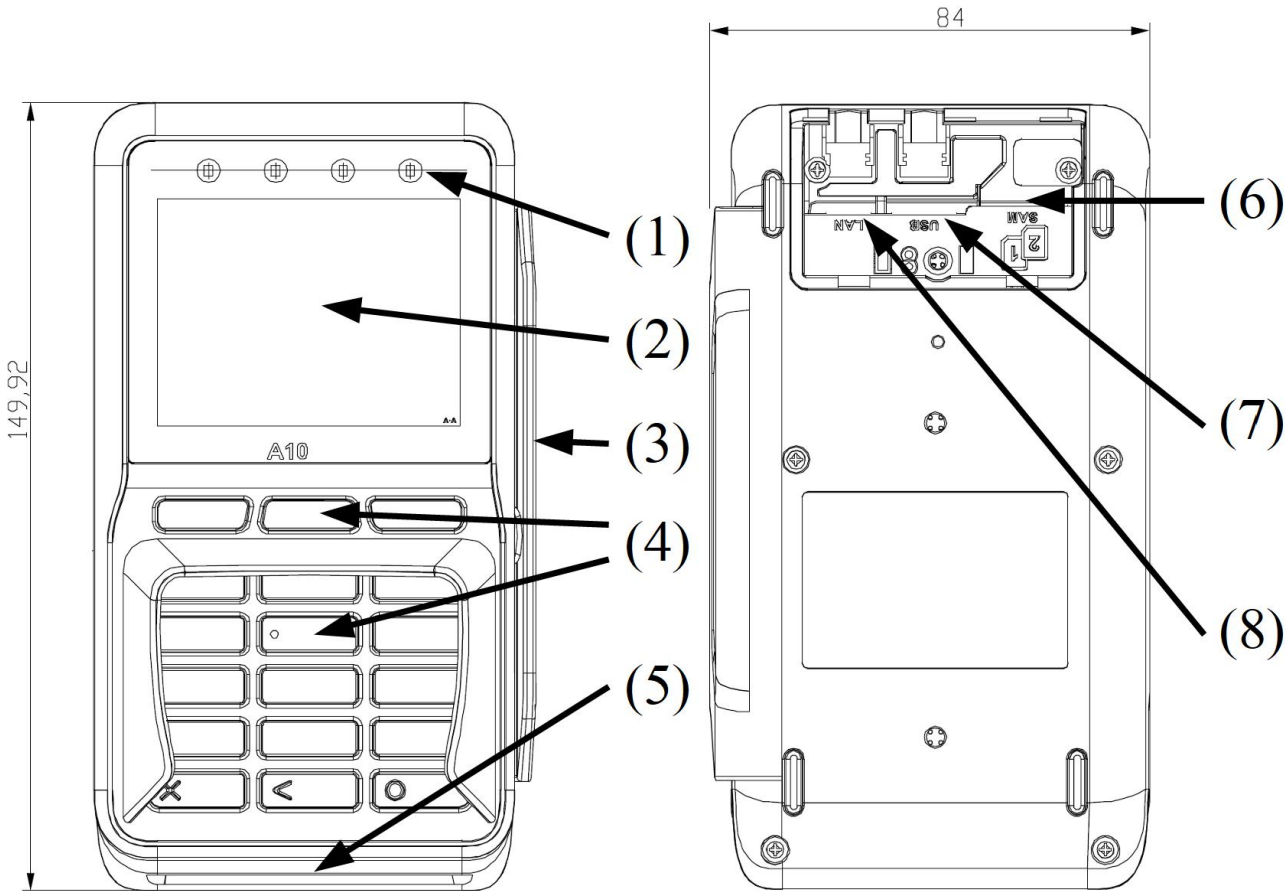
- RS232 Cable Model: DB9 with 5V power supply jack.
- USB Cable Model: Type A.

Dimensions and Components

A10-P Body Size	149.92mm(L) x 84.0mm(W) x 27.7mm(H)
Component Description	(1). Four Light-emitting diodes. (2). LCD display with 320 * 240 resolution. (3). Smart card reader inserting slot. (4). 18 key telephone-style keypad and 3 function keys. (5). Smart card reader inserting slot.

¹ A10-P cable type depends on individual model.

	(6). Two secure access module (SAM) slots, with protection cover. (7). Primary communication interface (USB), with protection cover. (8). Ethernet LAN Port.
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Components

Keypad

The PIN Pad uses its 18 keys to accept commands. For each key pressed, there will be a short beep to confirm that key is accepted. The following diagram shows the layout of the keys.

F1	F2	F3
1	2	3
4	5	6
7	8	9
*	0	#
CAN	CLR	ENT

[CAN]	(Cancel) button will abort PIN entering transactions or magnetic card swiping actions, and cause A10-P back to idle state.
[CLR]	(Clear) button will reset PIN input when doing PIN entry transactions.
[F1]	The F1 button is treated as “move up” when dealing with function menu (such as diagnostic mode or EMV Lv2 transaction menu). With customized application, system integrators can define its usage by themselves.
[F2]	The F2 button is treated as “go back” when dealing with function menu. With customized application, system integrators can define its usage by themselves.
[F3]	The F3 button is treated as “move down” when dealing with function menu. With customized application, system integrators can define its usage by themselves.

Section 2 Security Features

Physical security features

The secure processor of the A10-P has an embedded security module, which combines with external tamper sensors, environmental sensors, and battery backed-up SRAM area (BUSRAM).

- External tamper sensors are connected to the tamper detection grids and switches around the case.
- Environmental sensors are used for monitoring temperature, frequency, voltage, and processor's internal shield.
- BUSRAM area is scrambled with a randomly generated key, when security sensors detect a tamper event. This scramble key will be changed immediately to destroy the sensitive data inside, and then the BUSRAM controller will clear its contents.
- The security module of the processor will fire an unmaskable interrupt to inform the firmware to do other tamper responses.

Logical security features

Firmware integrity check

The firmware components (boot loader and operating system) of the A10-P are signed by RSA keys; and the certificate to verify the first stage boot loader has its hash burned into the one-time programmable flash (OTP) of the processor, as to provide immutable chain of trust.

Attempting to flash an unsigned or incorrectly signed firmware image will cause the processor refusing to boot; and trying to update an unsigned or incorrectly signed system files through firmware update mechanism will be rejected.

Reboot per 24 hours

Per PCI-PTS security requirements, A10-P should reboot itself every 24 hours to perform firmware integrity check and clear its memory.

The reboot mechanism is calculated by RTC, compare with the configured reboot time. If RTC run across the reboot time, A10-P will reboot itself.

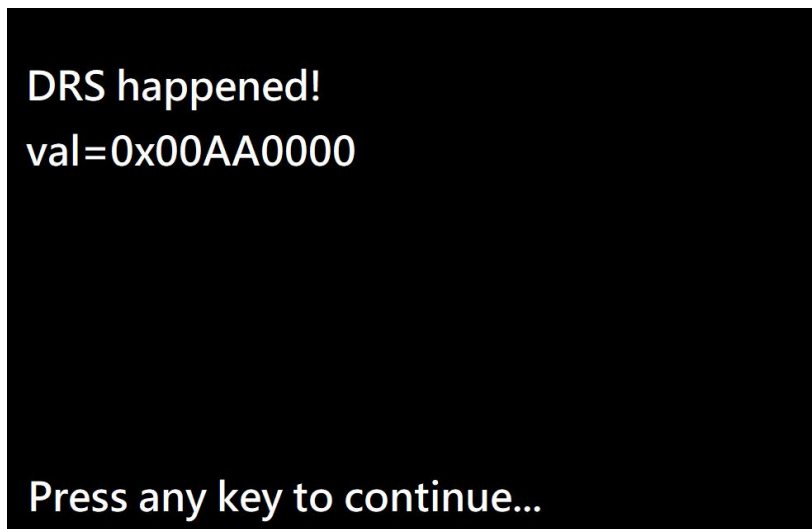
The reboot time can be set by the System Manager. If the administrator changes the RTC time or the reboot time, A10-P will reboot itself to enforce the new timer to take effect.

Active tamper response

A10-P will perform the following actions when tamper event is detected:

- All secret keys and key-injection passwords in the BUSRAM area will be erased.
- If A10-P is powered on, the operating system will reboot.
- A10-P will show the tamper evidence screen.

In such state, the payment application will not be launched. The system administrator must enter two passwords to release the tamper evidence. (If the tamper reason persists, it will trigger a new tamper evidence after reboot.)



The tamper evidence screen of the A10-P

Working Modes

The operating environment of the A10-P has 3 working modes, as described below.

Factory Test Mode:

This mode is the status of a A10-P just assembled in the factory and performing functional tests. It will show “FACTORY” on the upper right side of the screen and run the factory test application. Developers and users should not see any A10-P running in this mode.

All A10-P units will be “personalized” to enter the Developer Mode before leaving the factory.

Developer Mode:

This mode is the status of a A10-P with all security features enabled, but using “UIC default” passwords and certificates to facilitate application development. It will show “DEVELOPER” on the upper right side of the screen. End users should not see any A10-P running in this mode.

All A10-P units should be loaded with acquirer’s certificates and entered into Production Mode before deployment. This is done by the A10-P system manager.

Production Mode:

This mode is the status of a A10-P that is loaded with acquirer’s certificates, applications, and deployed into production environment. The screen will be controlled by the payment application.

Key management

Key management techniques for TDES keys

A10-P uses TDES keys to encrypt user PINs and account data for online transactions. The key management method can be Master / Session Key (MK/SK) or DUKPT, which is defined by the ANSI X9.24: 2009 specification.

A10-P has 21 slots to store master keys and 4 slots to store DUKPT keys. The 21th (#20) master key slot is fixed for the Key Transport Key (KTK) used for encrypted transportation of other master keys and DUKPT keys.

The first KTK is initialized by plain-text injection based on dual-password authentication (refer to **Appendix A** for detailed information.) After KTK is loaded, the other keys should be loaded by the encrypted key bundle which is defined by ANSI TR-31: 2018 specification.

Deterring the use of the device for exhaustive PIN determination: When doing transactions using the MK/SK scheme, the firmware of A10-P applies a DES calculation count limiter (only 10 transactions are allowed in 5 minutes period.) to comply with the PCI PED security requirement (average one transaction per 30 seconds.)

Backward Compatibility for Previous UIC PIN pads: The UIC A10-P PED application does the following mapping to keep the key ID compatible with UIC PP790SE and PP791:

A10-P Key Slot #	PED Key ID (mapped)	A10-P Key Slot #	PED Key ID (mapped)
00h ~ 09h	MK '0' ~ MK '9'	80h	DUKPT 0
0Ah	Not used	81h	DUKPT 1
0Bh ~ 0Eh	MK 'B' ~ 'E'	82h	Not used
0Fh	Not used	83h	Not used
10h	MK 'G'		
11h ~ 14h	Not used		
15h (KTK)	MK 'F'		

Key management techniques for RSA public keys

A10-P uses RSA public keys to verify user's application data, and validate the authenticity of the IC cards in complying with EMV specification.

The application data verification key is managed by the following chain of trust:

[Root Public key] → [Intermediate Public Key] → [Application Public Key] → [EMV Public Keys]

Root Public Key

The root public key of the A10-P is issued by UIC. It will be built in the Linux kernel as the root certificate. Since the kernel image will be verified by trusted boot loaders, A10-P can assure its authenticity.

Intermediate Public Key and Application Public Key

The intermediate and application public keys are saved in the file system of the A10-P as X.509 DER format certificate. Regarding to A10-P PED configuration, these two certificates are issued by UIC.

The system manager of the A10-P will verify and load these certificates during boot up.

EMV CA Public Keys

The EMV CA Public Keys are treated as a part of application configuration, so they should be signed by the Application Private Key. The A10-P PED configuration accepts X.509 PEM format CAPK by packet command T03, and use the application certificate to verify its authenticity.

UIC provides signing tool for users to sign their EMV CAPKs.

Cryptographic algorithms

A10-P applies the following cryptographic algorithms:

- RSA (for firmware authentication, application data authentication, and EMV data authentication.)
- SHA-256 (for firmware authentication and certificate authentication.)
- SHA-1 (for card data authentication during EMV transaction.)
- TDES (112 bits for DUKPT PIN or data encryption, 112/168 bits for MK/SK PIN or data encryption, and keys update.)

Cryptographic keys in A10-P

This section only lists out the keys that are controlled by the payment acquirer.

The key ID listed here is for A10-P PED configuration (UIC PIN pad compatibility mode).

Symmetric Keys

Key Name	Algorithm / Size (bits)	Purpose / Usage
Master keys (Shares MK ID: '0' ~ '9', 'B' ~ 'E', 'G')	TDES: 112 / 168	Decrypts session keys from external input. TR31 usage: K0, mode: D
PIN encrypting keys (Shares MK ID: '0' ~ '9')	TDES: 112 / 168	Encrypts PIN blocks TR31 usage: P0, mode: E
Data encrypting keys (shares MK ID 'G')	TDES: 168	Encrypts account / card data TR-31 usage: D0, mode: E
MAC calculating keys (shares MK ID 'B' ~ 'E')	TDES: 112	Generates or verifies message authentication codes. TR31 usage: M3, mode: G/V
Key Transfer Key (MK ID 'F')	TDES: 112 / 168	ANSI TR-31 key block protecting key. TR31 usage: K1, mode: D.
DUKPT initial keys (DUKPT ID '0' ~ '1')	TDES: 112	Derives DUKPT PIN, Data, and MAC keys. TR31 usage: B1, mode: X
Session key for PIN entry	TDES: 112 / 168	Encrypts PIN blocks.
Session key for Data encryption	TDES: 112 / 168	Encrypts account / card data.
Session key for MAC calculation	TDES: 112	Generates MAC for given transaction data.

Asymmetric Keys

Key Name	Algorithm / Size (bits)	Purpose / Usage
EMV CA public keys	RSA: 1024 ~ 1984	Verifies Issuer and ICC public key certificates.

Security policy for key management

1. The acquirer should manage the KTK and other secret keys in compliance with PCI security requirements. Improper use of the key management system will invalidate PCI approval to the A10-P.
2. Use FIPS 140-2 validated HSM to store secret keys to achieve proper security.
3. Keep “dual control” and “split of knowledge” principles for key injection or transportation.
4. Maintain KTK, passwords, and secret keys to be unique for each A10-P unit.
5. A10-P not only will reject the incorrect key bundle but also reject the insecure key update request defined by the PCI secure requirement:
 - Key length larger than KTK.
 - Repeated key value. (In other words, all symmetric keys in the A10-P should have different values, no matter what the key is.)
6. Any key should be replaced with a new key whenever the compromise of the original key is known or suspected; and whenever the time deemed feasible to determine the key by exhaustive attack elapses.

PIN security

A10-P protects user's PIN in accordance with the PCI PTS security requirements.

- A10-P supports ISO9564 format 0, 2, 4 PIN blocks. Refer to **Appendix B** for detailed information.
- A10-P deters the use of the device for exhaustive PIN determination: When doing transactions using the MK/SK scheme, the firmware of A10-P applies a DES calculation count limiter (only 10 transactions are allowed in 5 minutes period.) to comply with the PCI PED security requirement (average one transaction per 30 seconds.)

Account data security

A10-P provides two software configurations that have different account data security implementation, which can be confirmed from the firmware version string:

1. PCI PTS Only

Account data output messages (81, 83, QJ, T22, T26, T28, T2I, T64, T66) will output clear text account data. The IC card direct access messages (I01 ~ I07) are available. The MSR and RFID operation control messages (Q6 / QD) will not allow user to set encryption options.

2. PTS + SRED configuration

A10-P will obfuscate and encrypt the following data elements (that are not listed in the BIN exception table) from the MSR, ICC, and RFID before output.

- Primary account number (PAN), apply to EMV tag 0x5A
- Magnetic stripe track 1 data (TK1), apply to EMV tag 0x562
- Magnetic stripe track 2 data (TK2), apply to EMV tag 0x57
- Magnetic stripe track 3 data (TK3)

Payment Track Identification Rules

The following section defines how a track is considered a "payment card" track. If a track is not a payment card track, or if it is a payment card with a PAN that falls into the loaded BIN exception ranges, there is no obfuscation or encryption performed.

A given card could contain tracks that are both payment and non-payment – this could be indicative of incorrect or unexpected card encoding. We want to ensure if a payment card has had a track miscode (e.g., a single letter) we still protect what looks like payment data.

Note: The AAMVA card number range will not be identified as payment track.

Track 1

First byte in the track (after sentinel) is a 'B'

Next, there are 12-19 digits (or spaces) followed by a '^' separator

Next, there are 0 – 26 alphanumeric characters followed by a second '^' separator

Next, there are 4 digits (expiration date)

Remainder of track can be any data of any length

Examples of payment card track 1:

```
%B6011000993000201999^YOUR NAME GOES HERE^131200033ABCDEF123232?
```

```
%B0000000000000000^^1312?
```

Track 2 and Track 3

Track starts (after sentinel) with 12-19 numeric characters followed by a '=' separator

Next, there are 4 digits (expiration date)

Remainder of track can be any data of any length

Examples of payment card track 2/3:

```
;6011000993000201999=13120003334567890?
```

```
;0000000000000000=1312?
```

BIN Exception Rules

BIN (Bank Identification Number) refers to the leftmost 6 digits of PAN that identifies the issuer bank.

A10-P retains a 50-entry white list that can exclude specific PAN range from account data encryption.

² Not every IC card contains 0x56 or sufficient information to composite track 1 equivalent data.

The algorithm is:

IF (card has valid payment track) **AND** (PAN is within the range of any BIN exception entry)
THEN (output the track data in clear text form).

This feature can be used to exclude driver licenses, gift cards, or royalty cards from being encrypted.

The time stamp format of the BIN exception table is:

YYMMDDhhmmss<CR><LF> (First line as time stamp)

The data format of the white list entry is:

[Minimum PAN] . [Maximum PAN] <CR><LF>

Each PAN field must have 12 digit numbers.

Example:

400000000000 . 499999999999

510000000000 . 559999999999

352800000000 . 358999999999

601100000000 . 601199999999

622126000000 . 622925999999

502244000000 . 502244999999

The above data will make A10-P exclude the following account numbers from encryption:

Entry	Min PAN	Max PAN
Line 1	400000000000	499999999999999999
Line 2	510000000000	559999999999999999
Line 3	352800000000	358999999999999999
Line 4	601100000000	601199999999999999
Line 5	622126000000	622925999999999999
Line 6	502244000000	502244999999999999

MSR Track Obfuscation Rules

A valid payment track that does not fall into the loaded BIN exception ranges will be obfuscated by following rules:

Track 1

Start sentinel (%) and end sentinel (?) are present.

Spaces in the PAN before the first ^ delimiter remain as-is. These spaces should not be masked or obfuscated.

Any digits in PAN between first 6 and last 4 before first ^ delimiter are replaced by zeroes.

The 7th digit is changed to a 1 only if necessary to ensure the card number fails a Luhn check.

Any digits after the expiration date and service code (7 digits immediately after the second ^ delimiter) are replaced by zeroes.

Example:

Cleartext: %B3727 006 9925 1018^AMEX TEST CARD^2512990502700?

Obfuscated: %B3727 000 0000 1018^AMEX TEST CARD^2512990000000?

Track 2 and Track 3

Start sentinel (;) and end sentinel (?) are present.

Any digits in PAN between first 6 and last 4 before the first = delimiter are replaced by zeroes.

The 7th digit is changed to a 1 only if necessary to ensure the card number fails a Luhn check.

Any digits after the expiration date and service code (7 digits immediately after the = delimiter) are replaced by zeroes.

Example:

Cleartext: ;372700699251018=2512990502700?

Obfuscated: ;372700000001018=2512990000000?

MSR Track Encryption Rules

A valid payment track that does not fall into the loaded BIN exception ranges will be encrypted by the following rules:

Track 1 and 2

- Start sentinel (%) and end sentinel (?) are present.
- The byte array that contains track data will be padded by the rule in the specification chapter 6.1.1 or value with 0xFF to make the stream length divisible by 8 bytes.

For example, if the clear text track data is ;372700699251018=2512990502700? (31 bytes), so the padded data, express in hexadecimal form, will be:

3B3337323730303639393235313031383D323531323939303530323730303F01

(The last 01 is the padding byte in this case).

Or

3B3337323730303639393235313031383D323531323939303530323730303FFF

(The last FF is the padding byte in this case).

- Each padded track data is encrypted by a designated MSR working key (should be 24 bytes in length) , the initial vector (IV) is 8 bytes random number ;or DUKPT current key (data encryption variant) in code block chaining (CBC) mode, the initial vector (IV) is 0000000000000000.
- The encrypted track will be encoded into hexadecimal string (e.g. “3C” represents 0x3C) for all kinds of the output packets.
- All tracks in a single swipe shares the same data encryption key and the KSN.
- Data encryption and decryption can only be performed ten times in 300 seconds.

Track 3

The track 3 will not be encrypted (only obfuscated data will be output).

ICC Data Obfuscation Rules

For EMV transaction, A10-P TransArmor version will put obfuscated data into the following tags:

Tag	Description	Obfuscation Rule
0x5A	Primary Account Number (PAN)	Any digits in PAN between first 6 and last 4 are replaced by zeroes. The 7th digit is changed to a 1 only if necessary to ensure the card number fails a Luhn check. For example: PAN = 372700699251018 Obfuscated = 372700000001018 Tag 5A formatted = 37 27 00 00 00 01 01 8F
0x9F1F	Track 1 discretionary data	All digits are replaced by zero. For example: Tk1 discretionary data = 502700 Obfuscated = 000000 Tag 9F1F formatted = 30 30 30 30 30 30
0x9F20	Track 2 discretionary data	All digits are replaced by zero. For example: Tk1 discretionary data = 502700 Obfuscated = 000000 Tag 9F20 formatted = 00 00 00
0x56	Track 1 equivalent data	The same with MSR track 1. For example: Tk1 = B372700699251018^AMEX TEST CARD^2512990502700 Obfuscated = B372700000001018^AMEX TEST CARD^25129900000000 Tag 56 formatted data is the same ASCII string.
0x57	Track 2	The same with MSR track 2.

	equivalent data	For example: Tk2 = 372700699251018=2512990502700 Obfuscated = 372700000001018=2512990000000 Tag 57 formatted = 37 27 00 00 00 01 01 8D 25 12 99 00 00 00 0F (The right nibble 'D' of the 8th byte represents the field separator, the last 'F' is the padding nibble.)
--	-----------------	--

ICC Data Encryption Rules

Track 1 and 2 equivalent data

- The tag 56 (Track 1 equivalent data) and tag 57 (Track 2 equivalent data) will be converted to ASCII strings. Replace the track 2 field separator from 0xD to '=', and apply the same encryption rule with the MSR.
- The start and end sentinels like MSR will be added to the track equivalent data.
- The plaintext track equivalent data will be padded by PKCS#5 rule or value with 0xFF, then encrypted by the designated MSR working key (should be 24 bytes in length). The initial vector (IV) is 8 bytes random number ; or DUKPT current key (data encryption variant) in code block chaining (CBC) mode, the initial vector (IV) is 0000000000000000.
- The encrypted track + MAC will be encoded into hexadecimal string (e.g. "3C" represents 0x3C) for all kinds of the output packets.
- Data encryption and decryption can only be performed ten times in 300 seconds

Section 3 System Manager

Introduction

The system manager is a resident process launched automatically during the A10-P boot up process. It can manage the applications download, execution of individual application, and perform basic settings and self tests.

After the system boot-up and the startup application is launched, the user can press “<ENT> + [1]” (press <ENT> key and 1] at the same time) to enter the system manager menu.

The A10-P has three sets of the dual passwords to administer system manager functions. (The default passwords will be sent to the legal user by a secure method).

1. Personalization Passwords.
2. Key Injection Passwords.
3. Administration Passwords.

NOTE 1: The current application will be terminated after entering into the System Manager

NOTE 2: The System Manager will exit password page, if there is no input within 60 seconds.

Main menu of System Manager

DISPLAY	ACTION
About	Show firmware version, TCP/IP settings and certificate names.
Key Injection	Perform clear text key load. (Key Injection Passwords)
User Settings	Set user's options, ex: DHCP and mute.
Admin Settings	Set administration options, ex: update and RTC etc. (Administration Passwords)
Security Setting	Set up Security options. (Personalization Passwords)
Self Test	Perform basic hardware test. (Administration Passwords)

“About” menu

DISPLAY	ACTION
System info	Kernel A10-P Linux kernel version. RFS A10-P root file system version. SysMgr A10-P System Manager version. HW Model HW Model name of this device. Serial

	Serial number of this device. MAC MAC address of this device.
EMV Version	Display the ICC and PCD EMV version.
Cert. info	The certificate info has 3 pages: for vendor, intermediate, and root certificates. Cert. CN “Common Name” field of certificate Cert. Hash (partial) 4 most significant bytes of the SHA-1 hash of this certificate.

“User Settings”

DISPLAY	ACTION
Set DHCP / IP	Bring up the sub menu of the basic TCP/IP settings [Set DHCP] Enable or disable the DHCP client of the A10-P. This setting will take effect after reboot. [Set Local IP] Set fixed IP address of the A10-P if DHCP is disabled. [Set Gateway IP] Set the gateway IP address of the A10-P if DHCP is disabled. [Set Subnet Mask] Set the subnet mask of the A10-P if DHCP is disabled. [Set DNS Server] Set the DNS server address of the A10-P if DHCP is disabled.
Set Mute	Enable or disable the buzzer for keyboard event.
Display All Setting	Display the current settings of each “User Settings” menu.

“Admin Settings”

DISPLAY	ACTION
Update	Perform firmware and application updates.
Set Time Zone	- Enter the name of the local time zone. (3~6 bytes) - Input the time offsets from UTC. (use [F1] to add minus sign) - Enter the name of the Daylight Saving Time (DST) or press [CAN] to skip the related settings of DST. - Input the time offsets of the DST from UTC.

	6. Enter the start date and time of the DST. 7. Enter the date and time to set the end of the DST.
Set RTC	Set the current time zone, Current RTC time (UTC), and the Current Local time.
Set Reset Time	Enter the hour (24h format) and minute to setting reset time.
Set Console	Enable or disable the Linux command console.
Set Appl Launch Index	Set user's index path for A10-P starting up application.
Set Appl Update Index	Set user's index path for A10-P update application.
Config Setup Mode	Enter setup mode to wait for factory command packet.
Display All Setting	Display the current settings of each "Admin Settings" menu.

"Update" menu

DISPLAY	ACTION
xModem	Select the xModem method to wait for specific command to update A10-P. A10-P uses the 1K-Xmodem protocol on COM port to operate the downloading, using the 9600bps, none-parity, 8 data bits, 1 stop bit settings.
USB	Select the USB method to search for the updated compressed file in the USB to update A10-P. Kernel kernel.tarball.tar.gz RFS rfs.tarball.tar.gz SysMgr sysmgr.tarball.tar.gz User's Application app.tarball.tar.gz

"Security Settings" menu

DISPLAY	ACTION
Personalization	Reset the system to unlock the tampered state.
Enter Production Mode	Switch to product mode and change the authentication credentials to verify the startup application.
Reset to Developer	Reset to developer mode and change authentication credentials to verify the startup application.

“Self Test” menu

DISPLAY	ACTION
KBD Test	Test the keypad and show the pressed keys on the LCD.
LCD Test	Display color screens and text strings of different font sizes on the LCD.
MSR Test	Test by swiping MSR cards at different speeds; the PAN of the payment track will be partially masked out.
ICC Test	Test the smart card powerup operation and display its ATR.
MEM Test	Test writing and reading data to the memory.
SAM Test	Test the power up and selection commands for the SAM card.
RFID Test	Test by tapping an RFID credit card (such as the Visa payWave or Mastercard PayPass); the card type will be displayed on the LCD.
TCP Test	Test the ethernet connection by executing the ping command 4 times.
COM Test	Perform a loopback test to check whether the Rx and Tx functions of USB Virtual COM or RS232 are normal.
PRN Test (optional hardware)	Test the printer for normal operation.
RS232 Test (optional hardware)	Perform a loopback test to check whether the Rx and Tx functions of RS232 are normal.
USB Test (optional hardware)	Test by checking if there is a USB disk plugged in the USB host port.
DIA Test (optional hardware)	Perform a loopback test to check that the modem is operating properly.
LTE Test (optional hardware)	Test the LTE network connection by executing the ping command 4 times.

Section 4 Message Format

This chapter provides the details of the format of messages exchanged between the host application and the A10-P.

Notation Conventions

The following conventions are used to make the description of messages more readable:

Control Codes

Control codes (non-displayable codes) are represented by two to three capital letters enclosed in angled brackets "<>". This A10-P uses a total of 12 control codes. The actual code, when referenced, is represented by two hex digits enclosed in angled brackets, <0F> for example. The following table lists their usage and hex code values.

CODE	HEX VALUE	USAGE
STX	02	Denotes the beginning of a message frame
ETX	03	Denotes the ending of a message frame
ACK	06	Acknowledge of message received
SO	0E	Denotes the ending of a message frame
NAK	15	Indicates invalid message received
SUB	1A	Message parameter follows
FS	1C	Field separator
GS	1D	Message ID follows
DC1	11	Used for Z2 message, enable the inverse mode.
DC2	12	Used for Z2 message, disable the inverse mode.

[LRC]

Each message frame transmitted is appended with an LRC byte to detect communication error. This byte should be regarded as part of the message frame but comes after the ending delimiter character. [LRC] is used in each message frame description to represent this LRC byte.

LRC is calculated as an XORed value of every byte after the start code (<STX> or <SI>) in the message frame, all the way to and including the end code (<ETX> or <SO>).

[item]

A descriptive item name is enclosed in bracket. It denotes a string or data byte that has no fixed value.

Message frame summary

The data exchanged between the A10-P and the host is grouped into “message frames”. Each message frame has one of the two possible frame formats listed below:

- **<STX>**[message ID] [data] **<ETX>**[LRC]
- **<SI>**[message ID] [data] **<SO>**[LRC]

Each type of message has a unique value in its message ID field. In the following texts, we refer a message type by its message ID value, e.g. “message 70”.

Message type

Messages exchanged between the A10-P and the host can be divided into two categories.

One category is for administration and maintenance. In general administrative messages have the **<SI>** packet header and its return message sent back to the host is using the same message ID.

The other category is for various transactions. In general transaction messages are using the **<STX>** packet header. The messages work in pair. The odd number message sends the command and data to the A10-P, and the even number message returns the result.

Time-out

Whenever the A10-P sends out a message, a response (**<ACK>** character for acknowledgement or **<NAK>** character if LRC error occurs) from the host is expected. If the A10-P does not receive a response within 5 seconds, it will retransmit the last packet. If the A10-P does not receive **<ACK>** or **<NAK>** after two retransmit attempts, it will send out the **<EOT>** character and this communication session will be terminated.

Transmission Error

When the host decides that an invalid frame is received, it will send out the **<NAK>**. When the A10-P receives a **<NAK>**, it will retransmit its last message. If the message retransmitted is invalid again, then a **<NAK>** is sent by the host to request for another try. The A10-P will keep on retransmitting the message until an **<ACK>** or **<EOT>** is received.

Packet Error

When the A10-P receives a good transmission but an invalid packet (wrong message id), it will ignore the packet. If the received packet has an acceptable message id but having a wrong format, the A10-P will send out **<EOT>** as the error message. It will send out more detailed error codes if the messages are PIN entry functions.

Section 5 Administration and Maintenance Messages

Message 02 Load Master Key (Clear Text)

Format: <SI>02[Key ID][Key value]<SO>[LRC] (request frame)
<SI>02?[error code]<SO>[LRC] (error frame)

Message length: Variable (22 to 58 bytes).

Usage: Send the key value to the A10-P and save into designated master key slot. A10-P can store 21 master keys (A10-P application only uses 16 of them for backward compatibility); each with a one digit ID. These Master keys are divided into three groups of different functions. Refer to **Appendix A** for key usage and ID definition.

Clear text key injection is only accepted in the “Key Injection” session of the system manager; which is enabled by dual-password authentication. This function should be used only in security-controlled environment with HSM. Refer to **Appendix A** for detailed information.

The algorithm of the injected key is defined by the key management scheme, if the usage and mode are not designated. A10-P will apply default value for them. Refer to **Appendix A** for detailed information.

Change key management scheme will erase all existing keys in the A10-P.

All existing keys in the A10-P will be erased when the first message 02 or 90 is sent in a new key injection session.

Compatibility: The user of UIC PP790SE and PP791 should note:

1. The “key transport key” usage is changed to “K1” instead of “K0” in accordance to the new ANSI TR-31: 2018 standard.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<SI>	1	<0F>
02	2	Message ID
[Key ID]	1	['0' to '9', 'B' to 'G']
[Key value]	16 or 32 or 48	Hexadecimal string for key value. In DES scheme, 16 characters (single length) . In TDES scheme, 32 characters (double length) or 48 characters (triple length) keys are allowed.
<FS>	1	Field separator. (Optional, only available in clear text format frame if the following [Usage] and [Mode] exist)
[Usage]	2	(Optional: ANSI TR-31 key usage for clear text frame.) Available values are: “K0” for key encryption. (id 0 ~ 9, B ~ G) “K1” for key transport key. (id F) “P0” for PIN encryption. (id 0 ~ 9) “M3” for MAC calculation. (id B ~ E) “M1” for MAC calculation. (id B ~ E) “D0” for data encryption. (id G)
[Mode]	1	(Optional: ANSI TR-31 key mode for clear text frame.) Available values are: ‘D’ for decryption only. (K0 / K1 keys) ‘E’ for encryption only (P0 / D0 keys) ‘G’ for MAC generation only (M1 / M3 keys) ‘V’ for MAC verification only (M3 keys)
<SO>	1	<0E>
[LRC]	1	Checksum

Response frame – Error message (A10-P to HOST)

Field	Length	Value and description
<SI>	1	<0F>
02	2	Message ID

?	1	
[Err msg]	1	'2': Key duplicated with existing keys. '3': Internal fail: fail to allocate memory. '4': Internal fail: fail to read key structure. 'E': Incompatible key usage. 'F': Key injecton operating limit reached.
<SO>	1	<0E>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message 02 (request frame)	→	
	←	<ACK> /<NAK>/<EOT>
	←	Processing request. If format error, send <EOT> and end. If format and MAC correct, send Message 02 (echo of request frame).
Verify echo frame. If verify ok, send <ACK>. If packet LRC error, send <NAK>. If the host wants to cancel the key loading procedure, send <EOT>.	→	
	←	Save the key value and send <EOT>

Message 02 Load Master Key (TR-31 Bundle)

Format: <SI>02[Key ID][TR-31 Key Bundle]<SO>[LRC] (request frame)
<SI>02?[error code]<SO>[LRC] (error frame)

Message length: Variable.

Usage: Send the encrypted and MACed key bundle (ANSI TR-31: 2018 format) to the A10-P and save into the designated master key slot.
A10-P will use the key transport key (key slot #F) to validate and decrypt the key bundle. If key management scheme is set to TDES, A10-P will use TDES algorithm to do validation and decryption.

Compatibility: The users of UIC PP790SE and PP791 should note:

1. TR-31 Bundle single DES keys are no longer supported due to the evolution of the PCI security standards.
2. The “key transport key” usage is changed to “K1” instead of “K0” in accordance to the new ANSI TR-31: 2018 standard.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<SI>	1	<0F>
02	2	Message ID
[Key ID]	1	['0' to '9', 'B' to 'G']
[TR-31 Key Bundle]	Var.	ANSI TR-31: 2018 key bundle. Refer to Appendix A for detailed information.
<SO>	1	<0E>
[LRC]	1	Checksum

Response frame – Error message (A10-P to HOST)

Field	Length	Value and description
<SI>	1	<0F>
02	2	Message ID
?	1	
[Err msg]	1	'2': Key duplicate. '3': Internal fail: fail to allocate memory. '4': Internal fail: fail to read key structure. '7': Fail to decrypt the key value. 'A': TR31 format error. 'B': Insecure key inject. (New key is longer

		than the key used to protect it.) 'C': Fail to verify the MAC value. 'D': KKK does not exist. 'E': Incompatible key usage. 'F': Key injection operating limit reached. 'Z': Fatal Error.
<SO>	1	<OE>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message 02 (request frame)	→	
	←	<ACK> /<NAK>/<EOT>
	←	Processing request. If format error, send <EOT> and end. If format and MAC correct, send Message 02 (echo of request frame).
Verify echo frame. If verify ok, send <ACK>. If packet LRC error, send <NAK>. If the host wants to cancel the key loading procedure, send <EOT>.	→	
	←	Save the key value and send <EOT>

Example:

Refer to **Appendix A** for the detailed example of the TDES generation.

TDES TR-31 Examples

Message 04 Check Master Key

Format: <SI>04[key ID][get attr]<SO>[LRC] (request frame)
 <SI>04[response code][attr]<SO>[LRC] (response frame)

Message length: Variable (6 or 7) bytes for request frame.

Variable (6 or 22) bytes for response frame.

Usage: Check if the specified master key exists, and optionally get its TR31 attributes.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<SI>	1	<0F>
04	2	Message ID
[key ID]	1	Master key ID (0 to 9, B to G)
[get attr]	1	<Option>, 1: query key information
<SO>	1	<0E>
[LRC]	1	Checksum

Response frame (A10-P to HOST)

Field	Length	Value and description
<SI>	1	<0F>
04	2	Message ID
[response code]	1	0 Master key not loaded F Master key loaded
[attr]	6	2bytes key usage + <FS> + 1byte Mode of use + <FS> + 1byte Algorithm
<SO>	1	<0E>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message 04 (request)	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Check requested key slot Message 04 (response)
<ACK> (Good echo) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	
	←	<EOT>

Message 05 Load Serial Number

Format: <SI>05[string]<SO>[LRC] (request frame)

Message length: Variable.

Usage: Load the A10-P with the serial number given in the message frame. The A10-P will send the whole message frame back to the host as a confirmation of good reception. If the LRC is good but the echoed serial number is incorrect the host should then send an <ACK> to confirm or an <EOT> to cancel this serial number loading process. Follow the standard <NAK> process if an invalid LRC is detected.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<SI>	1	<0F>
05	2	Message ID
[string]	0 to 16	Alphanumeric string (0~9, A~Z, a~z)
<SO>	1	<0E>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message 05	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message 05 (echo frame) or <EOT> indicating error.
<ACK> (Good echo) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	
	←	(Stores serial number) <EOT>

Message 06 Get Serial Number

Format: <SI>06<SO>[LRC] (request frame)
 <SI>06[string]<SO>[LRC] (response frame)

Message length: Fixed 5 bytes for the requesting message, Variable for the response message.

Usage: This message is used to get the serial number of the A10-P. The A10-P will send the serial number previously loaded or a string of 16 '0's as the serial number if it has not been loaded. The serial number will be displayed on the LCD.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<SI>	1	<0F>
06	2	Message ID
<SO>	1	<0E>
[LRC]	1	Checksum

Response frame (A10-P to HOST)

Field	Length	Value and description
<SI>	1	<0F>
06	2	Message ID
[string]	0 to 16	String for serial number, or "0000000000000000" if not set.
<SO>	1	<0E>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message 06 (request)	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message 06 (response frame) or <EOT> if read error
<ACK> (Good echo) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	
	←	<EOT>

Message 07 Test Crypto Implementation

Format: <SI>07[**master key**][**clear text**][**cipher text**]<SO>[**LRC**]
(request frame)
<SI>07[**status**]<SO>[**LRC**] **(response frame)**

Message length: Fixed 53 bytes for DES test.

Usage: This message is used to validate the DES implementation of A10-P. Test result will be shown on the A10-P display and a response code will be returned for remote diagnostic.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<SI>	1	<0F>
07	2	Message ID
[Master key]	16	Key value used for test. (hexadecimal string)
[Clear text]	16	Clear text data used for test. (hexadecimal string)
[Cipher text]	16	Known enciphered data. (hexadecimal string)
<SO>	1	<0E>
[LRC]	1	Checksum

Response frame (A10-P to HOST)

Field	Length	Value and description
<SI>	1	<0F>
07	2	Message ID
[response code]	1	'0': Test Success 'F': Test Failed.
<SO>	1	<0E>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message 07 (request)	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message 07 (response) or <EOT> if format error
<ACK> (Good echo) <NAK> (Bad LRC) <EOT> (after 3 NAKs)		
	←	<EOT>

Message 08 Select Active Master Key

Format: <SI>08 [KeyID] <SO> [LRC] (request frame)

Message length: Fixed 6 bytes.

Usage: This message is used to select an “active” master key from one of the 10 possible PIN encrypting master keys previously loaded using message 02. The active master key will be used in the following transactions.

Note: **Confirm the key existence before issuing 08:** message 08 is not responsible to check if [KeyID] has a valid master key. Use message 04 before message 08.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<SI>	1	<0F>
08	2	Message ID
[KeyID]	1	0~9, one of Master key id.
<SO>	1	<0E>
[LRC]	1	Checksum

Message flow (compatible with UIC PP791 BV2 for A10-P BV2/VEL):

HOST	Direction	A10-P
Message 08	→	
	←	<ACK>/<NAK>/<EOT>
	←	Success: <SI>080<SO> Fail: <SI>081<SO> Key index > 9
<ACK> (Good echo) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	
	←	<EOT>

Message flow (compatible with UIC PP79SE BVA for A10-P BVP):

HOST	Direction	A10-P
Message 08	→	
	←	<ACK>/<NAK>/<EOT>
	←	Set active master key. If key ID error, error message will be displayed.
	←	<EOT>

Message 09 Communication Test

Format: <SI>09<SO>[LRC] (request frame)
 <SI>09<SUB>PROCESSING<SO>[LRC] (response frame)
 <SI>09[response code][LRC] (Result frame)

Message length: Fixed 5 bytes for the requesting message, fixed 16 bytes for the response message.

Usage: This message is used to test communication link between the host and the A10-P. Both host and A10-P can initiate a communication test. The initiating party should send the requesting message; the other party should respond with the response message that should be ACKed if received correctly. After verifying that the response message is correct, the initiating party should send back the same response message and the receiving party should acknowledge this message. The test results are shown on the A10-P display.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<SI>	1	<0F>
09	2	Message ID
<SO>	1	<0E>
[LRC]	1	Checksum

Response frame (A10-P to HOST)

Field	Length	Value and description
<SI>	1	<0F>
09	2	Message ID
<SUB>	1	<1A>
[Test string]	10	ASCII string "PROCESSING"
<SO>	1	<0E>
[LRC]	1	Checksum

Result frame (A10-P to HOST)

Field	Length	Value and description
<SI>	1	<0F>
09	2	Message ID
[response code]	1	'0': Test Success 'F': Test Failed.
<SO>	1	<0E>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message 09 (request)	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message 09 (response frame)
<ACK> (Good echo) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	
Message 09 (response)	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message 09 (result frame)
<ACK> (Good echo) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	
	←	<EOT>

Message 11 A10-P Device Connection Test

Format: <SI>11<SO>[LRC] (request frame)

Message length: Fixed 5 bytes.

Usage: This message is used to ensure that the A10-P attached to the host is working normally. The A10-P will respond an ACK (or NAK if LRC is incorrect) within one second.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<SI>	1	<0F>
11	2	Message ID
<SO>	1	<0E>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message 11	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)

Message 12 Select Prompt Language

Format: <SI>12[language index]<SO>[LRC] (request frame)
 <SI>12[status]<SO>[LRC] (response frame)

Message length: Fixed 6 bytes.

Usage: This message is used to select the A10-P prompt message table for different languages. The English prompt table is resident as default and its index number is '0'. Other prompts can be updated and selected and the A10-P will select the corresponding code page for display.

NOTE. The order of the supported language may be different from this manual as the prompt message module changes. Refer to the response from A10-P through message 1F.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<SI>	1	<0F>
12	2	Message ID
[language index]	1	Language code '0': English (default) '1': Spanish
<SO>	1	<0E>
[LRC]	1	Checksum

Response frame (A10-P to HOST)

Field	Length	Value and description
<SI>	1	<0F>
12	2	Message ID
[status]	1	Status: '0': Success 'F': prompt index value error.
<SO>	1	<0E>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message 12 (Request frame)	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message 12 (Response frame)
<ACK> (Good echo) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	
	←	<EOT>

Message 13 Adjust COM1 Baud Rate (RS-232 version only)

Format: <SI>13[baud code][mode]<SO>[LRC] (request frame)
<SI>13[status]<SO>[LRC] (response frame)

Message length: Variable, 6~7 bytes.

Usage: This message will change the communication baud rate and transmit mode of the A10-P for later operations. The setting is kept in the non-volatile memory. The baud rate will be changed after the message flow ends.

Note: If the [mode] parameter is not specified, the default transmit mode is N, 8, 1.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<SI>	1	<0F>
13	2	Message ID
[baud code]	1	ASCII character '1' = 1200bps '2' = 2400bps '3' = 4800bps '4' = 9600bps '5' = 19200bps '6' = 38400bps '7' = 57600bps '8' = 115200bps
[Transmit mode] (optional)	1	ASCII character '1' = None parity, 8-bit, 1 stop bit '2' = Even parity, 7-bit, 1 stop bit '3' = Odd parity, 7-bit, 1 stop bit 0x41 = None parity, 8-bit, 1 stop bit, with XON-XOFF flow control 0x42 = Even parity, 7-bit, 1 stop bit, with XON-XOFF flow control 0x43 = Odd parity, 7-bit, 1 stop bit, with XON-XOFF flow control 0x81 = None parity, 8-bit, 1 stop bit, with RTS flow control 0x82 = Even parity, 7-bit, 1 stop bit,

		with RTS flow control 0x83 = Odd parity, 7-bit, 1 stop bit, with RTS flow control 0xC1 = None parity, 8-bit, 1 stop bit, with RTS/XON-XOFF flow control 0xC2 = Even parity, 7-bit, 1 stop bit, with RTS/XON-XOFF flow control 0xC3 = Odd parity, 7-bit, 1 stop bit, with RTS/XON-XOFF flow control
<SO>	1	<OE>
[LRC]	1	Checksum

Response frame (A10-P to HOST)

Field	Length	Value and description
<SI>	1	<0F>
13	2	Message ID
[status]	1	ASCII character ‘0’ for success ‘1’ for parameter error
<SO>	1	<OE>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message 13 (request)	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message 13 (response)
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	
	←	<EOT>
		(Change the communication baud rate and save the setting)

Message 14 Enable/Disable Timer Display

Format: <SI>14[E/D]<SO>[LRC] (request frame)
 <SI>14[Status]<SO>[LRC] (response frame)

Message length: Fixed 6 bytes.

Usage: This message is used to enable/ disable the system timer display of the A10-P.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<SI>	1	<0F>
14	2	Message ID
[E/D]	1	'0': Disable '1': Enable
<SO>	1	<0E>
[LRC]	1	Checksum

Response frame (A10-P to HOST)

Field	Length	Value and description
<SI>	1	<0F>
14	2	Message ID
[Status]	1	'0': Disabled '1': Enabled '2': Format Error
<SO>	1	<0E>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message 14 (Request frame)	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message 14 (Response frame)
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message 15 Adjust LCD Backlight And Contrast Level

Format: <SI>15 [backlight] <SO> [LRC] (request frame)
 <SI>15 [backlight status] <SO> [LRC] (response frame)

Message length: Fixed 8 bytes for the request message.

Fixed 8 bytes for the response message.

Usage:

This message will set the backlight and contrast level of the LCD display.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<SI>	1	<0F>
15	2	Message ID
[backlight]	1	ASCII character ‘0’~‘7’: backlight ‘0’: darkest ‘7’: brightest (default)
<SO>	1	<0E>
[LRC]	1	Checksum

Response frame (A10-P to HOST)

Field	Length	Value and description
<SI>	1	<0F>
15	2	Message ID
[backlight status]	1	ASCII character ‘0’ ~ ‘7’: success ‘F’: save configuration fails
<SO>	1	<0E>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message 15 (Request frame)	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message 15 (Response frame) or <EOT> if [backlight] error"
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	
	←	<EOT>

Message 16 Remote self-test request

Format: <SI>16<SO>[LRC] (request frame)
 <SI>16[Response]<SO>[LRC] (response frame)

Message length: Fixed 5 bytes.

Usage: This message is reserved for backward compatibility, user of the A10-P should use message 20 to get the real meaningful remote diagnostic information.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<SI>	1	<0F>
16	2	Message ID
<SO>	1	<0E>
[LRC]	1	Checksum

Response frame (A10-P to HOST)

Field	Length	Value and description
<SI>	1	<0F>
16	2	Message ID
[Response]	1	0 – Healthy
<SO>	1	<0E>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message 16 (Request frame)	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message 16 (Response frame)
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	
	←	<EOT>

Message 17 Request random number

Format: <SI>17<SO>[LRC] (request frame)
 <SI>17[RndBlk]<SO>[LRC] (response frame)

Message length: Fixed 5 bytes.

Usage: This message is used to request the A10-P to generate an 8-byte random number block. This random number is generated by the hardware TRNG (True Random Number Generator) that is certified with sufficient security.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<SI>	1	<0F>
17	2	Message ID
<SO>	1	<0E>
[LRC]	1	Checksum

Response frame (A10-P to HOST)

Field	Length	Value and description
<SI>	1	<0F>
17	2	Message ID
[RndBlk]	16	Random number block generated by A10-P. Format: hexadecimal string.
<SO>	1	<0E>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message 17 (Request frame)	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message 17 (Response frame)
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	
	←	<EOT>

Message 18 Set PIN pad local time

Format: <SI>18 [YYYY] [MM] [DD] [W] [HH] [MM] [SS]<SO>[LRC] (request frame)
 <SI>18 [status]<SO>[LRC] (response frame)

Message length: Fixed 20 bytes.

Usage: This message sets the A10-P local time.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<SI>	1	<0F>
18	2	Message ID
[YYYY]	4	AD year, i.e. "2014"
[MM]	2	Month, "01"~"12"
[DD]	2	Day of month, "01"~"31"
[W]	1	Day of week, "0"=Sunday, ~ "6"=Saturday
[HH]	2	Hour, "00"~"23"
[MM]	2	Minute, "00"~"59"
[SS]	2	Second, "00"~"59"
<SO>	1	<0E>
[LRC]	1	Checksum

Response frame (A10-P to HOST)

Field	Length	Value and description
<SI>	1	<0F>
18	2	Message ID
[status]	1	0: Success F: Fail
<SO>	1	<0E>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message 18 (Request frame)	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message 18 (Response frame)
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	
	←	Processing and send <EOT>

Message 19 Query Firmware Version

Format: <SI>19[part]<SO>[LRC] (request frame)
 <SI>196[options]<SO>[LRC] (request frame)
 <SI>19.[Version].[SubVer].[hash]<SO>[LRC] (response frame)

Message length: Fixed 6 bytes (request frame) / Variable (response frame).

Usage: This message is used to query the A10-P firmware version number and the firmware check sum value.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<SI>	1	<0F>
19	2	Message ID
[part]	1	Firmware Part number 0: Full version 1: System Core (A10-PLSxx) 2: Service Routine (A10-PSRxx) 3: Transaction Term. (A10-PTTxx) 4: Prompt Message (A10-PSCxx)
<SO>	1	<0E>
[LRC]	1	Checksum

Request frame (HOST to A10-P)

Field	Length	Value and description
<SI>	1	<0F>
19	2	Message ID
6	1	Payment Kernel information
[options]	1	0: Entry Point version 1: Visa AP Kernel version 2: Mastercard Kernel version 3: Visa Kernel version 4: Amex Kernel version 5: Discover Kernel version 6: JCB Kernel version 7: CUP Kernel version 8: Interact Kernel version A: CL L1 Kernel version B: CT L1 Kernel version C: CT L2 Kernel version
<SO>	1	<0E>
[LRC]	1	Checksum

Response frame (A10-P to HOST)

Field	Length	Value and description
<SI>	1	<0F>
19	2	Message ID
.	1	<2E>, field separator
[Version]	Var.	Version string (ASCII string)
.	1	<2E>, field separator
[SubVer]	2	Firmware sub version ('0'~'9')
.	1	<2E>, field separator
[hash]	40	Firmware SHA-1 hash ('0'~'9', 'A'~'F')
<SO>	1	<0E>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message 19 (Request frame)	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message 19 (Response frame)
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	
	←	<EOT>

Message 1C Query Hardware Capability

Format: <SI>1C<SO>[LRC] (request frame)
 <SI>1C[string]<SO>[LRC] (response frame)

Message length: Fixed 5 bytes (request); variable (response).

Usage: This message is used to query the peripheral capability of the A10-P.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<SI>	1	<0F>
1C	2	Message ID
<SO>	1	<0E>
[LRC]	1	Checksum

Response frame (A10-P to HOST)

Field	Length	Value and description
<SI>	1	<0F>
1C	2	Message ID
[string]	Variable	<1C>[HW1]<1C>[HW2]..... Possible strings: “ICC”: Smart card reader. “SAM”: Security Access Module Slot. “MSR”: Magnetic Stripe Reader. “PCD”: RFID reader “0” Pure A10-P without peripherals.
<SO>	1	<0E>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message 1C (Request frame)	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message 1C (Response frame)
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	
	←	<EOT>

Message 1F Query Usable Prompt Table

Format: <SI>1F<SO>[LRC] (request frame)
 <SI>1F[Prompt_List]<SO>[LRC] (response frame)

Message length: Fixed 6 bytes (request frame); variable (response frame).

Usage: This message is used to query the usable A10-P prompt message table list. The list will be represented as 2-character country codes defined by ISO 3166.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<SI>	1	<0F>
1F	2	Message ID
<SO>	1	<0E>
[LRC]	1	Checksum

Response frame (A10-P to HOST)

Field	Length	Value and description
<SI>	1	<0F>
1F	2	Message ID
[Prompt List]	Var.	Format: [active country code] <FS> [country code0] ... <FS> [country code5]
<SO>	1	<0E>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message 1F (Request frame)	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message 1F (Response frame)
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	
	←	<EOT>

Message 1G Test Prompt Table

Format: <SI>1G[option]<SO>[LRC] (request frame)

Message length: Fixed 6 bytes.

Usage: This message is used to show prompt table message on LCD display.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<SI>	1	<0F>
1G	2	Message ID
[option]	1	'0': Standard messages defined by EMV 2000 Book 4 '1': Errors Messages '2': Administration Messages '3': Transaction Messages '4': Diagnostic Messages '6': EMV Lv2 related and FW Management Messages '7': Fixed prompt Messages table
<SO>	1	<0E>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message 1G	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
		Waiting for "ENTER" keyboard to show next message or "CANCEL" keyboard to leave the test.
	←	<ACK> (The table is finished) <EOT> (Leave the test or "option" error)

Message 1J Turn ON/OFF LCD Backlight

Format: <SI>1J[option]<SO>[LRC] (request frame)
 <SI>1J[status]<SO>[LRC] (response frame)

Message length: Fixed 6 bytes.

Usage: This message can control the global backlight on or off for the LCD display of the A10-P. By default, A10-P will turn on its LCD backlight when it receives PIN entry or a clear text entry message such as 70 or Z50, and turn it off when those functions exit. By sending message “1J1”, A10-P will keep the LCD backlight turned on until message “1J0” is issued.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<SI>	1	<0F>
1J	2	Message ID
[option]	1	ASCII character ‘0’: Turn off the LCD backlight ‘1’: Turn on the LCD backlight
<SO>	1	<0E>
[LRC]	1	Checksum

Response frame (A10-P to HOST)

Field	Length	Value and description
<SI>	1	<0F>
1J	2	Message ID
[status]	1	ASCII character ‘0’: Turn off the LCD backlight ‘1’: Turn on the LCD backlight
<SO>	1	<0E>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message 1J	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message 1J (Response frame)
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	
	←	<EOT>
		LCD backlight turned ON/OFF

Message 1K Turn ON/OFF LCD Power-save mode

Format: <SI>1K[option]<SO>[LRC] (request frame)
 <SI>1K[status]<SO>[LRC] (response frame)

Message length: Fixed 6 bytes.

Usage: For power consumption saving, this message can automatically adjust the backlight of the A10-P to darker level after several seconds. The backlight level will return to the level set before when a key strobe is pressed, a message of PIN entry or a key entry is received.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<SI>	1	<0F>
1K	2	Message ID
[option]	1	ASCII character ‘0’: Turn off the LCD power-save mode ‘1’: Turn on the LCD power-save mode
<SO>	1	<0E>
[LRC]	1	Checksum

Response frame (A10-P to HOST)

Field	Length	Value and description
<SI>	1	<0F>
1K	2	Message ID
[status]	1	ASCII character ‘0’: Turn off the LCD power-save mode ‘1’: Turn on the LCD power-save mode
<SO>	1	<0E>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message 1K	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message 1K (Response frame)
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	
	←	<EOT>
		LCD power-save mode turned ON/OFF

Message 1M Setup Keypad Beeper

Format: <SI>1M[option]<SO>[LRC] (request frame)
 <SI>1M[status]<SO>[LRC] (response frame)

Message length: Fixed 6 bytes.

Usage: This message is used to enable or disable the beeper in response to the key press.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<SI>	1	<0F>
1M	2	Message ID
[option]	1	ASCII character ‘0’: Disable keypad beeper. ‘1’: Enable keypad beeper.
<SO>	1	<0E>
[LRC]	1	Checksum

Response frame (A10-P to HOST)

Field	Length	Value and description
<SI>	1	<0F>
1M	2	Message ID
[status]	1	ASCII character ‘0’: Keypad beeper disabled. ‘1’: Keypad beeper enabled.
<SO>	1	<0E>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message 1M	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message 1M (Response frame)
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	
	←	<EOT>

Message 1P Control Beeper

Format: `<SI>1P<SUB>[count]<FS>[duration]<FS>[interval]<SO>[LRC]`
 (request frame)
`<SI>1P[status]<SO>[LRC]` (response frame)

Message length: Fixed 15 bytes for request frame, and 6bytes for response frame.

Usage: This message is used to customize the behavior of beeper.
 The command runs once, user has to issue this command whenever user needs sound.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<SI>	1	<0F>
1P	2	Message ID
<SUB>	1	<1A> as parameter delimiter.
[count]	1	Iteration count of following beeps. ASCII '1' to '9'.
<FS>	1	<1C> as field separator.
[duration]	3	Duration time of each beep, the unit is 10ms. ASCII "000" to "999".
<FS>	1	<1C> as field separator.
[interval]	3	The inactive interval between each beep, the unit is 10ms. ASCII "000" to "999".
<SO>	1	<0E>
[LRC]	1	Checksum

Response frame (A10-P to HOST)

Field	Length	Value and description
<SI>	1	<0F>
1P	2	Message ID
[status]	1	ASCII character '0': success. '1': fail.
<SO>	1	<0E>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message 1P	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message 1P (Response frame)
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	
	←	<EOT>

Message M03 Load Permanent Unit Serial Number

Format: **<SI>M03[PUSN][Slot]<SO>[LRC] (response frame)**
<SI>M03[status]<SO>[LRC] (response frame)

Message length: Variable.

Usage: This message is used to load the permanent unit serial number (PUSN) into the PIN pad. PIN pad supports 8 PUSN slots for customer.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<SI>	1	<0F>
M03	3	Message ID
[PUSN]	11	Permanent USN (ASCII string). The accepted characters are in range '0' ~ '9', 'A' ~ 'Z' and '-'.
[Slot Index]	1	<Optional> 0 ~ 7: slot 0 ~ 7. NULL: slot 0.
<SO>	1	<0E>
[LRC]	1	Checksum

Response frame (A10-P to HOST)

Field	Length	Value and description
<SI>	1	<0F>
M03	3	Message ID
[status]	1	0: no error 1: PUSN format error. Input PUSN is outside the range of '0'~'9', 'A'~'Z' and '-'. 2: PUSN slot is not empty. 3: PUSN slot is out of range.
<SO>	1	<0E>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message M03 (Request frame)	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message M03 (Response frame)
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	
	←	<EOT>

Message M04 Read Permanent Unit Serial Number

Format: <SI>M04[Slot]<SO>[LRC] (request frame)
 <SI>M04[Status]<SO>[LRC] (response frame)
 <SI>M04[PUSN]<SO>[LRC] (response frame)

Message length: Fixed 6(7) bytes.

Usage: This message is used to query the permanent USN inside the PIN pad.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<SI>	1	<0F>
M04	3	Message ID
[Slot]	1	<Optional> 0 ~ 7: slot 0 ~ 7. NULL: slot 0.
<SO>	1	<0E>
[LRC]	1	Checksum

Response frame (A10-P to HOST)

Field	Length	Value and description
<SI>	1	<0F>
M04	3	Message ID
[Status] or [PUSN]	1 or 11	If the [Slot] in the request frame is out of range (0~7), this field [Status] is assigned to '1'. Otherwise, this field [PUSN] shows the Permanent USN (ASCII string).
<SO>	1	<0E>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message M04	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message M04 or <EOT> if format error
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	
	←	<EOT>

Section 6 Contact EMV Level 2 Transaction Messages

EMV Level2 transaction messages are divided into 2 groups. One group are the EMV-configuration data operation messages (T01, T03, T05, T07, T09, T0B) and the other group are the EMV-transaction messages (T11, T13, T15, T17, T19, T1C, T21, T25, T27, T29).

The issuing sequence of the ICC EMV transaction messages is controlled by the A10-P. An invalid sequence will terminate that ICC EMV transaction. In the beginning of an ICC EMV transaction, the user has to issue message T11 to make the A10-P to negotiate with the card and generate a candidate list of EMV-applications that are supported both by the A10-P and the card. It will then automatically select the application with the highest priority or the one selected by the user (according to the terminal configuration data installed in the A10-P), and then return the EMV-application name. Message T15 is used for the terminal-side to transmit transaction information such as the amount to be used by the A10-P to complete the transaction with the card if the transaction does not need to be authorized online. The terminal can issue message T1D to transmit additional transaction data to the A10-P for ICC EMV transaction. Examples are online response data and magnetic stripe card track data. Message T17 is applied if the transaction needs to be authorized online. The terminal-side will transmit necessary information via this message to the A10-P to complete the rest of transactional steps. If the response from the host contains issuer script (see EMV), the terminal-side will issue message T19 to send these scripts into the A10-P. Then the A10-P will issue these scripts at appropriate time to the card. Message T1C is used to terminate an ICC EMV transaction. Finally, message T21 is used for terminal-side to get the transaction information throughout the EMV transaction.

The terminal can apply the same set of Txx messages for handling a complete ICC EMV transaction but for turning out the transaction must be changed to magnetic stripe card processing. According to EMV rule, if the terminal fails to read the IC card, the transaction could be switched to the magnetic stripe card transaction. Because of the different types of magnetic stripe cards, the magnetic stripe card processing should be taken by the terminal. The terminal can issue message Q1 provided by the A10-P to let the user to swipe his card and then issue message 70 to complete a magnetic stripe card transaction. In this situation, the terminal will get the T11 response message that indicates a failed IC card read. The terminal should then issue message T1D, T15 and T17 to the A10-P for batch data capture. The flow chart for changing to magnetic stripe card processing can be found in the "Overall Contact EMV level 2 transaction flow reference" section.

The meanings of the error codes in the [Err Message] are listed below:

Error Code	Error Description
00000003	Service not accepted.
00000004	POS execution.
00000008	Application selection.
0000F9B	Store configuration data error.
80000001	General error.
80000003	No information.
80000005	Timeout.
8000000C	Initial application error.
8000000E	Application not initial.
80000016	ICC function not supported.
80000019	Invalid terminal data.
8000001A	Invalid ICC data.
8000001B	Invalid issuer data.
8000001D	Invalid amount.
8000001E	Amount overflow.
8D000001	File exist.
8FFF0001	Out of memory.
8FFF0002	Parameter error.
8FFFFFF02	Tag's data format error.
8FFFFFF03	Some mandatory tags are not configured well.
8FFFFFF1A	Authentication key expired.
90000003	Crypto recognized error.
A10-P00001	CVM condition not recognized.
A10-P00002	CVM condition failed.
A10-P00003	CVM failed.
A10-P00004	CVM method not recognized.
A10-P00005	CVM method not supported.
A10-P00006	CVM forced failure.
A2000001	Application initial conditions are not satisfied.
A2000002	The generated cryptogram is not allowed.
A2000003	Exceeded PIN try limit.
A2000004	PIN not match.
A2000005	PIN match.
A2000006	TRM data not found.
A2000007	The instruction ID of the specified is not recognized.
A200000A	The type to request application cryptogram is incorrect.

A200000C	The status response (SW1 SW2) is other than '9000'.
A200000D	The Generate AC command is called more than 2 times in the current transaction.
A200000E	The AIP mandatory data is missing in response data from card.
A200000F	The AFL mandatory data is missing in response data from card.
A2000010	The response template from card isn't correct.
A2000011	The format of AFL is incorrect.
A2000012	Read record failed.
A2000014	Missing mandatory data in response data from card.
A2000015	EMV data not found.
A2000016	EMV conditions not satisfied.
A2000017	EMV function is missing for terminal type.
A2000018	EMV ICC data missing.
A2000020	No tag found.
A2000022	Data incompatible type.
E0	EMV level 2 state failure.
FFFFFFFF	EMV transaction cancel.
FFFF0001	Parse TLV list error.
F1111111	Transaction cancel.
F1111112	ICC Transaction cancel.

Message T01 Terminal Configuration Setup

Format: <STX>T01[Pkt No.][Total Pkts]<SUB>[DO]<ETX>[LRC] (request frame)

Message length: Variable.

Usage: Host can use this command to send the terminal configuration data to the A10-P. This command can be sent many times. The A10-P will save those data internally and utilize them during the transaction. The A10-P will send message T02 (Terminal Configuration Setup Response) to the host as a response. The functions of Message T01 and Message T51 are exactly the same and can be used alternately.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
T01	3	Message ID
Pkt No.	1	Decimal. Packet sequence number (1 ~ 9) (ex. 2)
Total Pkts	1	Decimal. Total packets (1 ~ 9) (ex. 8).
<SUB>	1	<1A>
DO	Var.	Data Object, format as below
<ETX>	1	<03>
[LRC]	1	Checksum

Data Object Format: each data object is delimited by <SUB>, and each field inside each data object is delimited by <FS>.

Data Format: (Please also refer to EMV 4.3 BOOK3, section 4.3)

Format	Description
1	a - Alphabetic data (a ~z, A~Z)
2	b - unsigned binary numbers or bit combinations
3	an - Alphanumeric data (a ~z, A~Z, 0~9)
4	ans - Alphanumeric Special data (Characters defined in ISO8859)
5	cn - Compressed numeric data (0~9, left justify, pad hexadecimal 'F's. Ex. 12 34 56 12 3F FF)
6	n - Numeric data (0~9, right justify, pad leading hexadecimal zeroes. Ex. 00 00 00 01 23 45)
7	var - Variable data (Any bit combination)

Note. User has to follow the restriction specified in the EMV 4.3 BOOK3, Annex A and Appendix C of this document to load the configuration data. The A10-P will check if the length of each configuration data item is consistent. Any inconsistent data item will make the data loading fail.

Note. If the data format is '2' (binary), '5' (compressed numeric), '6' (numeric) or '7' (variable data), it cannot be allocated in message T01 directly. It should be transferred into a hexadecimal string and then allocated in message T01.

Example: (Clear the terminal configuration data and then setup the new data.)

Merchant Category Code: 0000 (Numerical)

Terminal ID: SmartPOS (Ascii)

UI Capability: 0x01 (binary)

```
<STX>T0111<SUB>9f15<FS>6<FS>0000<SUB>9f1c<FS>3<FS>SmartPOS<SUB>
FFFF8101<FS>2<FS>01<ETX> [LRC]
```

Special Tag for A10-P: 0xFFFF8101, 0xFFFF8103

Name	Description	Format	Tag	Length
Terminal UI Capability	0: Make A10-P selects the highest priority application and ask for the user's confirmation. 1: Make A10-P provides a list of candidate applications.	b	FFFF8101	1
MSR Processing Batch Tag List	A10-P will store transaction data according to the tag list while the transaction is performed as MSR processing. Ex, for tag list (829F365F309F399F029F03), the A10-P will store the transaction data of tag numbers 0x82, 0x9F36, 0x5F30, 0x9F39, 0x9F02 and 0x9F03.	b	FFFF8103	var

These data objects are defined in EMV 4.3 BOOK3, Annex A without tag values. The tag values are defined by UIC.

The A10-P will check if the terminal downloads the minimum set of terminal-related information into itself. The download process will fail if there is not enough data in this message. Please refer to [Appendix D](#) for the minimum set of terminal-related data

Message flow:

HOST	Direction	A10-P
1 st Message T01	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	1 st Message T02
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	
2 nd Message T01	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	2 nd Message T02
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	
...	...	...
Last one Message T01	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Last one Message T02
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message T02 Terminal Configuration Setup Response

Format: <STX>T02 [Result] [Reason] [Err Msg] [Err Tag Number] <ETX> [LRC]
(request frame)

Message length: Variable.

Usage: The response message of message T01. The functions of Message T02 and Message T52 are exactly the same and can be used alternately.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
T02	3	Message ID
T02	3	Message ID
[Result]	1	'0': OK, '1': Fail
[Reason]	1	<Optional, if Res = '1'> '1': Fatal Error '2': Format Error '3': Invalid Data Object format. '4': Invalid Tag value
[Err Message]	8	Optional, if Reason = '1', Hex decimal string
[Err Tag Number]	Var.	Optional, if Reason = '3' or '4', Hex decimal string
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow: Please refer to message T01.

Message T03 Certification Authority Public Key Setup

Format: Plain Text (Key Format: 1)

```
<STX>T03[Op code][RID][PKI][Hash Algo][Hash][PK Algo]  
[PK Leng][PK Exponent]<ETX>[LRC] (request frame)  
<STX>T03[Op code][PK Modulus]<ETX> (request frame)
```

Format: Certificate (Key Format: 2(PEM) or 3(DER))

```
<STX>T03[PktType][SeqNo]<SUB>[RID][PKI]<FS>[Cert]<ETX>[LRC]  
(request frame)  
<STX>T03[PktType][SeqNo]<SUB>[Cert]<ETX>[LRC] (request frame)
```

Message length: Variable.

Usage: Host can use this command to send the Certification Authority Public key data to the A10-P. Each command can only setup one key but this command can be sent many times. The A10-P will save those key data inside and use those data during the transaction. A10-P will send message T04 (Certification Authority Public Key Setup Response) to the host. The data is installed into the A10-P via this message and it will then be saved into its internal storage structure with a name as the concatenation of the values in the [RID] and [PKI] fields. For example, the value in the [RID] field is "A000000003" and the value in the [PKI] field is "90", then the A10-P will save these data and give an ID as "A00000000390". The functions of Message T03 and Message T53 are exactly the same and can be used alternately.

Key Format: There are three types of key format. '1' is plain text format, which is [RID][PKI][Hash Algo][Hash][PK Algo][PK Leng][PK Exponent][PK Modulus]. '2' is PEM certificate format, which is [RID][PKI][PEM certificate]. '3' is DER certificate format, which is [RID][PKI][DER certificate with HEX String format].

Message element:

1st Packet (Plain Text Public Key):

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
T03	3	Message ID
[Op code]	1	1: Load first part of RSA public key
[RID]	10	Hexadecimal string, the left 5 bytes of Registered Application Provider ID
[PKI]	2	Public Key Index, hexadecimal string. (Refer to EMV 4.3, tag '9F22')
[Hash Algorithm]	2	Hash Algorithm Index, hexadecimal string '01': SHA-1. Now, A10-P accepts only '01'.
[Hash]	40	Hash checksum, hexadecimal SHA1(RID+PKI+PKModules+PKExp)
[PK Algorithm]	2	Public Key Algorithm, hexadecimal string '01': RSA digital signature. Now, A10-P only accepts '01'.
[PK Leng]	2	Public Key Modulus length, hexadecimal string, for example: '80' = 128 bytes = 1024 bits
[PK Exponent]	1	Public Key Exponent '1': 3 '2': 216+1
<ETX>	1	<03>
[LRC]	1	Checksum

2nd Packet (Plain Text Public Key):

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
T03	3	Message ID
[Op code]	1	2: Load second part of RSA public key
[PK Modulus]	Var	Public Key Modulus, presented in hexadecimal, data length = 2*[PK Leng bytes]
<ETX>	1	<03>
[LRC]	1	Checksum

PEM certificate:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
T03	3	Message ID
[PktType]	1	'0': Initial or middle packet of CA public keys. '1': Single or last packet of CA public keys.
[SeqNo]	1	The packet sequence number, starting from '0'.
<SUB>	1	<1A> as delimiter
[RID]	10	Hexadecimal string, the left 5 bytes of Registered Application Provider ID
[PKI]	2	Public Key Index, hexadecimal string. (Refer to EMV 4.3, tag '9F22')
<FS>	1	<1C> as delimiter
[certificate]	Var.	Base64 encoded string.
<ETX>	1	<03>
[LRC]	1	Checksum

DER certificate:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
T03	3	Message ID
[PktType]	1	'4': Initial or middle packet of CA public keys. '5': Single or last packet of CA public keys.
[SeqNo]	1	The packet sequence number, starting from '0'.
<SUB>	1	<1A> as delimiter
[RID]	10	Hexadecimal string, the left 5 bytes of Registered Application Provider ID
[PKI]	2	Public Key Index, hexadecimal string. (Refer to EMV 4.3, tag '9F22')
<FS>	1	<1C> as delimiter
[certificate]	Var.	DER certificate content with Base64 encoded string format.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
1 st Message T03	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	1 st Response Message T04
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	
2 nd Message T03	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	2 nd Response Message T04
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message T04 Certification Authority Public Key Setup Response

Format: <STX>T04 [Sequence] [Result] [Reason] [Err Msg] <ETX> [LRC]
(response frame)

Message length: Variable.

Usage: The response message of message T03. The functions of Message T04 and Message T54 are exactly the same and can be used alternately.

Message element:

1st, 2nd Packet:

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
T04	3	Message ID
[Sequence]	1	Same sequence number of the corresponding T03 message.
[Result]	1	'0': OK, '1': Fail
[Reason]	1	'1': Fatal Error '2': Format Error '3': Authentication Fail
[Err Message]	8	Optional, if Reason = '1' Hex String
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow: Please refer to message T03.

Message T05 EMV Application Configuration Setup

Format: <STX>T05 [Pkt No.] [Total Pkts] <SUB> [AID] <SUB> [DO] <ETX> [LRC]
(request frame)

<STX>T05 [Pkt No.] [Total Pkts] <SUB> [DO] <ETX> [LRC] (request frame)

Message length: Variable.

Usage: Host can use this command to send out the EMV application configuration data to the A10-P. This command can be sent many times but each command is only used for one application. A10-P will save those data internally and use those data during the transaction. A10-P will send message T06 (EMV Application Configuration Setup Response) to the host. For the data installed into A10-P via this message will be saved in the internal storage structure with a name same as in the [AID] field.

Message element:

1st Message:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
T05	3	Message ID
[Pkt No.]	1	Decimal. Packet sequence number (1)
[Total Pkts]	1	Decimal. Total packets (1 ~ 9) (ex. 8).
<SUB>	1	<1A>
[AID]	10~32	EMV Application ID, refer to EMV 4.3
<SUB>	1	<1A>
[DO]	Var.	Data Object, format as below
<ETX>	1	<03>
[LRC]	1	Checksum

Rest of Message (If there are 2 more messages):

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
T05	3	Message ID
[Pkt No.]	1	Decimal. Packet sequence number (2 ~ 9) (ex. 2).
[Total Pkts]	1	Decimal. Total packets (2 ~ 9) (ex. 8).
<SUB>	1	<1A>
[DO]	Var.	Data Object, format as below
<ETX>	1	<03>
[LRC]	1	Checksum

Data Format: (Please also refer to EMV 4.3 BOOK3, section 4.3)

Format	Description
1	a - Alphabetic data (a ~z, A~Z)
2	b - unsigned binary numbers or bit combinations
3	an - Alphanumeric data (a ~z, A~Z, 0~9)
4	ans - Alphanumeric Special data (Characters defined in ISO8859)
5	cn - Compressed numeric data (0~9, left justify, pad hexadecimal 'F's. Ex. 12 34 56 12 3F FF)
6	n - Numeric data (0~9, right justify, pad leading hexadecimal zeroes. Ex. 00 00 00 01 23 45)
7	var - Variable data (Any bit combination)

Note. User has to follow the restriction specified in EMV 4.3 BOOK3, Annex A and Appendix C of this document to load the configuration data. A10-P will check if the length of each configuration data item is consistent. Any inconsistent data item will make the data loading fail.

Note. If the data format is '2' (binary), '5' (compressed numeric), '6' (numeric) or '7' (variable data), it cannot be allocated in message T05 directly. It should be transferred into a hexadecimal string and then allocated in message T05.

Example:

Default TDOL: 97 07 9f 02 06 95 05 9b 02 (binary)

Threshold Value for Biased Random Selection: 00 00 00 00 40 00(numerical)

Max. Target percentage to be used for Biased Random selection: 100 (decimal) / 0x46 (binary)

```
<STX>T0511<SUB>A00000031010<SUB>97<FS>2<FS>97079f020695059b02
<SUB>40000004<FS>6<FS>000000004000 <SUB>40000006<FS>2<FS>46<ETX> [LRC]
```

A10-P saves these data and give an ID as "A00000031010" to this group of data.

Special Tag for PIN pad: 0x40000001, 0x40000004, 0x40000005, 0x40000006, 0x40000007, 0x40000008, 0x40000009.

Name	Description	Format	Tag	Length
Application Selection Indicator	See below	n	40000001	1
Threshold Value for Biased Random Selection	See below	n	40000004	6
Target Percentage to be used for Biased Random Selection	See below	b	40000005	1
Maximum Target Percentage to be used for Biased Random Selection	See below	b	40000006	1
Terminal Action Code - Default	See below	b	40000007	5
Terminal Action Code - Denial	See below	b	40000008	5
Terminal Action Code - Online	See below	b	40000009	5
Data Tags required in Online message (ARQC)	See below	b	4000000A	var.
Data tags required in reversal message	See below	b	4000000D	var.
Data tags for batch data capture	See below	b	40000010	var.
ARC Approve	See below	b	4000001A	var.
ARC Decline	See below	b	4000001B	var.
ARC Referral	See below	b	4000001C	var.

These data objects are defined in EMV 4.3 BOOK3, Annex A without tag values. The tag values are defined by the A10-P.

Message flow:

HOST	Direction	A10-P
1 st Message T05	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	1 st Message T06
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	
2 nd Message T05	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Application Select Response. 2 nd Message T06
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	
...	...	...
Last one Message T05	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Application Select Response. Last one Message T06
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

The A10-P will check if the terminal downloads the minimum set of EMV Application-related information into itself. The download process will fail if there is not enough data in this message. Please refer to **Appendix D** for the minimum set of EMV Application-related data.

Message T06 EMV Application Configuration Setup Response

Format: <STX>T06[Resp][Reason][Err Msg]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: The response message of message T05.

Message element:

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
T06	3	Message ID
[Resp]	1	'0': OK, '1': Fail
[Reason]	1	'1': Fatal Error '2': Format Error '3': Authentication Fail '4': Invalid Tag value
[Err Message]	8	Optional, if Reason = '1' Hex String
[Err Tag Number]	Var.	Optional, if Reason = '3' or '4', Hex decimal string
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow: Please refer to message T05.

Message T07 Data Format Table Setup

Format: <STX>T07 [Clear] <SUB> [DO] <ETX> [LRC] (request frame)

Message length: Variable.

Usage: A10-P will check the consistency of the TLV object from the terminal via messages T01 and T05. The terminal can add more TLV format checking rules into the A10-P. A10-P will send message T08 (Data Format Setup Response) to the host as the response.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
T07	3	Message ID
[Clear]	1	'1': Clear existing Data Format file
<SUB>	1	<1A>, Optional, only if [DO] exist
[DO]	Var.	Data Object, format as below
<ETX>	1	<03>
[LRC]	1	Checksum

Format of Data Objects:

The Data Objects for T07 shall include five data fields delimited by <FS>: [Tag#] <FS> [Format] [Min Len] [Max Len] [Flag]. Each data object is delimited by a <SUB> to construct multiple <DO>s. If the [Flag] is '0', the data object will accept the length of value within [Min Len] and [Max Len]; if the [Flag] is '1', the data object will only accept either the [Min Len] or [Max Len] as valid length.

Example: (setup new data format.)

Tag	Data Format	Min Len	Max Len	Or Flag
5F88	4	02	1A	0 (data length is 2 ~ 26 bytes)
5F99	2	08	0B	1 (data length is 8 bytes or 11 bytes)

<STX>T071<SUB>5F88<FS>4021A0<SUB>5F99<FS>2080B1<ETX> [LRC]

In this example, A10-P will check if the length of data with tag value 5F99 is 8-byte or 11-byte. It will check if the length of data with tag value 5F88 is more than 1-byte and less than 27-byte.

Message flow:

HOST	Direction	A10-P
Message T07	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message T08
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message T08 Data Format Table Setup Response

Format: <STX>T08 [Resp] [Reason] [Err Msg]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: The response message of message T07.

Message element:

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
T08	3	Message ID
Resp	1	'0': OK, '1': Fail
Reason	1	<Optional, if Resp = 1> '1': Fatal Error '2': Format Error
Err Message	8	Optional, if Reason = '1' Hex String
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow: Please refer to message T07.

Message T09 EMV Config Data Query Message

Format: <STX>T09[Config Type]<ETX>[LRC] (request frame)

Message length: Fixed 7 bytes.

Usage: Get the group ID of the EMV application data or the CA public key stored in the A10-P.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
T09	3	Message ID
[Config Type]	1	1: All the IDs of the CA public key. 2: All the IDs of the EMV application data.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message T09	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message T0A
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message T0A Response of EMV Config Data Query Message

Format: <STX>T0A[Config Type][Status]<SUB>[ID List]<ETX>[LRC]
 (response frame)

Message length: Variable.

Usage: The response message of message T09.

Message element:

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
T0A	3	Message ID
Config Type	1	Return the input data in the field [Config Type] of T09.
Status	1	0: There is data in identified data type. 1: No data in identified data type. 2: Arrangement exceeded.
<SUB>	1	<1A>, Optional if Status is 0
ID List	Var	Optional, if Status is 0.
[LRC]	1	Checksum

[ID List]: The concatenation of IDs. There is a <FS> between each ID.

Ex. [ID List] = A00000000390<FS> A000000004F8<FS> A000000004F5 (Config Type = 1)

[ID List] = A0000000041010<FS>B0123456781234<FS> A000000010 (Config Type = 2)

Message flow: Please refer to message T09.

Message T0B Delete EMV Configuration Data Message.

Format: <STX>T0B[Config Type]<SUB>[ID List]<ETX>[LRC] (request frame)

Message length: Variable.

Usage: Host can use this command to delete the EMV application configuration data or the CA public key in the A10-P. This command can be sent many times.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
T0B	3	Message ID
[Config Type]	1	1: CA public key. 2: EMV application data.
<SUB>	1	<1A>
[ID List]	Var	List of IDs that will be erased.
<ETX>	1	<03>
[LRC]	1	Checksum

[ID List]: The concatenation of IDs. There is a <FS> between each ID.

Ex. [ID List] = A00000000390<FS> A000000004F8<FS> A000000004F5 (Config Type = 1)

[ID List] = A0000000041010<FS>B0123456781234<FS> A000000010 (Config Type = 2)

Message flow:

HOST	Direction	A10-P
Message T0B	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message T0C
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message T0C Response of Delete EMV Configuration Data Message.

Format: <STX>T0C[Config Type][Status]<SUB>[Del Result]<ETX>[LRC]
(response frame)

Message length: Variable.

Usage: The response message of message T0B.

Message element:

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
T0C	3	Message ID
[Config Type]	1	1: CA public key. 2: EMV application data.
[Status]	1	0: Delete OK 1: Fatal error. 2: Format error. 3: No CA public key or EMV application data exists.
<SUB>	1	<1A>, Optional, if Status is 0.
[Del Result]	Var.	Optional, if Status is 0. 0: the ID deletion is successful. 1: the ID deletion is failure.
<ETX>	1	<03>
[LRC]	1	Checksum

[Del Result]: It's the concatenation of the results for the delete operation on the indicated ID. There is a <FS> between each result. If the delete operation is successful (when A10-P finds out the matched one and delete it successfully), the result is 0. If there is no such ID as the terminal specified, the result is 1.

Ex. Terminal issues message T0B to erase the EMV application data with ID "A0000000102020", "A0000020" and "A0000000103030". Now the A10-P keeps the EMV application data with ID "A0000000103030" only.

The data in the [Del Result] field will be "1<FS>1<FS>0".

Message flow: Please refer to message T0B.

Message T11 Application Select

Format: <STX>T11<ETX>[LRC] (request frame)

Message length: Fixed 6 bytes.

Usage: A10-P performs an application select on the active smart card. A10-P will prompt the user to insert its card if the card has not yet be engaged. A10-P will send message T12 (Application Selection Response) to host as a response.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
T11	3	Message ID
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message T11	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Application Select Response. <i>Message T12</i>
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message T12 Application Select Response

Format: <STX>T12 [Status] [Application Name] <ETX> [LRC] (response frame)

 <STX>T12 [Status] [Reason] [ErrorMessage] <ETX> [LRC] (response frame)

Message length: Variable, depending on the length of returned application name.

Usage: The message contains the name of the final selected application on the smart card to be sent to the host. In general, the application name conforms to the EMV 4.3 level 2 specification. Refer to these standards for more detailed information.

Message element:

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
T12	3	Message ID
Status	1	0:OK
[Application Name]	10-32	EMV application is selected successfully. Format: Hexadecimal string. The content of this field does not exist if no EMV application is selected.
<ETX>	1	<03>
[LRC]	1	Checksum

OR

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
T12	3	Message ID
Status	1	1:Fail
[Reason]	1	1:Fatal Error 2:Command Format Error 3:Transaction Canceled 4: MSR processing.
[Err Message]	8	Optional, if Reason = '1' Hex String
<ETX>	1	<03>
[LRC]	1	Checksum

NOTE. According to the EMV Level2's rule, if the IC card cannot be read, A10-P will return message "T1214" to indicate that the terminal should apply magnetic stripe card processing. In this situation, the terminal should issue message Q1 to get the magnetic stripe data and issue message T1D to inform the A10-P on the magnetic stripe card data element. Message T15 is used to transmit the necessary data records (amount, transaction type, and so on...) for the A10-P to complete this transaction.

Message flow: Please refer to message T11.

Message T13 Application Select Next

Format: <STX>T13<ETX>[LRC] (request frame)

Message length: Fixed 6 bytes.

Usage: If the selected EMV application is blocked (the terminal will know that through message T16), the terminal can issue this message to ask the A10-P to display the rest of the EMV applications for cardholder selection.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
T13	3	Message ID
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message T13	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Application Select Response. <i>Message T12</i>
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message T15 Start Transaction

Format: <STX>T15<SUB>[AmtAuth]<SUB>[AmtOther]<SUB>
[CurExponent][CurCode]<SUB>[TranType]<SUB>[TranInfo]<SUB>
[Account Type]<SUB>[Force Online]<SUB>
[Encrypted Session key]<ETX>[LRC] (request frame)

Message length: Variable.

Usage: A10-P performs a completed EMV transaction flow (via 'Initiate Application' through 'Completion', see EMV 4.3, book 3, chap 8.2, figure 6 - transaction flow example) based on the selected EMV application that has the corrected application name on the T12 (Application Select response). The A10-P will prompt the user to arrange the appropriate entries when informed, like confirmation or enter the PIN code. A10-P will send message T16 (Start Transaction Response) to the host as a response. If the IC card cannot be read (switch to magnetic stripe card processing), the A10-P will arrange the information provided from this message and add the magnetic stripe card data to finish the EMV transaction (The rest of operation on magnetic stripe card is outside the scope of EMV). The terminal could have extra operation on magnetic stripe card transaction (by issuing message 70 for PIN, and so on.).

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
T15	3	Message ID
<SUB>	1	<1A>
[AmtAuth]	12	Hexadecimal, Amount Authorized tag '0x9F02'
<SUB>	1	<1A>
[AmtOther]	12	Hexadecimal, Amount Other, tag '0x9F03'
<SUB>	1	<1A>
[CurExponent]	1	Hexadecimal, tag '0x5F36'
[CurCode]	3	Hexadecimal, tag '0x5F2A'. For example, USD\$ = 0x840
<SUB>	1	<1A>
[TranType]	2	Hexadecimal, Transaction Type, tag '0x9c'
<SUB>	1	<1A>
[TranInfo]	2	Transaction Info, tag '0xFFFF8109'
<SUB>	1	<1A>
[Account	2	Account Type, tag '0x5F57'

Type]		
<SUB>	1	<1A>
[Force Online]	1	'1': Force this transaction to be authorized online. Valid only if the terminal configuration enables the capability to support online authorization. '0': Run the default EMV authorization flow
<SUB>	1	<1A> (Optional, if the Encrypted Session key exists)
[Encrypted Session key]	16 or 32 or 48	(Optional, if the transaction needs the user to enter the PIN for online authorization. If no session key exists, the A10-P will use PIN encryption key (Usage P0) or DUKPT key exist in A10-P). Only DES session key / TDES session key are used for EMV online PIN entry.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message T15	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Start the Transaction Response. <i>Message T16</i>
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message T16 Transaction Response

Format: <STX>T16[Status][Reason][Err Message][Result]<SUB>
 <Advice Need><Reversal Need><Financial Need><ETX>[LRC]
 (response frame)

Message length: Variable.

Usage: The message contains the transaction result on the smart card to be sent to terminal.

Message element:

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
T16	3	Message ID
[Status]	1	0:OK; 1:Fail
[Reason]	1	Optional. (If Status = 1) 1:Fatal Error 2:Command Format Error
[Err Message]	8	Optional, if Reason = '1'Hex String
[Result]	2	Optional. (If Status = 0) 'Y1': Offline Approved, 'Z1': Offline Declined 'Y3': Unable to go online, Offline Approved 'Z3': Unable to go online, Offline Decline. 'Y4': Online Approved 'Z4': Online Decline 'A1': Online Authorize Request, 'A4': Application reselection.
<Reversal Need>	1	0: Terminal does not need to send a reversal to the host for this transaction 1: Terminal should send a reversal to the host for this transaction.
<ETX>	1	<03>
[LRC]	1	Checksum

Note: If the [Result] is 'A1', then the terminal should send the online authorization request to the issuer host. After that is done, send message T17 to the A10-P to continue the transaction. (See EMV 4.3, book 3, chap 9, figure 7)

Note: If the transaction is switched to MSR processing, T16 will always return "A1" in the [Result] field.

Note: If the previous selected application on the IC card cannot perform the transaction (for example, this application has blocked) but has another application ID within this IC card, the A10-P will respond 'A4' to let the terminal knows and the terminal can issue message T13 to select another application ID and issue message T15 to re-start the transaction. Please refer to the paragraph of "Ref. 5 Packet command flow for first EMV application is blocked" in the section of "Overall EMV Level 2 transaction flow reference".

Message flow: Please refer to message T15.

Message T17 Send Online Authorized Code

Format: <STX>T17[OnlineRes]<ETX>[LRC] (request frame)
<STX>T17[OnlineRes]<SUB>[ARC]<SUB>[IAD]<ETX>[LRC]
(request frame)

Message length: Variable.

Usage: A10-P continues to perform the EMV transaction flow after receiving this online response from the host when the transaction response in message T16 is 'A1' (online authorized request, see EMV 4.3, book 3, chap 8.2, figure 6 – transaction flow example). A10-P will send the message T16 (Start Transaction Response) to the host.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
T17	3	Message ID
Online Res	1	'0': Unable to go online
<ETX>	1	<03>
[LRC]	1	Checksum

OR

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
T17	3	Message ID
Online Res	1	'1' : Get Online Response,
<SUB>	1	<1A>
ARC	2	Optional, if there is ARC from host. Authorisation Response code, ASCII (0~9, A~Z),.
<SUB>	1	<1A>
[IAD]	16~32	Optional, if there is IAD from host. Issuer Authentication Data , Hex string.
<ETX>	1	<03>
[LRC]	1	Checksum

OR (If the transaction is changed to Magnetic stripe card processing)

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
T17	3	Message ID
Online Res	1	'3': MSR Online Approve Other: MSR Online Decline,
<ETX>	1	<03>
[LRC]	1	Checksum

Note: If [Online Res] is '4', then A10-P will decide to approve or decline by the rule of TAC/IAC default.

Message flow:

HOST	Direction	A10-P
Message T17	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Start Transaction Response. <i>Message T16</i>
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message T19 Send Issuer Script Command

Format: <STX>T19[IS]<ETX>[LRC] (request frame)

Message length: Variable.

Usage: A10-P performs the Issuer script processing as in the EMV transaction flow. After receiving this command from the host as the response message upon performing online authorization. This command can be sent many times if too many script commands needed to be processed. But the last one should be sent before message T17.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
T19	3	Message ID
[IS]	Var.	Issuer Script, format as below. Hex string.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message T19	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message T20
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Issuer Script Format: (see EMV 4.3, book 3, chap 10, Figure 14)

T	L	T	L	Script ID	Commands
'71' or '72'	Including Script ID, tags, lengths	'9F18'	'04'	Identifier (4 bytes)	Issuer Script Command Format (see below)

Issuer Script Command Format: (see EMV 4.3, book 3, chap 10.10, Figure 15)

T1	L1	V1	T2	L2	V2	T3	L3	V3	Tx	Lx	Vx
'86'	L(V1)	cmd	'86'	L(V2)	cmd	'86'	L(V3)	cmd	'86'	L(Vx)	cmd

Message T20 Send Issuer Script Command Response

Format: <STX>T20 [Status] [Reason] [Err Message] <ETX> [LRC] (response frame)

Message length: Variable.

Usage: The message response to message T19.

Message element:

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
T20	3	Message ID
Status	1	0:OK ; 1:Fail
Reason	1	Optional. (if Status = 1) 1:Fatal Error 2:Command Format Error
Err Message	8	Optional, if Reason = '1' Hex String
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow: Please refer to message T19.

Message T1C Terminal Cancel Transaction

Format: <STX>T1C<ETX>[LRC] (request frame)

Message length: Fixed 6 bytes.

Usage: This message is used to cancel the transaction.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
T1C	3	Message ID
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message T1C	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)

Message T1D Transaction/Configuration Data Overwrite

Format: <STX>T1D<SUB>[DO]<ETX>[LRC] (request frame)

Message length: Variable.

Usage: The terminal can use this command to update transaction/configuration data of the A10-P in run time. This command can be sent multiple times if needed. A10-P will save those data internally and use those data during the transaction. As a response the A10-P will send the message T1E to the terminal.

[Notes]

1. T1D will temporarily override the transaction/configuration data (which were set by the previous configuration commands) within this transaction. Those previously configurations will be resumed in the new transaction.
2. Any erroneous transaction data update may fail the transaction process.
3. For the issuer returned authorization data such as ARC, IAD, or issuer scripts should use message T17 to send to the A10-P in order to complete the EMV transaction.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
T1D	3	Message ID
<SUB>	1	<1A>, Optional, only if [DO] exists
DO	Var.	Data Object, format as below
<ETX>	1	<03>
[LRC]	1	Checksum

Data Object:

Each <DO> shall include three data fields: [Tag#] || [Format] || [Value], and each field shall delimit with a <FS>. Each data object is delimited by a <SUB> to construct multiple <DO>s. The [tag#] defined in EMV 4.3 Book3 Annex A and specific [tag#] defined at **Appendix C** of this manual have the pre-defined data format and length range. These [tag#] must follow up with these rules, otherwise the A10-P will reject this data setup. Any customer defining [tag#] number shall use message T07 to pre-define the data format and length range before using message T1D to load the data.

Data Format: (Please also refer to EMV 4.3 BOOK3, section 4.3)

Format	Description
1	a - Alphabetic data (a ~z, A~Z)
2	b - unsigned binary numbers or bit combinations
3	an - Alphanumeric data (a ~z, A~Z, 0~9)
4	ans - Alphanumeric Special data (Characters defined in ISO8859)
5	cn - Compressed numeric data (0~9, left justify, pad hexadecimal 'F's. Ex. 12 34 56 12 3F FF)
6	n - Numeric data (0~9, right justify, pad leading hexadecimal zeroes. Ex. 00 00 00 01 23 45)
7	var - Variable data (Any bit combination)

Example:

(If the IC card cannot be read, the terminal will switch to MSR processing and then load these data into the A10-P.)

Service Code: 0211 (Numeric)

PAN: 123456789022334455(Compressed Numeric)

Terminal sends,

<STX>T1D<SUB>5F30<FS>6<FS>0211<SUB>5A<FS>5<FS>123456789022334455<ETX> [LRC]

Message flow:

HOST	Direction	A10-P
Message T1D	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message T1E
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message T1E Transaction Data loading Response

Format: <STX>T1E[Res][Reason][Err Msg]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: The response message of message T1D.

Message element:

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
T1E	3	Message ID
Res	1	'0': OK, '1': Fail
Reason	1	<Optional, if Res = '1'> '1': Fatal Error '2': Format Error '3': Invalid Data Object format. '4': Invalid Tag value
Err Message	8	Optional, if Reason = '1', Hex decimal string
Err Tag Number	Var.	Optional, if Reason = '3' or '4', Hex decimal string
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow: Please refer to message T1D.

Message T21 Get Transaction Result's Data

Format: <STX>T21 [DOL] <ETX> [LRC] (request frame)

Message length: Variable.

Usage: A10-P will retrieve the data listed on the DOL after the EMV transaction is done. A10-P will send message T22 (Get Transaction Result's Data Response) to the host as a response.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
T21	3	Message ID
DOL	Var.	Data Object List: each object is expressed by tag number and <SUB> is used to delimit each object. For example, 9F12<SUB>9A<SUB>9F02<SUB>....
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message T21	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> after 3 NAKs
	←	Get Transaction Result's Data Response. <i>Message T22</i>
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message T22 Get Transaction Result's Data Response

Format: <STX>T22[DO]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: This message contains the data of the transaction result to be sent to the host.

Message element:

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
T22	3	Message ID
[DO]	Var.	Data Object: each data object is expressed by the TLV format with an <FS> delimit in each field, and <SUB> is used to delimit each object. For example: 9F12<FS>0F<FS>CREDITO DE VISA<SUB>9A<FS>06<FS>050806<SUB>
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow: Please refer to message T21.

Message T23 Erase EMV Transaction Log

Format: **<STX>T23<ETX>[LRC] (request frame)**

Message length: Fixed 6 bytes.

Usage: The message is used to purge the memory (flash) area for transaction logs.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
T23	3	Message ID
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message T23	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)

Message T25 Get Batch Data

Format: <STX>T25<ETX>[LRC] (request frame)

Message length: Fixed 6 bytes.

Usage: This message is used to get the batch data (EMV).

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
T25	3	Message ID
<ETX>	1	<03>
[LRC]	1	Checksum

In each issuing of message T25, the A10-P will return one record in the batched data via message T26. The terminal can keep issuing message T25 until the A10-P returns a message to indicate that there is no more record in the batch data.

User can load a tag list of batch data into the A10-P, and the A10-P will search the corresponding values according to the tag list and store it as a transaction record. Please refer to **Appendix C** for more details.

Note. If there is no more memory space for batch data, the EMV transaction will turn out to be failed.

Please issue message T23 to release the memory space.

Note. A10-P can store at most 1000 transaction records.

Message flow:

HOST	Direction	A10-P
Message T25	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message T26 (T262)
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	
Message T25	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message T26 (T262)
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	
...	...	...
Message T25	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message T26 (T261)
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	A10-P erases batch data.

Message T26 Response of Get Batch Data message

Format: <STX>T26[Status] [record of Batch data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Return the batch data to the terminal.

Message element:

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
T26	3	Message ID
Status	1	0: No more record. 1: It is the last record inside the A10-P. 2: There is still record existing inside the A10-P.
[Batch data]	Var.	Hex string.
<ETX>	1	<03>
[LRC]	1	Checksum

Format of Batch data:

According to the tag list of the batch data (refer to **Appendix C**), A10-P will search the corresponding values of the tag list and store it as a transaction record.

[EMV Tag Number (1 ~ 4 byte)] || [Length (1byte)] || [Value] || [EMV Tag Number (1 ~ 4 byte)] || [Length (1byte)] || [Value] || ...[EMV Tag Number (1 ~ 4 byte)] || [Length (1byte)] || [Value]. A10-P can save up to 256 bytes of data as one record. The right part of data will be ignored if the length record is greater than 256 bytes.

A10-P will unpack the binary data into hex string and return to the terminal. For example, if the data of one record is 0x9F02 || 06 || 000000001100 || 5A || 10 || 1122334455667 7889900AABBCCDDEEFF, then the terminal will see "9F02060000000011005A10-P11223344556677889900AABBCCDDEEFF" in the [Batch data] field.

Message flow: Please refer to message T25

Message T27 Get Online authorization Data

Format: <STX>T27<ETX>[LRC] (request frame)

Message length: Fixed 6 bytes.

Usage: Issue this message to get the (EMV) data for online authorization.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
T27	3	Message ID
<ETX>	1	<03>
[LRC]	1	Checksum

While the EMV transaction must be authorized online, message T27 is issued to get the necessary data for online authorization from the A10-P. A10-P will wait for the terminal to send message T17 as to inform the go online authorization result.

During the application configuration setup (using message T05) the user shall load a data object's tag list in tag# 4000000A into the A10-P to be used as the online authorization data. After receiving message T27 from the terminal, A10-P will search the corresponding values according to this tag list and return that online authorization data with message T28 . Please refer to **Appendix C** for more details.

Message flow:

HOST	Direction	A10-P
Message T27	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message T28
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message T28 Response of Get Online authorization Data message

Format: <STX>T28[online authorization data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Return the online authorization data to the terminal.

Message element:

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
T28	3	Message ID
Online authorization data	Var.	Hex string. Optional, if this transaction needs to be authorized online.
<ETX>	1	<03>
[LRC]	1	Checksum

Online authorization data:

According to the tag list of online authorization data (refer to **Appendix C**), A10-P will search the corresponding values according to the tag list and return it as the transaction record.

[EMV Tag Number (1 ~ 4 byte)] || [Length (1byte)] || [Value] || [EMV Tag Number (1 ~ 4 byte)] || [Length (1byte)] || [Value] || ...[EMV Tag Number (1 ~ 4 byte)] || [Length (1byte)] || [Value]. A10-P can return up to 256 bytes of data as one record. The right part of data will be ignored if the length record is greater than 256 bytes.

A10-P will unpack the binary data into hex string and return to the terminal. For example, the data of one record is 0x9F02 || 06 || 000000001100 || 5A || 10 || 11223344556677889900AABBC

CDDEEFF, the terminal will see "9F02060000000011005A1011223344556677889900AABBCDDEEFF" in the [Batch data] field.

Message flow: Please refer to message T27

Message T29 Get Reversal Data

Format: <STX>T29<ETX>[LRC] (request frame)

Message length: Fixed 6 bytes.

Usage: Issue this message to get the (EMV) data for reversal.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
T29	3	Message ID
<ETX>	1	<03>
[LRC]	1	Checksum

If the EMV transaction needs a reversal, issuing message T29 to get the necessary data from the A10-P for reversal. User can load a tag list for reversal data into the A10-P and the latter will search the corresponding values according to the tag list and return it as a transaction record. Please refer to **Appendix C** for more details.

Message flow:

HOST	Direction	A10-P
Message T29	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message T2A
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Note. Terminal can always get reversal data even if message T16 indicates that the reversal is not needed.

Message T2A Response of Get Reversal Data message

Format: <STX>T2A[Reversal data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Return the reversal data to the terminal

Message element:

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
T2A	3	Message ID
Reversal data	Var.	Hex string. Optional, if this transaction needs a reversal.
<ETX>	1	<03>
[LRC]	1	Checksum

Reversal data:

According to the tag list of the reversal data (refer to **Appendix C**), A10-P will search the corresponding values according to the tag list and return it as a transaction record.

[EMV Tag Number (1 ~ 4 byte)] || [Length (1byte)] || [Value] || [EMV Tag Number (1 ~ 4 byte)] || [Length (1byte)] || [Value] || ...[EMV Tag Number (1 ~ 4 byte)] || [Length (1byte)] || [Value]. A10-P can return up to 256 bytes of data as one record. The right part of data will be ignored if the record length is greater than 256 bytes.

A10-P will unpack the binary data into hex string and return to the terminal. For example, the data of one record is 0x9F02 || 06 || 000000001100 || 5A || 10 || 11223344556677889900AABBC

CDDEEFF, the terminal will see
"9F02060000000011005A1011223344556677889900AABBCDDEEFF" in the [Batch data] field.

Message flow: Please refer to message T29

Message T2B Get Online authorization Data with multi packets

Format: <STX>T2B[control]<ETX>[LRC] (request frame)

Message length: Fixed 6 or 7 bytes.

Usage: Issue this message to get the (EMV) data for online authorization.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
T2B	3	Message ID
[control]	1	Optional, as acknowledgment of multiple packets '0' = ready for next packet
<ETX>	1	<03>
[LRC]	1	Checksum

If the EMV transaction must be authorized online, issue message T2B from the A10-P to get the necessary data for online authorization. If the transaction must be authorized online, the user can load a tag list of online authorization data into the A10-P. The latter will search for the corresponding values according to the tag list and returns it as the transaction record.

Message flow:

HOST	Direction	A10-P
Message T2B (1st packet)	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> if LRC error after 2 retries.
	←	Message T2C (1st packet)
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	
Message T2B (2nd packet) with [control] = '0'	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message T2C (2nd packet)
...	...	...
Message T2B (final)	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message T2C (final packet) with [PktType] = '1'.
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message T2C Response of Get Online authorization Data message with multi packets

Format: <STX>T2C[PktType][SeqNo]<SUB>[record of online authorization data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Return the online authorization data to the terminal.

Message element:

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
T2C	3	Message ID
[PktType]	1	'0' = There is still remaining data. '1' = The last packet. '2' = sequence error. '3' = Format error.
[SeqNo]	1	The packet sequence number, start from '0'.
<SUB>	1	<1A> as delimiter
[online authorization data]	Var.	Hex string. Optional, if this transaction needs to be authorized online.
<ETX>	1	<03>
[LRC]	1	Checksum

Online authorization data:

According to the tag list of online authorization data which is defined at tag# 4000000A, A10-P will search the corresponding values according to this tag list and returns a series of data objects with TLV (tag || length|| value) format as below:

[EMV Tag Number (2 ~ 8 byte)] || [Length (2byte)] || [Value] <SUB>

[EMV Tag Number (2 ~ 8 byte)] || [Length (2byte)] || [Value] <SUB>

.....

[EMV Tag Number (2 ~ 8 byte)] || [Length (2byte)] || [Value]

When the A10-P responds to these data objects, it will convert their values from binary value to a hex decimal string if the data object format is "b" or "cn" or "n".

For example:

The data of one record:

9F02 || 06 || 000000001100 || 5A || 10 || 11223344556677889900AABBCCDDEEFF

The terminal will see data:

"9F0206000000001100<SUB>5A1011223344556677889900AABBCCDDEEFF"

Message Flow:

Refer to message T2B.

Message T2H Get Transaction Data with Multiple Packets

Format: <STX>T2H[Request Type][control]<ETX>[LRC] (request frame)

Message length: Variable.

Usage: This message is used to query the following types of transaction data:

- Online Authorization Data. (Equal to T27)
- Batch Data (Ref. EMV book4 12.1.5; Only for the latest transaction).
- Reversal Data. (Equal to T29)

If the output data objects are longer than the maximum packet length, A10-P will divide them into multiple response packets

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
T2H	3	Message ID
[Request Type]	1	'1' = Online authorization data (object list defined by application config 4000000A) '2' = (Not supported) '3' = Batch data (application config 40000010) '4' = Reversal data (application config 4000000D)
[control]	1	Optional, as acknowledgment of multiple packets '0' = ready for next packet
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message T2H (1st packet)	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> if LRC error after 2 retries.
	←	Message T2I (1st packet)
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	
Message T2H (2nd packet) with [control] = '0'	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message T2I (2nd packet)
...	...	...
Message T2H (final)	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message T2I (final packet) with [PktType] = '1'.
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message T2I Response of Transaction Data with Multiple Packets

Format: <STX>T2I[PktType][SeqNo]<SUB>[TLV records]<ETX>[LRC]
(response frame)
<STX>T2I[err code]<ETX>[LRC] (error frame)

Message length: Variable.

Usage: This is the response message of message T2H.

Message element:

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
T2I	3	Message ID
[PktType]	1	'0' = There is still remaining data. '1' = The last packet.
[SeqNo]	1	The packet sequence number, start from '0'.
<SUB>	1	<1A> as delimiter
[Condensed TLV records]	Var.	The TLV (tag-length-value) records of the data objects listed in the application configuration.
<ETX>	1	<03>
[LRC]	1	Checksum

Details of Condensed TLV records:

- TLV record for a single data object is omitting field separator <FS>:
[Tag Number (2 ~ 8 bytes hex)][Length (2bytes hex)][Value].
- Each data object is delimited by the <SUB> character, including the 1st one.
- The value with format codes 'a', 'an', 'ans' will be expressed as an ASCII string, and format codes 'b', 'cn', 'n', 'var' will be expressed as a hexadecimal string.

Message element:

Error frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
T2I	3	Message ID
[ErrCode]	1	'2' = Packet sequence error. '3' = Command format error.
<ETX>	1	<03>
[LRC]	1	Check value

Message T31 Extract Information on Smart Card (non-EMV transactions)

Format: <STX>T31<ETX>[LRC] (request frame)

Message length: Fixed 6 bytes.

Usage: PIN Pad performs the application selection, initiate application processing and read application data on the active smart card. All information will be stored in RAM temporarily but it is not retrievable until host sends out the message T3C to force the transaction declined or host sends out the message T35 to complete the transaction, and then the host will be able to get the information by message T21.

PIN Pad will also prompt the user to insert a smart card if the card has not yet presented. PIN Pad will send the message T12 (Application Selection Response) to host.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
T31	3	Message ID
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message T31	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Application Select Response. Message T12
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message T33 Start Quick Chip Transaction

Format: <STX>T33<SUB> [AmtAuth] <SUB> [AmtOther] <SUB> [CurExponent] [CurCode]
 <SUB> [TranType] <SUB> [TranInfo] <SUB> [Account Type] <SUB> [Timeout]
 <SUB> [DataObjects] <ETX> [LRC] (request Frame)
 <STX>T330<ETX> (acknowledge frame)

Message length: Variable.

Usage: This message is used to start Quick Chip transaction and acknowledge the response message in case of multiple packets scenario.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
T33	3	Message ID
<SUB>	1	<1A> as delimiter
[AmtAuth]	12	“Amount Authorized” field (EMV tag ‘0x9f02’) Refer to message T15.
<SUB>	1	<1A> as delimiter
[AmtOther]	12	“Amount Other” field (EMV tag ‘0x9f03’) Refer to message T15.
<SUB>	1	<1A> as delimiter
[CurExponent]	1	Hexadecimal, tag ‘0x5F36’
[CurCode]	3	Hexadecimal, tag ‘0x5F2A’. For example, USD\$ = 0x840
<SUB>	1	<1A> as delimiter
[TranType]	2	Hexadecimal, Transaction Type, tag ‘0x9c’
<SUB>	1	<1A> as delimiter
[TranInfo]	2	Transaction Info, tag ‘0xFFFF8109’
<SUB>	1	<1A> as delimiter
[Account Type]	2	Account Type, tag ‘0x5F57’
<SUB>	1	<1A> as delimiter
[Timeout]	3	Decimal string to indicate how many seconds does reader wait for ICC insertion (001 ~ 255).
<SUB>	1	<1A> as delimiter
[Data Object]	Var.	Optional. Additional data objects to be used in this transaction. Refer to message T1D for its

		detailed format.
<ETX>	1	<03>
[LRC]	1	Checksum

Acknowledge frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
T33	3	Message ID
0	1	Acknowledge
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message T33	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC)
		If ICC already inserted, start transaction immediately, else waitfor ICC insert until times out.
	←	Message T34 (1st packet)
<ACK> (Good LRC) <NAK> (Bad LRC)	→	
Message T33 (acknowledge)	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC)
	←	Message T34 (2nd packet)
...	...	...
Message T33 (acknowledge)	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC)
	←	Message T34 (final packet)
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message T34 Quick Chip Transaction Result

Format: <STX>T34 [Status] [Result] <SUB> [Control] [Seq] [ARQ_Data] <ETX> [LRC]
 (response frame)
 <STX>T34 [Status] [Reason] [ErrCode] <ETX> [LRC] (error frame)

Message length: Variable.

Usage: The response of the Quick Chip transaction.

Message element:

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
T34	3	Message ID
[Status]	1	Optional, will be omitted for subsequent packets. '0' for OK.
[Result]	2	Optional, will be omitted for subsequent packets. Transaction result codes: "Y1": Offline Approved, "Z1": Offline Declined "Y3": Unable to go online, Offline Approved "Z3": Unable to go online, Offline Decline.
<SUB>	1	<1A> as delimiter
[Control]	1	'0': more data packets follow. '1': the last data packet.
[Seq]	1	'0' ~ '9': data sequence number.
[ARQ_Data]	Var.	Pre-defined EMV data objects output for online authorization. Data object list defined in private tag 4000000A (see message T27.)
<ETX>	1	<03>
[LRC]	1	Check value

Detail of ARQ_Data (Condensed TLV records):

- TLV record for a single data object is omitting field separator <FS>:
 [Tag Number (2 ~ 8 bytes hex)][Length (2bytes hex)][Value].
- Each data object is delimited by the <SUB> character, including the 1st one.
- The value with format codes 'a', 'an', 'ans' will be expressed as an ASCII string, and format codes 'b', 'cn', 'n', 'var' will be expressed as a hexadecimal string.

Message element:

Error frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
T34	3	Message ID
[Status]	1	'1' for Fail.
[Reason]	1	'1': Fatal Error '2': T33 Command Format Error '3': Transaction Canceled or timed out. '4': ICC error, fallback to MSR processing.
[ErrCode]	8	(Optional, if reason = '1'), overall error code.
<ETX>	1	<03>
[LRC]	1	Check value

Message Flow:

Refer to message T33.

Example:

If an ARQ data needs two T34 packets to transmit, the first packet format will be:

<STX>T340Z3<SUB>01 [Tag1] [Len1] [Value1] <SUB> [Tag2]<ETX> [LRC]

The second and (the last) packet format will be:

<STX>T34<SUB>12 [TagN] [LenN] [ValueN] <SUB> [TagN+1]<ETX> [LRC]

Message T35 Start Transaction (non-EMV transactions)

Format: <STX>T35<SUB>[AmtAuth]<SUB>[AmtOther]<SUB>[CurExponent][CurCode]
<SUB>[TranType]<SUB>[TranInfo]<SUB>[Account Type]<SUB>[Force
Online]<SUB>[Encrypted Session key]<ETX>[LRC] (request frame)

Message length: Variable.

Usage: After receiving this message command T35, PIN Pad will perform a complete non-EMV transaction but processed in the same order as EMV transaction flow (the flow will cover 'Data Authentication' through 'Completion', see EMV 4.3, Book 3, chap 8.2, figure 6 - transaction flow example) based on the selected EMV application that has corrected application name on the T12 (Application Select response). PIN pad will also prompt the user to do the appropriated entry when presented, like confirm or enter PIN code. PIN pad will send the message T16 (Start Transaction Response) to the host.

If the IC card can't be read (which has known from T12 response code with "T1214" when in message T31 processing and terminal changed to do magnetic stripe card processing; please refer the Notes in message T12), then after receiving this message T35, PIN pad will not process the EMV transaction flow. But in place of just storing the information provided from this message and returns the message T16 with result code "A1" to ask the terminal to go online to get the authorization, then send back a message T17 to finish this transaction. Terminal could have extra operation on magnetic stripe card transaction (like issue message Q1 to read track data and issue message 70 for PIN entry, and so on.) before issuing this message T35.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
T35	3	Message ID
<SUB>	1	<1A>
[AmtAuth]	12	Hexadecimal, Amount Authorized tag '0x9F02'
<SUB>	1	<1A>
[AmtOther]	12	Hexadecimal, Amount Other, tag '0x9F03'
<SUB>	1	<1A>
[CurExponent]	1	Hexadecimal, tag '0x5F36'
[CurCode]	3	Hexadecimal, tag '0x5F2A'. For example, USD\$ = 0x840
<SUB>	1	<1A>

[TranType]	2	Hexadecimal, Transaction Type, tag '0x9c'
<SUB>	1	<1A>
[TranInfo]	2	Transaction Info, tag '0xFFFF8109'
<SUB>	1	<1A>
[Account Type]	2	Account Type, tag '0x5F57'
<SUB>	1	<1A>
[Force Online]	1	'1': Force this transaction to be authorized online. Valid only if the terminal configuration enables the capability to support online authorization. '0': Run the default EMV authorization flow
<SUB>	1	<1A> (Optional, if the Encrypted Session key exists)
[Encrypted Session key]	16 or 32 or 48	(Optional, if the transaction needs the user to enter the PIN for online authorization. If no session key exists, the A10-P will use PIN encryption key (Usage P0) or DUKPT key exist in A10-P). Only DES session key / TDES session key are used for EMV online PIN entry.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message T35	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Start Transaction Response. Message T16
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message T37 EMV PIN Management

Format: <STX>T37<SUB>[Operation]<ETX>[LRC] (request frame)

Message length: Variable.

Usage: After receive this message command T37, PIN Pad will perform the corresponding operation for PIN Change, PIN Unblock, PIN Verification process.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
T37	3	Message ID
<SUB>	1	<1A>
[Operation]	1	'1': Change PIN. '2': Unblock PIN. '3': Verify PIN.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message T37	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) (<EOT> after 3 NAKs)
	←	EMV PIN Management Response. <i>Message T38</i>
<ACK> (Good LRC) <NAK> (Bad LRC) (<EOT> after 3 NAKs)	→	

Message T38 EMV PIN Management Response

Format: <STX>T38 [Status] [Reason] [Err Message] [Result] <ETX> [LRC]
(response frame)

Message length: Variable.

Usage: The message contains the transaction result on the smart card to be sent to terminal.

Message element:

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
T38	3	Message ID
Status	1	0:OK; 1:Fail
Reason	1	Optional. (If Status = 1) 1:Fatal Error 2:Command Format Error 3:Transaction Canceled
[Err Message]	8	Optional, if Reason = '1' Hex String
[Result]	2	Optional. (If Status = 0) 'A1': Online Authorize Request, 'A4': Application reselection. 'V0': PIN verification succeed. 'V1': PIN verification failed.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow: Please refer to message T37.

Message T3C Force to Complete the Transaction (non-EMV transactions)

Format: <STX>T3C<ETX>[LRC] (request frame)

Message length: Fixed 6 bytes.

Usage: The message used to complete the non-EMV transaction.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
T3C	3	Message ID
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message T3C	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Respond results as Message T16. The Message ID (T16) is replaced by T3C.

Overall Contact EMV Level 2 transaction flow reference

Ref. 1 Packet command flow for transaction with offline

HOST	Direction	A10-P
T11 (Application Select)	→	
	←	T12 (Application Select Response)
T15 (Start Transaction)	→	
	←	T16 (Start Transaction Response) [Result] = 'Y1' or 'Z1'
T21 (Get Transaction Result's Data)	→	
	←	T22 (Get Transaction Result's Data Response)

Ref. 2 Packet command flow for transaction with online

HOST	Direction	A10-P
T11 (Application Select)	→	
	←	T12 (Application Select Response)
T15 (Start Transaction)	→	
	←	T16 (Start Transaction Response) [Result] = 'A1'
T27 (Get online data from A10-P)	→	
		T28 (Online authorization data response)
Go Online		
T1D (Send necessary online response data to A10-P.)	→	
	←	T1E
T17 (Send Online Authorized Code)	→	
	←	T16 (Start Transaction Response) [Result] = 'Y3' or 'Y4' or 'Z3' or 'Z4'
T21 (Get Transaction Result's Data)	→	
	←	T22 (Get Transaction Result's Data Response)

Ref. 3 Packet command flow for transaction with MSR.

HOST	Direction	A10-P
T11 (Application Select)	→	
	←	T12 (Application Select Response indicating that A10-P fails to read the IC card)
Issue magnetic stripe card commands		
T1D (Send necessary magnetic stripe card data to A10-P.)	→	
	←	T1E
T15 (Start Transaction)	→	
	←	T16 (Start Transaction Response) [Result] = 'A1'
Go Online		
T1D (Send necessary online response data to A10-P.)	→	
	←	T1E
T17 (Send Online Authorized Code)	→	
	←	T16 (Start Transaction Response) [Result] = 'Y4' or 'Z4'
T21 (Get Transaction Result's Data)	→	
	←	T22 (Get Transaction Result's Data Response)

Ref. 4 Packet command flow for online transaction and Issuer Script command Processing

HOST	Direction	A10-P
T11 (Application Select)	→	
	←	T12 (Application Select Response)
T15 (Start Transaction)	→	
	←	T16 (Start Transaction Response) [Result] = 'A1'
T27 (Get online data from A10-P)	→	
	←	T28 (Online authorization data response)
Go Online		
T19 (Send Issuer Script command) (if many commands)	→	
	←	T20 (Send Issuer Script command Response, continued)
T1D (Send necessary online response data to A10-P.)	→	
	←	T1E
T17 (Send Online Authorized Code)	→	
	←	T16 (Start Transaction Response) [Result] = 'Y3' or 'Y4' or 'Z3' or 'Z4'
T21 (Get Transaction Result's Data)	→	
	←	T22 (Get Transaction Result's Data Response)

Ref. 5 Packet command flow if the first EMV application is blocked

HOST	Direction	A10-P
T11 (Application Select)	→	
	←	T12 (Application Select Response)
T15 (Start Transaction)	→	
	←	T16 (Start Transaction Response) [Result] = 'A4'
T13 (Application Select Next)	→	
	←	T12 (Application Select Response)
T15 (Start Transaction)	→	
	←	T16 (Start Transaction Response) [Result] = 'Y1' or 'Z1'
T21 (Get Transaction Result's Data)	→	
	←	T22 (Get Transaction Result's Data Response)

Section 7 Contactless EMV Level 2 Transaction Messages

Contactless EMV Level2 transaction messages are divided into 2 groups. One group are the EMV-configuration data operation messages (T51, T53, T55, T75, T77) and the other group are the EMV-transaction messages (T61, T63, T65, T6C, T71 and T73).

The issuing sequence of the PCD EMV transaction messages is controlled by the A10-P. An invalid sequence will terminate that PCD EMV transaction. Message T61 is used for the terminal-side to transmit transaction information such as the amount to be used by the A10-P to complete the transaction with the card if the transaction does not need to be authorized online. Message T71 is applied if the transaction needs to be authorized online. The terminal-side will transmit necessary information via this message to the A10-P to complete the rest of transactional steps. If the response from the host contains issuer script (see EMV Book), the terminal-side will issue message T73 to send these scripts into the A10-P. Then the A10-P will issue these scripts at appropriate time to the card. Message T6C is used to terminate an EMV transaction. Finally, message T65 is used for terminal-side to get the transaction information through out the PCD EMV transaction. The transaction flow can be referred in the “Overall Contactless EMV level 2 transaction flow reference” section.

The meanings of error codes in the [Err Message] are listed below:

Error Code	Error Description
00000003	Service not accepted.
00000F9B	Store configuration data error.
8FFF0001	Out of memory.
8FFF0002	Parameter error.
8FFFFFF02	Tag's data format error.
8FFFFFF03	Some mandatory tags are not configured well.
8FFFFFFF0	Command format error.
8FFFFFFF1	The sequence of EMV transaction command is erroneous.
8FFFFFFF2	Terminal fundamental data is missing.
8FFFFFFF3	Authentication failed.
8FFFFF1A	Authentication key expired.
8FFFFF1B	Terminal configuration data is missing.
8FFFFF1C	No configuration data of EMV application or the data is missing.
8FFFFF4	The storage space of batch data capture is full.
8FFFFF5	No terminal configuration data, EMV application configuration data or CA public key.
A2000001	Application initial conditions are not satisfied.
A2000002	The generated cryptogram is not allowed.
A2000007	The specified instruction ID is not recognized.
A200000A	The type of the requested application cryptogram is incorrect.
A200000C	The status response code (SW1 SW2) is not '9000'.
A200000D	The Generate AC command is called more than 2 times in the current transaction.
A200000E	The AIP mandatory data is missing in the card response data .
A200000F	The AFL mandatory data is missing in the card response data .
A2000010	The response template from the card is incorrect.
A2000011	The format of AFL is incorrect.
A2000013	A redundant data output from the card is not allowed.
A2000014	Missing mandatory data in the card response data .
EEEEFFFF	EMV transaction cancelled.
DFFFFFFF	EMV forced abort.
FFFF0001	There is an error when parsing the TLV list data.
FFFFFFF	The contactless card does not respond data.

Message T51 PCD Terminal Configuration Setup

Format: <STX>T51[Pkt No.][Total Pkts]<SUB>[DO]<ETX>[LRC] (request frame)

Message length: Variable.

Usage: Host can use this command to send terminal configuration data to the A10-P. This command can be sent many times. A10-P will save those data internally and apply them during the transaction. A10-P will send the message T52 (Terminal Configuration Setup Response) to the host as the response. The functions of Message T51 and Message T01 are exactly the same and can be used alternately.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
T51	3	Message ID
[Pkt No.]	1	Decimal. Packet sequence number (1 ~ 9)(ex. 2)
[Total Pkts]	1	Decimal. Total packets (1 ~ 9)(ex. 8).
<SUB>	1	<1A>
[DO]	Var.	Data Object, format as below
<ETX>	1	<03>
[LRC]	1	Checksum

Data Object Format: each data object is delimited by <SUB>, and each field inside each data object is delimited by <FS>.

Data Format: (Please also refer to the EMV **Contactless Specifications v2.4**.)

Format	Description
1	a - Alphabetic data (a ~z, A~Z)
2	b - unsigned binary numbers or bit combinations
3	an - Alphanumeric data (a ~z, A~Z, 0~9)
4	ans - Alphanumeric Special data (Characters defined in ISO8859)
5	cn - Compressed numeric data (0~9, left justify, pad hexadecimal 'F's. Ex. 12 34 56 12 3F FF)
6	n - Numeric data (0~9, right justify, pad leading hexadecimal zeroes. Ex. 00 00 00 01 23 45)
7	var - Variable data (Any bit combination)

Note. User has to follow the restriction specified in the EMV Contactless Specifications v2.4, and Appendix E of this document to load the configuration data. A10-P will check if the length of each configuration data item is consistent. Any inconsistent data item will make the data loading fail.

Note. If the data format is '2' (binary), '5'(compressed numeric), '6'(numeric) or '7'(variable data), it cannot be allocated in message T51 directly. It should be transferred into a hexadecimal string and then allocated in message T51.

Example: (Clear the terminal configuration data and then setup the new data.)

Merchant Category Code: 0000 (Numerical)

Terminal ID: SmartPOS (Ascii)

<STX>T5111<SUB>9f15<FS>6<FS>0000<SUB>9f1c<FS>3<FS>SmartPOS<ETX> [LRC]

A10-P will check if the terminal downloads the minimum set of terminal-related information into itself. The download process will fail if there is not enough data in this message. Please refer to Appendix F for the minimum set of terminal-related data.

Message flow:

HOST	Direction	A10-P
1 st Message T51	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	1 st Message T52
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	
2 nd Message T51	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	2 nd Message T52
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	
...	...	...
Last one Message T51	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Last one Message T52
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message T52 PCD Terminal Configuration Setup Response

Format: <STX>T52 [Result] [Reason] [Err Message] [Err Tag Number] <ETX> [LRC]
 (response frame)

Message length: Variable.

Usage: The response message of message T51. The functions of Message T52 and Message T02 are exactly the same and can be used alternately.

Message element:

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
T52	3	Message ID
[Result]	1	'0': OK, '1': Fail
[Reason]	1	<Optional, if Res = '1'> '1': Fatal Error '2': Format Error
[Err Message]	8	Optional, if Reason = '1', Hex decimal string
[Err Tag Number]	Var.	Optional, if Reason = '3' or '4', Hex decimal string
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

Refer to message T51.

Message T53 PCD Certification Authority Public Key Setup**Usage:**

In A10-P, Message T53 is as same as Message T03. They have the same format, but only Message ID is different. In other words, PCD CAPK and ICC CAPK are the same. Using Message T53 to setup CAPK will have the same result by using Message T03. For the detailed format and usage, please refer to Section 6 Message T03.

Message T54 PCD Certification Authority Public Key Setup Response

Format: <STX>T54 [Sequence] [Result] [Reason] [Err Message] <ETX> [LRC]
(response frame)

Message length: Variable.

Usage: The response message of message T53. The functions of Message T54 and Message T04 are exactly the same and can be used alternately.

Message element:

1st, 2nd Packet:

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
T54	3	Message ID
[Sequence]	1	Same sequence number of the corresponded T53 message.
[Result]	1	'0': OK, '1': Fail
[Reason]	1	'1': Fatal Error '2': Format Error '3': Authentication Fail
[Err Message]	8	Optional, if Reason = '1' Hex String
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

Refer to Section 6 Message T03.

Message T55 PCD Application Configuration Setup

Format: <STX>T55 [Pkt No.] [TotalPkts] <SUB> [TXN] <SUB> [KID] <SUB> [AID] <SUB> [DO] <ETX> [LRC] (request frame)

<STX>T55 [Pkt No.] [Total Pkts] <SUB> [DO] <ETX> [LRC] (request frame)

Message length: Variable.

Usage: Host can use this command to send out the EMV application configuration data to the A10-P. This command can be sent many times but each command is only used for one application. A10-P will save those data inside and use them during the transaction. A10-P will send message T56 (EMV Application Configuration Setup Response) to the host. The data installed into the A10-P via this message will be saved in the internal storage structure with a name same as in the [AID] field.

Message element:

1st Message:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
T55	3	Message ID
[Pkt No.]	1	Decimal. Packet sequence number (1)
[Total Pkts]	1	Decimal. Total packets (1 ~ 9)(ex. 8).
<SUB>	1	<1A>
[TXN]	2	Transaction type, refer to EMV Contactless Specifications v2.4.
<SUB>	1	<1A>
[KID]	6	Kernel ID, refer to Appendix E .
<SUB>	1	<1A>
[AID]	10~32	EMV Application ID, refer to EMV Contactless Specifications v2.4.
<SUB>	1	<1A>
[DO]	Var.	Data Object, format as below
<ETX>	1	<03>
[LRC]	1	Checksum

Rest of Message (If there are 2 more messages):

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
T55	3	Message ID
[Pkt No.]	1	Decimal. Packet sequence number (2 ~ 9) (ex. 2).
[Total Pkts]	1	Decimal. Total packets (2 ~ 9)(ex. 8).
<SUB>	1	<1A>
[DO]	Var.	Data Object, format as below
<ETX>	1	<03>
[LRC]	1	Checksum

Format of Data Objects:

Each data object shall include three data fields delimited by <FS>: [Tag#]<FS>[Format Code]<FS>[Value]. Each data object is delimited by a <SUB> to construct multiple <DO>s. The [tag#] is defined in the EMV Contactless Specifications v2.4. There are also some private tags defined in **Appendix E**.

Data Format: (Please also refer to EMV **Contactless Specifications v2.4**)

Format	Description
1	a - Alphabetic data (a ~z, A~Z)
2	b - unsigned binary numbers or bit combinations
3	an - Alphanumeric data (a ~z, A~Z, 0~9)
4	ans - Alphanumeric Special data (Characters defined in ISO8859)
5	cn - Compressed numeric data (0~9, left justify, pad hexadecimal 'F's. Ex. 12 34 56 12 3F FF)
6	n - Numeric data (0~9, right justify, pad leading hexadecimal zeroes. Ex. 00 00 00 01 23 45)
7	var - Variable data (Any bit combination)

Note. User has to follow the restriction specified in EMV Contactless Specifications v2.4 and **Appendix E** of this document to load the configuration data. A10-P will check if the length of each configuration data item is consistent. Any inconsistent data item will make the data loading fail.

Note. If the data format is '2' (binary), '5'(compressed numeric), '6'(numeric) or '7'(variable data), it cannot be allocated in message T55 directly. It should be transferred into a hexadecimal string and then allocated in message T55.

Example:

Zero Amount Allowed: 01 (binary)
Contactless Floor Limit: 00 00 00 00 80 00 (numerical)
VLP Support Indicator: 01 (VLP supported) (binary)

<STX>T5511<SUB>00<SUB>030000<SUB>A00000031010<SUB>40000042<FS>2<FS>01
<SUB>40000043<FS>6<FS>000000008000<SUB>9F7A<FS>2<FS>01<ETX> [LRC]

A10-P saves these data and give an ID as "A00000031010" to this group of data.

The A10-P will check if the terminal downloads the minimum set of EMV Application-related information into itself. The downloading process will fail if there is not enough data in this message. Please refer to **Appendix F** for the minimum set of EMV Application-related data.

Message flow:

HOST	Direction	A10-P
1 st Message T55	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	1 st Message T56
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	
2 nd Message T55	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Application Select Response. 2 nd Message T56
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	
...	...	...
Last one Message T55	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Application Select Response. Last one Message T56

Message T56 PCD Application Configuration Setup Response

Format: <STX>T56[Resp][Reason][Err Msg][Err Tag Number]<ETX>[LRC]
(response frame)

Message length: Variable.

Usage: The response message of message T55.

Message element:

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
T56	3	Message ID
[Resp]	1	'0': OK, '1': Fail
[Reason]	1	Option if [Resp] is '1', '1': Fatal Error '2': Format Error '3': Invalid Data Object format. '4': Invalid Tag value
[Err Message]	8	Optional, if Reason = '1' Hex String
[Err Tag Number]	Var.	Optional, if Reason = '3' or '4', Hex decimal string
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

Refer to message T55.

Message T59 PCD Config Data Query Message

Format: <STX>T59[Config Type][control]<ETX>[LRC] (request frame)

Message length: Fixed 7 or 8 bytes.

Usage: Get the group ID of the PCD application data or the CA public key stored in the A10-P.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
T59	3	Message ID
[Config Type]	1	1: All the IDs of the PCD public key. 2: All the IDs of the PCD application data of Purchase.
[control]	1	(option)'0': Ready for next packet.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message T59	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message T5A(Result = 3)
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	
Message T59	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
...	...	...
	←	Message T5A(Result = 0)
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message T5A Response of PCD Config Data Query Message

Format: <STX>T5A[Config Type][Result]<SUB>[ID List]<ETX>[LRC]
(response frame)

Message length: Variable.

Usage: The response message of message T59.

Message element:

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
T5A	3	Message ID
[Config Type]	1	0: Unknow type. 1: All the IDs of PCD public key. 2: All the IDs of PCD application data of Purchase.
[Result]	1	0: There is data in identified data type. 1: No data in identified data type. 2: Wrong configuration type. 3: Arrangement exceeded.
<SUB>	1	<1A>, Optional if Result is 0, 3.
[ID List]	Var.	Optional, if Result is 0, 3.
<ETX>	1	<03>
[LRC]	1	Checksum

[ID List]: The concatenation of IDs. There is a <FS> between each ID.

Ex. [ID List] = A000000000390<FS> A0000000004F8<FS> A0000000004F5 (Config Type = 1)

[ID List] = A00000000041010<FS>B0123456781234<FS> A0000000010 (Config Type = 2)

Message flow:

Refer to message T59.

Message T5B Delete PCD Configuration Data Message.

Format: <STX>T5B[Config Type]<SUB>[ID List]<ETX>[LRC] (request frame)

Message length: Variable.

Usage: Host can use this command to delete the PCD application configuration data or the CA public key in the A10-P. This command can be sent many times.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
T5B	3	Message ID
[Config Type]	1	1: CA public key. 2: EMV application data.
<SUB>	1	<1A>
[ID List]	Var	List of IDs that will be erased.
<ETX>	1	<03>
[LRC]	1	Checksum

[ID List]: The concatenation of IDs. There is a <FS> between each ID.

Ex. [ID List] = A00000000390<FS> A000000004F8<FS> A000000004F5 (Config Type = 1)

[ID List] = A0000000041010<FS>B0123456781234<FS> A000000010 (Config Type = 2)

Message flow:

HOST	Direction	A10-P
Message T5B	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message T5C
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message T5C Response of Delete PCD Configuration Data Message.

Format: <STX>T5C[Config Type][Status]<SUB>[Del Result]<ETX>[LRC]
 (response frame)

Message length: Variable.

Usage: The response message of message T5B.

Message element:

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
T5C	3	Message ID
[Config Type]	1	0: Unknow type. 1: CA public key. 2: EMV application data.
[Status]	1	0: Delete OK 1: Fatal error. 2: Format error.
<SUB>	1	<1A>, Optional, if Status is 0.
[Del Result]	var	The results of the delete operation on the indicated ID
<ETX>	1	<03>
[LRC]	1	Checksum

[Del Result]:

It's the concatenation of the results for the delete operation on the indicated ID. There is a <FS> between each result. If the delete operation is successful (when A10-P finds out the matched one and delete it successfully), the result is 0. If there is no such ID as the terminal specified, the result is 1.

Ex. Terminal issues message T5B to erase the EMV application data with ID "A0000000102020", "A0000020" and "A0000000103030". Now the A10-P keeps the EMV application data with ID "A0000000103030" only.

The data in the [Del Result] field will be "1<FS>1<FS>0".

Message flow:

Refer to message T5B.

Message T5D PCD House Keeping Message.

Format: <STX>T5D<SUB>[Config Type]<ETX>[LRC] (request frame)

Message length: Variable.

Usage: Host can use this command to call PCD housekeeping in the A10-P. This command can be sent many times.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
T5D	3	Message ID
<SUB>	1	<1A>
Config Type	1	1: Paypass.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message T5D	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message T5E10
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	
Message T5D	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message T5E10 or T5E1F
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message T5E Response of PCD House Keeping Message.

Format: <STX>T5E<SUB>[Status][Result]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: The response message of message T5D.

Message element:

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
T5E	3	Message ID
<SUB>	1	<1A>
Status	1	0: Unkown type 1: Paypass
Result	var	0: Empty or not expired. 1: Clear data 2: Fatal error. F: Format error.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

Refer to message T5D.

Message T5F PCD DRL Configuration Setup.

Format: <STX>T5F<SUB>[AID]<SUB>[DO]<ETX>[LRC] (request frame)

Message length: Variable.

Usage: Host can use this command to send DRL(Dynamic Reader Limit) configuration data to the A10-P. This command can be sent many times. A10-P will save those data internally and apply them during the transaction. A10-P will send the message T5G (PCD DRL Configuration Setup Response) to the host as the response.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
T5F	3	Message ID
<SUB>	1	<1A>
AID	2~32	EMV Application ID, refer to EMV Contactless Specifications v2.4, Hex decimal string.
<SUB>	1	<1A>
[DO]	Var.	Data Object, format as below
<ETX>	1	<03>
[LRC]	1	Checksum

Format of Data Objects:

Each data object shall include three data fields delimited by <FS>: [Tag#]<FS>[Format Code]<FS>[Value] Each data object is delimited by a <SUB> to construct multiple <DO>s. The [tag#] defined in EMV Contactless Specifications v2.4, section 4.3.

Data Format: (Please also refer to EMV **Contactless Specifications v2.4**)

Format	Description
1	a - Alphabetic data (a ~z, A~Z)
2	b - unsigned binary numbers or bit combinations
3	an - Alphanumeric data (a ~z, A~Z, 0~9)
4	ans - Alphanumeric Special data (Characters defined in ISO8859)
5	cn - Compressed numeric data (0~9, left justify, pad hexadecimal 'F's. Ex. 12 34 56 12 3F FF)
6	n - Numeric data (0~9, right justify, pad leading hexadecimal zeroes. Ex. 00 00 00 01 23 45)
7	var - Variable data (Any bit combination)

Message flow:

HOST	Direction	A10-P
Message T5F	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	<i>Message T5G</i>
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message T5G PCD DRL Configuration Setup Response.

Format: <STX>T5G[Res] [Reason]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: The response message of message T5F.

Message element:

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
T5G	3	Message ID
Res	1	0: OK 1: Fail
Reason	1	<Optional, if Res = '1'> '1': Fatal Error '2': Format Error
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

Refer to message T5F.

Message T5H Delete PCD DRL Configuration .

Format: <STX>T5H<ETX>[LRC] (request frame)

Message length: Variable.

Usage: Delete All PCD DRL Configurations.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
T5H	3	Message ID
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message T5H	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message T5I
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message T5I Delete PCD DRL Configuration Response.

Format: <STX>T5I [Res] [Reason] <ETX> [LRC] (response frame)

Message length: Variable.

Usage: The response message of message T5H.

Message element:

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
T5I	3	Message ID
Res	1	0: OK 1: Fail
Reason	1	<Optional, if Res = '1'> '1': Fatal Error
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

Refer to message T5H.

Message T61 Start Transaction

Format: <STX>T61<SUB>[AmtAuth]<SUB>[AmtOther]<SUB>
[CurExponent][CurCode]<SUB>[TranType]<SUB>[TranInfo]<SUB>
[Account Type]<SUB>[Force Online]<SUB>
[Encrypted Session key]<ETX>[LRC] (request frame)

Message length: Variable.

Usage: After receiving message T61, the A10-P will perform a complete EMV transaction (the flow will cover from 'Initiate Application' through 'Completion', see EMV Contactless Specifications 2.4, book A, chap 5.3, figure 5-2 - Logical Architecture). A10-P will send message T62 (Start Transaction Response) to the host as the response.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
T61	3	Message ID
<SUB>	1	<1A>
[AmtAuth]	12	Optional: Numerical string (include digits after decimal point) to describe the "Amount Authroized" used in the purchase transaction. Map to EMV tag '0x9f02' Ex: "000000012345" means 123.45 if [CurExponent] is set to '2'.
<SUB>	1	<1A>
[AmtOther]	12	Optional: Numerical string (include digits after decimal point) to describe the "Amount, Other" used in the special transaction, such as the "cashback" amount. Map to EMV tag '0x9f03'
<SUB>	1	<1A>
[CurExponent]	1	Numerical digit ('0' to '3') to describe the "Transaction Currency Exponent", which is the count of digits after the decimal point. Map to EMV tag '0x5f36'
[CurCode]	3	Numerical string to describe the "Transaction Currency Code". Map to EMV tag '0x5f2a'. Ex: USD = "840"

<SUB>	1	<1A>
[TranType]	2	Numerical string to indicate the type of transaction, presented by the first two digits of the ISO 8583:1987 field 3 (processing code.) Map to EMV tag '0x9c' Ex. "00" – purchase, "01" – cash advance, "09" – purchase with cashback, "20" – refund. The actual values to be used are defined by the relevant payment system.
<SUB>	1	<1A>
[TranInfo]	2	Hexadecimal string to indicate the private parameter "Transaction Info". Map to private tag '0xFFFF8109' Allowable values: "80" - Cash (cash advance) "40" - Goods "20" - Services "10" - Cashback
<SUB>	1	<1A>
[Account Type]	2	Account Type, Map to EMV tag '0x5f57' Allowable values (ref. EMV v4.3 book3 Annex G) "00" – Default "10" – Savings "20" – Cheque / Debit "30" – Credit.
<SUB>	1	<1A>
[Force Online]	1	'1': Force this transaction to be authorized online. Valid only if the terminal configuration enables the capability to support online authorization. '0': Run the default EMV authorization flow
<SUB>	1	<1A>
[Encrypted Session key]	16 or 32 or 48	Optional, DES or TDES session key that is used to encrypt the PIN block when CVM asks for online PIN verification. If this session

		key is not entered, A10-P will use an activated PIN encryption key (the key usage is P0) or an activated DUKPT to encrypt the PIN block.
<SUB>	1	<1A>,Optional, only if [DO] exists
DO	Var.	Data Object, format as below
<ETX>	1	<03>
[LRC]	1	Checksum

Format of Data Objects:

Each data object shall include three data fields delimited by <FS>: [Tag#]<FS>[Format Code]<FS>[Value]. Each data object is delimited by a <SUB> to construct multiple <DO>s. The [tag#] is defined in the EMV Contactless Specifications v2.4. There are also some private tags defined in **Appendix E**.

Data Format: (Please also refer to EMV Contactless Specifications v2.4)

Format	Description
1	a - Alphabetic data (a ~z, A~Z)
2	b - unsigned binary numbers or bit combinations
3	an - Alphanumeric data (a ~z, A~Z, 0~9)
4	ans - Alphanumeric Special data (Characters defined in ISO8859)
5	cn - Compressed numeric data (0~9, left justify, pad hexadecimal 'F's. Ex. 12 34 56 12 3F FF)
6	n - Numeric data (0~9, right justify, pad leading hexadecimal zeroes. Ex. 00 00 00 01 23 45)
7	var - Variable data (Any bit combination)

Note. User has to follow the restriction specified in EMV Contactless Specifications v2.4 and **Appendix E** of this document to load the configuration data. A10-P will check if the length of each configuration data item is consistent. Any inconsistent data item will make the data loading fail.

Note. If the data format is '2' (binary), '5'(compressed numeric), '6'(numeric) or '7'(variable data), it cannot be allocated in message T61 directly. It should be transferred into a hexadecimal string and then allocated in message T61.

Example:

Balance Read Before Gen AC: 000000 (Numeric)

Balance Read After Gen AC: 000000 (Numeric)

[DO]:

DF8104<FS>6<FS>000000<SUB>DF8105<FS>6<FS>000000

Message flow:

HOST	Direction	A10-P
Message T61	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Start Transaction Response. <i>Message T62</i>

Message T62 Start Transaction Response

Format: <STX>T62 [Status] [Reason] [Err Message] [Result] <ETX> [LRC]
(response frame)

Message length: Variable.

Usage: The message contains the transaction result on the smart card to be sent to the terminal.

Message element:

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
T62	3	Message ID
[Status]	1	0:OK; 1:Fail
[Reason]	1	Optional. (If Status = 1) 1:Fatal Error 2:Command Format Error
[Err Message]	8	Optional, if Reason = '1' Hex String '00000007': Please see note.
[Result]	2	Optional. (If Status = 0) 'Y1': Offline Approved, 'Z1': Offline Declined 'Y3': Unable to go online, Offline Approved 'Z3': Unable to go online, Offline Declined. 'Y4': Online Approved 'Z4': Online Declined 'Y7': Offline Approved With Signature 'Y8': Unable to go online, Offline Approved With Signature 'Y9': Online Approved With Signature 'A1': Online Authorize Request 'B0': Try another interface.
<ETX>	1	<03>
[LRC]	1	Checksum

Note: If the [Result] is 'A1', the terminal should send the online authorization request to the issuer host and After that, the terminal shall send message T71 to the A10-P to continue the transaction.

Note: For special error code, if the [Status] is '1', [Reason] is '1', [Err Message] is '00000007', the terminal shall send message T61 to Start Transaction.

Message flow:

Refer to message T61.

Message T63 Get Transaction Result's Data

Format: <STX>T63[control] [DOL]<ETX>[LRC] (request frame)

Message length: Variable.

Usage: A10-P to retrieve the data listed on the DOL after the EMV transaction is done. The A10-P will send message T64 (Get Transaction Result Data Response) to the host as a response.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
T63	3	Message ID
[control]	1	'0': Ready for next packet
[DOL]	Var	Data Object List each object is expressed by a tag number, and <SUB> is used to delimit each object. For example, 9F12<SUB>9A<SUB>9F02<SUB>....
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message T63	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> after 3 NAKs
	←	Get the Transaction Result Data Response. <i>Message T64</i>
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message T64 Get the Transaction Result Data Response

Format: <STX>T64[Pkt NO.][Total Pkt][DO]<ETX>[LRC] (response frame)
<STX>T64[Status]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: The message contains the transaction result data to be sent to the host.

Message element:

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
T64	3	Message ID
[Pkt NO.]	1	Decimal. Packet sequence number (1 ~ 9).
[Total Pkt]	1	Decimal. Total packets (1~9).
[DO]	Var.	Data Object: each data object is expressed in the TLV format with an <FS> delimit in each field, and <SUB> is used to delimit each object. For example: 9F12<FS>0F<FS>CREDITO DE VISA<SUB>9A<FS>06<FS>050806<SUB>
<ETX>	1	<03>
[LRC]	1	Checksum

or

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
T64	3	Message ID
[Status]	1	F: Query fail or data empty.
<ETX>	1	<03>
[LRC]	1	Checksum

Data Object:

A10-P will return a series of data objects listed on the [DOL] fields in the message T64 with TLV (tag || length|| value) format as below:

[EMV Tag Number (2 ~ 8 byte)] <FS> [Length (2byte)] <FS> [Value] <SUB>

[EMV Tag Number (2 ~ 8 byte)] <FS> [Length (2byte)] <FS> [Value] <SUB>

.....

[EMV Tag Number (2 ~ 8 byte)] <FS> [Length (2byte)] <FS> [Value].

Note: When the A10-P responds these data objects, it will convert from binary values to a hex decimal string if the data object format is “b” or “cn” or “n”.

Message flow:

Refer to message T63.

Message T65 Get Online authorization Data

Format: <STX>T65[Data type][control]<ETX>[LRC] (request frame)

Message length: Variable 7 to 8 bytes.

Usage: Issue this message to get the (EMV) data for online authorization.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
T65	3	Message ID
[Data type]	1	0 : Data record 1 : Discretionary data
[control]	1	(optional) '0': Ready for next packet.
<ETX>	1	<03>
[LRC]	1	Checksum

If the EMV transaction must be authorized online, the terminal shall issue message T65 to get the necessary data from the A10-P for online authorization. And the A10-P will wait for the terminal to send message T71 to inform the A10-P on the go online authorization result

A10-P will search for the corresponding values according to this tag list and return message T66 after receiving message T65 from the terminal. Please refer to **Appendix F** for more details.

Message flow:

HOST	Direction	A10-P
Message T65	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message T66
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message T66 Response of Get Online authorization Data message

Format: <STX>T66[Pkt NO.] [Total Pkt] [online authorization data]<ETX>[LRC]
(response frame)
<STX>T66[status] <ETX>[LRC] (response frame)

Message length: Variable

Usage: Return the online authorization data to the terminal.

Message element:

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
T66	3	Message ID
[Pkt NO.]	1	Decimal. Packet sequence number (1 ~ 9)
[Total Pkt]	1	Decimal. Total packets (1~9).
[Online authorization data]	Var.	Hex string.
<ETX>	1	<03>
[LRC]	1	Checksum

OR

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
T66	3	Message ID
[status]	1	F: Query failed or data empty.
<ETX>	1	<03>
[LRC]	1	Checksum

Online authorization data:

A10-P returns a series of data objects with the TLV (tag || length|| value) format as below:

[EMV Tag Number (2 ~ 8 byte)] || [Length (2byte)] || [Value]

[EMV Tag Number (2 ~ 8 byte)] || [Length (2byte)] || [Value]

.....

[EMV Tag Number (2 ~ 8 byte)] || [Length (2byte)] || [Value].

When the A10-P responds these data objects, it will convert them from binary values to a hex decimal string if the data object format is “b” or “cn” or “n”. For example, the data of one record is 0x9F02 || 06 || 000000001100 || 5A || 10 || 11223344556677889900AABBCCDDEEFF, the terminal will see “9F02060000000011005A10-P11223344556677889900AABBCCDDEEFF”.

Message flow:

Refer to message T65

Message T67 Get Transaction Payment Scheme

Format: <STX>T67<ETX>[LRC] (request frame)

Message length: Fixed 6 bytes.

Usage: A10-P to retrieve the payment scheme after the EMV transaction is done. The A10-P will send message T68 (Get Transaction Payment Scheme Response) to the host as a response.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
T67	3	Message ID
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message T67	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message T68
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message T68 Get Transaction Payment Scheme Response

Format: <STX>T68 [Scheme] <ETX> [LRC] (response frame)

Message length: Variable 7 to 8 bytes.

Usage: The message contains the transaction payment scheme to be sent to the host.

Message element:

Request frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
T68	3	Message ID
[Scheme]	1 or 2	F : Format error. 21 : MASTER MAGSTRIPE 22 : MASTER MCHIP 31 : VISA MSD CVN17 32 : VISA qVSDC 33 : VISA AP 34 : VISA MSD LEGACY 41 : AE MAGSTRIPE 42 : AE EMV 51 : JCB MAGSTRIPE 52 : JCB LEGACY 53 : JCB EMV 61 : DISCOVER ZIP 62 : DISCOVER DPAS 71 : PBOC CREDIT DEBIT 72 : PBOC QPBOC
<ETX>	1	<03>
[LRC]	1	Checksum

Message T6C Cancel PCD Transaction

Format: <STX>T6C<ETX>[LRC] (request frame)

Message length: Fixed 6 bytes.

Usage: This message is used to cancel the transaction for the contactless card.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
T6C	3	Message ID
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message T6C	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)

Message T71 Send PCD Online Authorized Code

Format: <STX>T71[Online Res]<SUB>[ARC]<SUB>[IAD]<ETX>[LRC]
(request frame)

Message length: Variable.

Usage: After receiving this message T71, the A10-P will continue to perform the EMV transaction if the previous T62 response result is 'A1' (online authorized request, see EMV Contactless Specifications v2.4). A10-P will respond message T62 (Start Transaction Response) to this T71 to the host.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
T71	3	Message ID
[Online Res]	1	'0': Unable to go online
<ETX>	1	<03>
[LRC]	1	Checksum

OR

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
T71	3	Message ID
[Online Res]	1	'1': Get Online Authorize Response,
<SUB>	1	<1A>
[ARC]	2	Optional, Authorisation Response code, ASCII (0~9, A~Z).
<SUB>	1	<1A>
[IAD]	16~32	Optional: Issuer Authentication Data, Hex string. If there is an IAD response from the remote host , the terminal shall send this to the A10-P.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message T71	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Start Transaction Response. <i>Message T62</i>
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message T73 Send Issuer Script Command

Format: <STX>T73[IS]<ETX>[LRC] (request frame)

Message length: Variable.

Usage: A10-P performs the Issuer script processing as in the EMV transaction flow after receiving this command from the host as the response message after performing online authorization. This command can be sent many times if too many script commands need to be processed. But the last one should be sent before message T71.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
T73	3	Message ID
IS	Var.	Issuer Script, format as follows. Hex string.
<ETX>	1	<03>
[LRC]	1	Checksum

Issuer Script Format: (see EMV Contactless Specifications v2.4)

T	L	T	L	Script ID	Commands
'71' or '72'	Including Script ID, tags, lengths	'9F18'	'04'	Identifier (4 bytes)	Issuer Script Command Format (see below)

Issuer Script Command Format: (see EMV Contactless Specifications v2.4)

T1	L1	V1	T2	L2	V2	T3	L3	V3	Tx	Lx	Vx
'86'	L(V1)	Cmd	'86'	L(V2)	Cmd	'86'	L(V3)	Cmd	'86'	L(Vx)	cmd

Message flow:

HOST	Direction	A10-P
Message T73	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message T74
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message T74 Send Issuer Script Command Response

Format: <STX>T74[Status]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: The message response of message T73.

Message element:

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
T74	3	Message ID
Status	1	0:OK ; 1:Fail
Reason	1	Optional. (if Status = 1) 1:Fatal Error 2:Command Format Error
Err Message	8	Optional, if Reason = '1' Hex String
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

Refer to message T73.

Message T75 Revocation List Setup

Format: <STX>T75<SUB>[RID] [SN] [PKI]<ETX>[LRC] (request frame)

Message length: Fixed 25 bytes

Usage: Host can use this command to send the revocation key information to the A10-P. This command can be sent many times. A10-P will save this information internally and check that during the transaction. The A10-P will send message T76 (Revocation List Setup Response) to the host as the response.

Message elements:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
T75	3	Message ID
<SUB>	1	Separator
[RID]	10	The RID for the revoked public key. Present in hexstring.
[SN]	6	The serial number of the revoked public key. Present in hexstring.
[PKI]	2	The public key index of the revoked key. Present in hexstring.
<ETX>	1	<03>
[LRC]	1	Checksum

Example:

Revocation list information

RID: A0 00 00 00 03

SN: 00 00 01

PKI: 51

<STX>T75<SUB> A00000000300000151<ETX>[LRC]

Message flow:

HOST	Direction	A10-P
Message T75	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message T76
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message T76 Revocation List Setup Response

Format: <STX>T76[Res][Reason][Err Msg]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: The response message of message T75.

Message element:

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
T76	3	Message ID
[Res]	1	'0': Ok, '1': Fail
[Reason]	1	<Optional, if Res = '1'> '1': Fatal error '2': Invalid Data format '3': Revocation list is full '4': The added info exists
[Err Message]	8	Optional, if Reason = '1', Hex decimal string
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

Refer to message T75.

Message T77 Exception List Setup

Format: <STX>T77<SUB>[PAN] [<SUB>[PAN Seq. No.]]<ETX>[LRC] (request frame)

Message length: Variable.

Usage: Host can use this command to send the exception PAN to the A10-P. A10-P will save the information inside and check it during the transaction. Once the transaction PAN is found on the exception list, the transaction will be terminated. The A10-P will send message T78 (Exception List Setup Response) to the host when it adds an exception file.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
T77	3	Message ID
<SUB>	1	Separator
[PAN]	up to 19	Primary Account Number in numeric format
<SUB>	1	Separator. This is optional, exists if PAN Seq. No. exists.
[PAN Seq. No.]	2	PAN Sequence Number, this is an optional data.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message T77	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Exception List Setup Response Message T78
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Command Example:

PAN: 47 61 73 90 01 01 00 10

<STX>T77<SUB>4761739001010010<ETX><LRC>

PAN: 37 37 37 34 56 78 90 4

PAN Seq. No.: 00

<STX>T77<1A>373737345678904<1A>00<ETX><LRC>

Message T78 Exception List Setup Response

Format: <STX>T78 [Resp] [Reason] [Err Msg]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: The response message of message T77.

Message element:

Response frame (A10-P to HOST)

HOST	Direction	A10-P
<STX>	1	<02>
T78	3	Message ID
[Resp]	1	'0': OK, '1': Fail
[Reason]	1	Option if [Resp] is '1', '1': Fatal Error '2': Format Error '3': Exception List is full '4': PAN exists
[Err Message]	8	Optional, if Reason = '1', Hex String
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

Refer to message T77

Message T81 Start EMV Contact/Contactless Transaction

Format: <STX>T81<SUB>[AmtAuth]<SUB>[AmtOther]<SUB>[CurExponent][CurCode]
 <SUB>[TranType]<SUB>[TranInfo]<SUB>[Account Type]<SUB>[Force
 Online]<SUB>[Encrypted Session key]<SUB>[DO]<ETX>[LRC]
 (request frame)

Message length: Variable.

Usage: After receiving this message T81, Reader will perform a complete EMV transaction (the flow will cover from 'Initiate Application' through 'Completion', see EMV Contactless Specifications v2.4, book A, chap 5.3, figure 5-2 - Logical Architecture, or see EMV 4.3, book 3, chap 8.2, figure 6 - transaction flow example). Reader will send message T12/T16/T62 (Contact/Contactless Transaction Response) to the host as a response.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
T81	3	Message ID
[AmtAuth]	12	Optional: Numerical string (include digits after decimal point) to describe the "Amount Authroized" used in the purchase transaction. Map to EMV tag '0x9f02' Ex: "000000012345" means 123.45 if [CurExponent] is set to '2'.
<SUB>	1	<1A>
[AmtOther]	12	Optional: Numerical string (include digits after decimal point) to describe the "Amount, Other" used in the special transaction, such as the "cashback" amount. Map to EMV tag '0x9f03'
<SUB>	1	<1A>
[CurExponent]	1	Numerical digit ('0' to '3') to describe the "Transaction Currency Exponent", which is the count of digits after the decimal point. Map to EMV tag '0x5f36'
[CurCode]	3	Numerical string to describe the "Transaction Currency Code". Map to EMV tag '0x5f2a'. Ex: USD = "840"

<SUB>	1	<1A>
[TranType]	2	Numerical string to indicate the type of transaction, presented by the first two digits of the ISO 8583:1987 field 3 (processing code.) Map to EMV tag '0x9c' Ex. "00" – purchase, "01" – cash advance, "09" – purchase with cashback, "20" – refund. The actual values to be used are defined by the relevant payment system.
<SUB>	1	<1A>
[TranInfo]	2	Hexadecimal string to indicate the private parameter "Transaction Info". Map to private tag '0xFFFF8109' Allowable values: "80" - Cash (cash advance) "40" - Goods "20" - Services "10" - Cashback
<SUB>	1	<1A>
[Account Type]	2	Account Type, Map to EMV tag '0x5f57' Allowable values (ref. EMV v4.3 book3 Annex G) "00" – Default "10" – Savings "20" – Cheque / Debit "30" – Credit.
<SUB>	1	<1A>
[Force Online]	1	'1': Force this transaction to be authorized online. Valid only if the terminal configuration enables the capability to support online authorization. '0': Run the default EMV authorization flow
<SUB>	1	<1A>
[Encrypted Session key]	16 or 32 or 48	Optional, DES or TDES session key that is used to encrypt the PIN block when CVM asks for online PIN verification. If this session

		key is not entered, the A10-P will use an activated PIN encryption key (the key usage is P0) or an activated DUKPT to encrypt the PIN block.
<SUB>	1	<1A>,Optional, only if [DO] exists
DO	Var.	Data Object, format as below
<ETX>	1	<03>
[LRC]	1	Checksum

Format Codes for Data Object Elements:

Format	Description
1	a - Alphabetic data (a ~z, A~Z)
2	b - unsigned binary numbers or bit combinations
3	an - Alphanumeric data (a ~z, A~Z, 0~9)
4	ans - Alphanumeric Special data (Characters defined in ISO8859)
5	cn - Compressed numeric data (0~9, left justify, pad hexadecimal 'F's. Ex. 12 34 56 12 3F FF)
6	n - Numeric data (0~9, right justify, pad leading hexadecimal zeroes. Ex. 00 00 00 01 23 45)
7	var - Variable data (Any bit combination)

Notes on Data Objects field:

If the data format is '2' (binary), '5' (compressed numeric), '6' (numeric) or '7' (variable data), the [value] corresponding to each format cannot be sent with the format directly in message T81. It shall be converted to a hexadecimal string. A pad character has to be added at the end if this tag# has odd characters. The pad char 'F' is used for format 'cn', and '0' is used for formats 'b' or 'n'.

Example:

Balance Read Before Gen AC: 000000 (Numeric)

Balance Read After Gen AC: 000000 (Numeric)

[DO]:

DF8104<FS>6<FS>000000<SUB>DF8105<FS>6<FS>000000

Message flow:

HOST	Direction	A10-P
Message T81	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message T12/T16/T62
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message SIG Signals to the Terminal

Format: <STX>SIG[Control] [PktNo] [SeqID] [DataType] [Timeout]
<SUB>[Data]<ETX>[LRC] (request frame)

Message length: Variable.

Usage: Initiated by Device during EMV transaction for external user interface request or send DET signal. The terminal should handle the request with Display and Keypad, or receive and match DET to get DEK, then return the result to the Device within the timeout.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
SIG	3	Message ID
[Control]	1	'0': (transmitter side) more packets follow. '1': (transmitter side) the last packet. '2': (receiver side) ready for the next packet. '3': (receiver side) acknowledge the last packet.
[PktNo]	1	'0' ~ '9', The packet number (increment on transmitter side).
[SeqID]	2	"00" ~ "FF", the sequence number increments on each SIG packet on both sides for handshaking purpose. Reset on power-on, rollover to "00" after reaching "FF".
[DataType]	2	"00": Clear screen. "03": Display text and get function key. "04": Function key input (response of 03.) "07": Display and select Application ID. "08": Selected AID (response of 07.) "09": Display and select language. "0A": Selected language (response of 09.) "0B": Data Exchange Kernel (DEK Signal) "0C": Data Exchange Terminal (DET Signal, response of 0B) "FF": Host error or times out.
[Timeout]	2	Hex string represents the waiting time for

		response, in seconds. Available value: “01” ~ “FE”. “00”: No waiting time and no need response SIG packet. “FF”: Wait forever for response.
<SUB>	1	<1A> as delimiter
[Data]	Var.	Data field, see description below.
<ETX>	1	<03>
[LRC]	1	Check value

[Data] format for type “00” (clear screen) (reserved word):

Field	Length	Value and description
[Data]	0	The “clear screen” have an empty data field. The terminal should clear its display.

[Data] format for type “03” (display text and get function key) (reserved word):

Field	Length	Value and description
[Data]	Var.	The message field includes ASCII text messages to be shown on external UI. The terminal should display the message and collect a single function key (ENTER / CANCEL,) or responds a “type FF” packet for error situation.

[Data] format for type “04” (function key input) (reserved word):

Field	Length	Value and description
[Data]	1	‘0’: ENTER key pressed. ‘1’: CANCEL key pressed.

[Data] format for type “07” (display and select Application ID) (reserved word):

Field	Length	Value and description
[Data]	Var.	[AppName1]<FS>[AppName2]<FS>... Each AppName is a string that identifies application in the IC card. The terminal should parse the AID list and allow the user to select one AID for following EMV transaction, or responds a “type FF” packet for error situation.

[Data] format for type “08” (selected AID) (reserved word):

Field	Length	Value and description
[Data]	Var	The exact AID string that user chose.

[Data] format for type “09” (display and select language) (reserved word):

Field	Length	Value and description
[Data]	Var.	[LangID1]<FS>[LangID2]<FS>... Each LangID is a two-character ISO639-1 language code, or “Default” that allows the terminal to determine its preferred language. The terminal should parse the LangID list and allow the user to select one supported language for following EMV transaction, or responds a “type FF” packet for error situation.

[Data] format for type “0A” (selected language) (reserved word):

Field	Length	Value and description
[Data]	Var	The exact language ID string that user chose.

[Data] format for type “0B” (Data Exchange Kernel (DEK Signal)):

Field	Length	Value and description
[Data]	Var	<SUB>[Tag#]<FS>[Length Code]<FS>[Value]...

[Data] format for type “0C” (Data Exchange Terminal (DET Signal)):

Field	Length	Value and description
[Data]	Var	<SUB>[Tag#]<FS>[Length Code]<FS>[Value]...

[Data] format for type “FF” (host error or timed out) (reserved word):

Field	Length	Value and description
[Data]	1	‘0’: Timeout happened. ‘1’: User aborted. ‘2’: Other error on the host side.

Message Flow (signals that does not require response):

HOST	Direction	A10-P
T61	→	
	←	Message SIG
<ACK> if LRC ok. <NAK> if LRC error.	→	

Message Flow (signals that require response):

HOST	Direction	A10-P
T61	→	
	←	Message SIG (request)
<ACK> if LRC ok. <NAK> if LRC error.	→	
Message SIG (response)	→	
	←	<ACK> / <NAK>

Example:

This example is the captured transmission for AID / language selection by card holder. The terminal configuration needs to be modified to enable these features. <ACK> / <NAK> characters are omitted. For MCL data exchange:

HOST	Direction	A10-P
T61	→	
	←	SIG10000BFF<1C>FF810400DF8 10603DF8112
SIG10010C00<1C>DF8 1120182	→	
	←	SIG10000BFF<1C>FF810404820 20000DF810600
SIG10010C00<1C>DF8 1100100DF81100100DF 81100100DF81100100D F81100100DF81100100 DF81100100DF8110010 0DF81100100DF811001 00DF81100101DF81120 55A5F209F12	→	
	←	SIG10000BFF<1C>FF81040D5A 0854133390000015139F1200
SIG10010C00<1C>	→	
	←	T620A1

Note: SIG occurs only between T61 and T62. The rest of the transaction process after this is according to other situations (e.g. T71, T63, T65).

Overall Contactless EMV Level 2 transaction flow reference

A10-P provides one complete EMV Level 2 transaction with online or offline approval by various possible command flows. To force the A10-P to leave the command flow, you can either press the [CANCEL] key or send message T6C to the A10-P.

Ref. 1 Packet command flow for transaction with offline approval

HOST	Direction	A10-P
T61 (Start Transaction)	→	
	←	T62 (Start Transaction Response) [Result] = 'Y1' or 'Z1'
T65 (Get online data from A10-P)	→	
	←	T66 (Online authorization data response)
T63 (Get Transaction Result's Data)	→	
	←	T64 (Get Transaction Result Data Response)

Ref. 2 Packet command flow for transaction with online approval

HOST	Direction	A10-P
T61 (Start Transaction)	→	
	←	T62 (Start Transaction Response) [Result] = 'A1'
T65 (Get online data from A10-P)	→	
	←	T66 (Online authorization data response)
T63 (Get Transaction Result's Data)	→	
	←	T64 (Get Transaction Result Data Response)
Go Online		
T71 (Send Online Authorized Code)	→	
	←	T62 (Start Transaction Response)
T63 (Get Transaction Result's Data)	→	
	←	T64 (Get Transaction Result Data Response)

Section 8 MIFARE Card Messages

This chapter describes the commands of the MIFARE cards. The A10-P supports the MIFARE classic, Ultralight and DESFire cards. All MIFARE cards are compliant to the ISO/IEC 14443-1, 2 and 3 standards. The MIFARE DESFire card supports the ISO/IEC 14443-4 protocol (also called “T=CL” protocol).

Message P01 Enable/Disable MIFARE

Format: <STX>P01[Flag]<EXT>[LRC] (request frame)
<STX>P01[Result][Reason]<ETX>[LRC] (response frame)

Message length: Fixed 7 bytes for the request frame and variable for the response frame.

Usage: Enable or Disable the MIFARE.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
P01	3	Message ID
[Flag]	1	'0': Disable MIFARE. '1': Enable MIFARE.
<ETX>	1	<03>
[LRC]	1	Checksum

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
P01	3	Message ID
[Result]	1	'0': Success. '1': Fail.
[Reason]	1	Optional: if [Result] = '1' '1': Format error. '2': Sequence error.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message P01 (request frame)	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message P01 (response frame)
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message P02 Query MIFARE Presence

Format: <STX>P02<ETX>[LRC] (request frame)

 <STX>P02 [Result] [ATQA] [Reason]<ETX>[LRC] (response frame)

Message length: Fixed 6 bytes for the request frame and variable for the response frame.

Usage: Send the WUPA command to the contactless card.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
P02	3	Message ID
<ETX>	1	<03>
[LRC]	1	Checksum

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
P02	3	Message ID
[Result]	1	'0': Success. '1': Fail.
[ATQA]	4	Optional: if [Result] = '0' ATQA: 4 bytes. Format: hexadecimal string.
[Reason]	1	Optional: if [Result] = '1' '1': Format error. '2': Sequence error. '3': Card not exists.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message P02 (request frame)	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message P02 (response frame)
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message P03 MIFARE Anti-collision

Format: <STX>P03<ETX>[LRC] (request frame)
 <STX>P03[Result][UID][Reason]<ETX>[LRC] (response frame)

Message length: Fixed 6 bytes for the request frame

Variable for the response frame.

Usage: This command is used to get the card's UID. When many cards are in the RFID field, the A10-P will only return the UID from one card.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
P03	3	Message ID
<ETX>	1	<03>
[LRC]	1	Checksum

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
P03	3	Message ID
[Result]	1	'0': Success. '1': Fail.
[UID]	Var.	Optional: if [Result] = '0' The UID length is 8, 14 or 20 bytes. Format: hexadecimal string.
[Reason]	1	Optional: if [Result] = '1' '1': Format error. '2': Sequence error. '3': Anti collision fails.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message P03 (request frame)	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message P03 (response frame)
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message P04 MIFARE Selection

Format: <STX>P04<ETX>[LRC] (request frame)
 <STX>P04[Result][SAK][Reason]<ETX>[LRC] (response frame)

Message length: Fixed 6 bytes for the request frame

Variable for the response frame.

Usage: Select one card to activate.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
P04	3	Message ID
<ETX>	1	<03>
[LRC]	1	Checksum

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
P04	3	Message ID
[Result]	1	'0': Success. '1': Fail.
[SAK]	2	Optional: if [Result] = '0' The response of the select command. Format: hexadecimal string.
[Reason]	1	Optional: if [Result] = '1' '1': Format error. '2': Sequence error. '3': Select fail.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message P04 (request frame)	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message P04 (response frame)
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message P05 MIFARE Card Activation

Format: <STX>P05<ETX>[LRC] (request frame)

 <STX>P05 [Result] [ATQA]<FS>[SAK]<FS>[UID] <ETX>[LRC]

 (response frame1)

 <STX>P05 [Result] [Reason]<ETX>[LRC] (response frame2)

Message length: Fixed 6 bytes for the request frame

Variable for the response frame.

Usage: Activate MIFARE card. This message will send the WUPA(Anti-collision and Selection) command to the card.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
P05	3	Message ID
<ETX>	1	<03>
[LRC]	1	Checksum

Response frame1 (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
P05	3	Message ID
[Result]	1	'0': Success.
[ATQA]	4	ATQA: 4 bytes. Format: hexadecimal string.
<FS>	1	Field separator.
[SAK]	2	The response of the select command. Format: hexadecimal string.
<FS>	1	Field separator.
[UID]	Var.	The card Unique Identifier. The UID length is 8, 14 or 20 bytes. Format: hexadecimal string.
<ETX>	1	<03>
[LRC]	1	Checksum

Response frame2 (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
P05	3	Message ID
[Result]	1	'1': Fail.
[Reason]	1	'1': Format error. '2': Sequence error. '3': Card activate fail.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message P05 (request frame)	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message P05 (response frame)
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message P06 MIFARE Card HALT

Format: <STX>P06<ETX>[LRC] (request frame)
 <STX>P06[Result][Reason]<ETX>[LRC] (response frame)

Message length: Fixed 6 bytes for the request frame

Fixed 7 or 8 bytes for the response frame.

Usage: Send the HALT command will deactivate the card.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
P06	3	Message ID
<ETX>	1	<03>
[LRC]	1	Checksum

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
P06	3	Message ID
[Result]	1	'0': Success. '1': Fail.
[Reason]	1	Optional: if [Result] = '1' '1': Format error. '2': Sequence error. '3': Card deactivate fail.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message P06 (request frame)	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message P06 (response frame)
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message P07 MIFARE Classic Card Authentication

Format: <STX>P07[Sector number][Key type][key value]<ETX>[LRC]
 (request frame 1)
 <STX>P07[Sector number][Key number][key type]<ETX>[LRC]
 (request frame 2)
 <STX>P07[Result][Reason]<ETX>[LRC] (response frame)

Message length: variable.

Usage: Before any memory operation can be carried out, the card has to be selected and authenticated.

NOTE: If the [Key type] and [Key value] is empty in the request frame 1, the A10-P will use key value with "FFFFFFFFFFFF" and key type with 'A'.

Message element:

Request1 frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
P07	3	Message ID
[Sector number]	2	The sector number. Decimal string: 00~39.
[Key type]	1	(optional) Determine which card type to authenticate. Format: 'A' or 'B'.
[Key value]	12	(optional) The card will be authenticated with this key value. Format: hexadecimal string.
<ETX>	1	<03>
[LRC]	1	Checksum

Request2 frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
P07	3	Message ID
[Sector number]	2	The sector number. Decimal string: 00~39.
[Key number]	3	The key stored in the A10-P. Decimal string: 000~255.
[Key type]	1	Determine which card type to authenticate. Format: 'A' or 'B'.
<ETX>	1	<03>
[LRC]	1	Checksum

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
P07	3	Message ID
[Result]	1	'0': Success. '1': Fail.
[Reason]	1	Optional: if [Result] = '1' '1': Format error. '2': Sequence error. '3': Key authentication fail.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message P07 (request frame)	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message P07 (response frame)
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message P08 MIFARE Ultralight Card Read Page

Format: <STX>P08[page number]<ETX>[LRC] (request frame)
 <STX>P08[Result][Page data][Reason]<ETX>[LRC] (response frame)

Message length: Fixed 8 ytes for the request frame

Fixed 8 or 15 bytes for the response frame.

Usage: Read the block data.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
P08	3	Message ID
[page number]	2	The page number. Decimal string: 00~47.
<ETX>	1	<03>
[LRC]	1	Checksum

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
P08	3	Message ID
[Result]	1	'0': Success. '1': Fail.
[page data]	8.	Optional: if [Result] = '0' The page data. Format: hexadecimal string.
[Reason]	1	Optional: if [Result] = '1' '1': Format error. '2': Sequence error. '3': Read data fail.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message P08 (request frame)	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message P08 (response frame)
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message P09 MIFARE Ultralight Card Write Page

Format: <STX>P09[page number][page data]<ETX>[LRC] (request frame)
 <STX>P09[Result][Reason]<ETX>[LRC] (response frame)

Message length: Fixed 16 ytes for the request frame

Fixed 7 or 8 bytes for the response frame.

Usage: Write the data to the specific block.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
P09	3	Message ID
[page number]	2	The page number. Decimal string: 00~47.
[page data]	8	The page data will be written into the MIFARE card. Format: hexadecimal string.
<ETX>	1	<03>
[LRC]	1	Checksum

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
P09	3	Message ID
[Result]	1	'0': Success. '1': Fail.
[Reason]	1	Optional: if [Result] = '1' '1': Format error. '2': Sequence error. '3': Write data fail.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message P09 (request frame)	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message P09 (response frame)
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message P10 MIFARE Classic/Ultralight Card Read Block

Format: <STX>P10[Block number]<ETX>[LRC] (request frame)
 <STX>P10[Result][Block data][Reason]<ETX>[LRC] (response frame)

Message length: Fixed 9 bytes for the request frame

Fixed 8 or 39 bytes for the response frame.

Usage: Read the block data.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
P10	3	Message ID
[Block number]	3	The block number. Decimal string: 000~255.
<ETX>	1	<03>
[LRC]	1	Checksum

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
P10	3	Message ID
[Result]	1	'0': Success. '1': Fail.
[Block data]	32.	Optional: if [Result] = '0' The block data. Format: hexadecimal string.
[Reason]	1	Optional: if [Result] = '1' '1': Format error. '2': Sequence error. '3': Read data fail.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message P10 (request frame)	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message P10 (response frame)
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message P11 MIFARE Classic/Ultralight Card Write Block

Format: <STX>P11[Block number][Block data]<ETX>[LRC] (request frame)
 <STX>P11[Result][Reason]<ETX>[LRC] (response frame)

Message length: Fixed 41 bytes for the request frame

Fixed 7 or 8 bytes for the response frame.

Usage: Write the data to the specific block.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
P11	3	Message ID
[Block number]	3	The block number. Decimal string: 000~255.
[Block data]	32	The block data will be written into the MIFARE card. Format: hexadecimal string.
<ETX>	1	<03>
[LRC]	1	Checksum

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
P11	3	Message ID
[Result]	1	'0': Success. '1': Fail.
[Reason]	1	Optional: if [Result] = '1' '1': Format error. '2': Sequence error. '3': Write data fail.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message P11 (request frame)	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message P11 (response frame)
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message P12 MIFARE Classic/Ultralight Card Read Sector

Format: <STX>P12[Sector number]<ETX>[LRC] (request frame)

<STX>P12[Result][Sector data][Reason]<ETX>[LRC] (response frame)

Message length: Fixed 8 bytes for the request frame

Variable for the response frame.

Usage: Read a sector data from the MIFARE card.

NOTE: For MIFARE Classic 4K cards, each sector contains 4 blocks in sectors 0~31; and each sector contains 16 blocks in sectors 32~39 .

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
P12	3	Message ID
[Sector number]	2	The sector number. Decimal string: 00~39.
<ETX>	1	<03>
[LRC]	1	Checksum

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
P12	3	Message ID
[Result]	1	'0': Success. '1': Fail.
[Sector data]	128 or 512	Optional: if [Result] = '0' The Sector data. Format: hexadecimal string.
[Reason]	1	Optional: if [Result] = '1' '1': Format error. '2': Sequence error. '3': Read data fail.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message P12 (request frame)	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message P12 (response frame)
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message P13 MIFARE Classic/Ultralight Card Write Sector

Format: <STX>P13[Sector number][Sector data]<ETX>[LRC] (request frame)
 <STX>P13[Result][Reason]<ETX>[LRC] (response frame)

Message length: Variable for the request frame

Fixed 7 or 8 bytes for the response frame.

Usage: Write a sector data to the MIFARE card.

NOTE: ultra light card -

sector 0: 48 bytes (96 bytes hexadecimal string).

ultra light C card -

sector 1: 64 bytes. (128 bytes hexadecimal string).

sector 2: 32 bytes. (64 bytes hexadecimal string).

classic 1K card and classic 4K card -

sector 0: 32 bytes. (64 bytes hexadecimal string).

sector 1-31: 48 bytes. (96 bytes hexadecimal string).

classic 4K card -

sector 32-39: 240 bytes. (480 bytes hexadecimal string).

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
P13	3	Message ID
[Sector number]	2	The sector number. Decimal string: 00~39.
[Sector data]	64 or 96 or 128 or 480	The data will be written into the MIFARE card. Format: hexadecimal string.
<ETX>	1	<03>
[LRC]	1	Checksum

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
P13	3	Message ID
[Result]	1	'0': Success. '1': Fail.
[Reason]	1	Optional: if [Result] = '1' '1': Format error. '2': Sequence error. '3': Write sector fail.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message P13 (request frame)	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message P13 (response frame)
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message P14 MIFARE Classic Card Value Operation

Format: <STX>P14[Block number][Op mode][Value][Transfer block]
 <ETX>[LRC] (request frame)
 <STX>P14[Result][Reason]<ETX>[LRC] (response frame)

Message length: Fixed 13, 18 or 21 bytes for the request frame

Fixed 7 or 8 bytes for the response frame.

Usage: Value operation with block data.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
P14	3	Message ID
[Block number]	3	The block number. Decimal string: 000~255.
[op mode]	1	Operation mode: '0' = Create MIFARE Value block. '1' = Decrement. '2' = Increment. '3' = Copy block.
[Value]	8	(Optional), if [op mode] != '3' Value blocks allow performing electronic purse functions. Format: hexadecimal string (signed integer). Ex : 1234567 _d = 00 12 D6 87
[Transfer block]	3	(Optional), if [op mode] != '0' The block number. Decimal string: 000~255.
<ETX>	1	<03>
[LRC]	1	Checksum

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
P14	3	Message ID
[Result]	1	'0': Success. '1': Fail.
[Reason]	1	Optional: if [Result] = '1' '1': Format error. '2': Sequence error. '3': Create value block fail. '4': Value restore fail. '5': Value operation fail. '6': Value transfer fail.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message P14 (request frame)	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message P14 (response frame)
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message P15 Load MIFARE key

Format: <STX>P15[Key number][Key A value][Key B value]<ETX>[LRC]
 (request frame)
 <STX>P15[Result][Reason]<ETX>[LRC] (response frame)

Message length: Fixed 33 bytes for the request frame

Fixed 7 or 8 bytes for the response frame.

Usage: A10-P can save up to 256 key sets for the MIFARE Classic card application.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
P15	3	Message ID
[Key number]	3	The key unique number. Range: '000'~'255'
[Key A value]	12	The key A data will be stored in the A10-P.
[Key B value]	12	The key B data will be stored in the A10-P.
<ETX>	1	<03>
[LRC]	1	Checksum

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
P15	3	Message ID
[Result]	1	'0': Success. '1': Fail.
[Reason]	1	Optional: if [Result] = '1' '1': Format error. '2': Load MIFARE key fail.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message P15 (request frame)	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message P15 (response frame)
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message P16 Identify MIFARE Card Type

Format: <STX>P16<ETX>[LRC] (request frame)
 <STX>P16[Result][Card type][Reason]<ETX>[LRC] (response frame)

Message length: Fixed 6 bytes for the request frame

Fixed 8 bytes for the response frame.

Usage: This command reports the MIFARE card type.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
P16	3	Message ID
<ETX>	1	<03>
[LRC]	1	Checksum

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
P16	3	Message ID
[Result]	1	'0': Success. '1': Fail.
[Card type]	1	Optional: if [Result] = '0' '0': Not MIFARE card or card not supported . '1': MIFARE Ultralight. '2': MIFARE Classic 1K. '3': MIFARE Classic 4K. '4': MIFARE DESFire. '5': MIFARE Plus '6': MIFARE Mini. '7': MPCOS Gemplus. '8': Jewel for Innovision. '9': JCOP31.
[Reason]	1	Optional: if [Result] = '1' '1': Format error. '2': Sequence error. '3': Card detect fail.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message P16 (request frame)	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message P16 (response frame)
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message P17 MIFARE DESfire Card Activation

Format: <STX>P17<ETX>[LRC] (request frame)
 <STX>P17[Result][ATS/PUPI][Reason]<ETX>[LRC] (response frame)

Message length: Fixed 6 bytes for the request frame

Variable for the response frame.

Usage: This command will activate the PICC card.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
P17	3	Message ID
<ETX>	1	<03>
[LRC]	1	Checksum

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
P17	3	Message ID
[Result]	1	'0': Success. '1': Fail.
[ATS/PUPI]	Var.	Optional: if [Result] = '0'
[Reason]	1	Optional: if [Result] = '1' '1': Format error. '2': Sequence error. '3': Card activate fail.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message P17 (request frame)	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message P17 (response frame)
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message P18 MIFARE DESfire Card Deselect

Format: <STX>P18<ETX>[LRC] (request frame)
 <STX>P18[Result][Reason]<ETX>[LRC] (response frame)

Message length: Fixed 6 bytes for the request frame

Fixed 7 or 8 bytes for the response frame.

Usage: Sends the ISO/IEC 14443-4 DESELECT command to the card. After the card receives the DESELECT command, the card will be brought to the HALT state.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
P18	3	Message ID
<ETX>	1	<03>
[LRC]	1	Checksum

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
P18	3	Message ID
[Result]	1	'0': Success. '1': Fail.
[Reason]	1	Optional: if [Result] = '1' '1': Format error. '2': Sequence error. '3': DESELECT execute fail.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message P18 (request frame)	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message P18 (response frame)
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message P19 I/O to MIFARE card with APDU format

Format: <STX>P19[C-APDU]<ETX>[LRC] (request frame)
 <STX>P19[Result][R-APDU][Reason]<ETX>[LRC] (response frame)

Message length: variable.

Usage: The command is used to send an APDU to the card where both data and an ISO status are expected in the response.

Note: The APDU (Application Protocol Data Unit) format is defined in ISO/IEC 7816-4 chapter 5. Below briefly describes the command and response of the APDU.

C-APDU format

CLA	INS	P1	P2	Lc	Data	Le
-----	-----	----	----	----	------	----

R-APDU format

Data (Optional)	SW1~SW2
-----------------	---------

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
P19	3	Message ID
[C-APDU]	Var.	Command APDU. The C-APDU format please see Note.
<ETX>	1	<03>
[LRC]	1	Checksum

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
P19	3	Message ID
[Result]	1	'0': Success. '1': Fail.
[R-APDU]	Var.	Optional: if [Result] = '0' The R-APDU format please see Note.
[Reason]	1	Optional: if [Result] = '1' '1': Format error. '2': Sequence error. '3': APDU fail.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message P19 (request frame)	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message P19 (response frame)
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message P20 I/O to MIFARE card for block data exchange

Format: <STX>P20[CRC mode][Wait time][Block data]<ETX>[LRC]
 (request frame)
 <STX>P20[Result][Block data][Reason]<ETX>[LRC] (response frame)

Message length: variable.

Usage: The command is used to pass a block of data to the MIFARE card.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
P20	3	Message ID
[CRC mode]	1	'0' : No CRC in [block data]. '1' : [Block data] contain CRC.
[Wait time]	4	The time out value for the card response.
[Block data]	Var.	Format: hexadecimal string. The command uses the block format as defined in ISO 14443-4 chapter 7.
<ETX>	1	<03>
[LRC]	1	Checksum

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
P20	3	Message ID
[Result]	1	'0': Success. '1': Fail.
[Block data]	Var.	Optional: if [Result] = '0' The card response data with block format as defined in ISO 14443-4 chapter 7.
[Reason]	1	Optional: if [Result] = '1' '1': Format error. '2': Sequence error. '3': Data exchange fail.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message P20 (request frame)	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message P20 (response frame)
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Overall MIFARE operation flow reference

Before any memory operation for the MIFARE card, the user has to activate the card as follows.

Ref. 1 Activate and authenticate for MIFARE classic card.

HOST	Direction	A10-P
P01 (Enable Mifare)	→	
	←	P01 (Response Frame)
P02 (Query MIFARE Presence)	→	
	←	P02 (Response Frame)
P03 (Anti-collision)	→	
	←	P03 (Response Frame)
P04 (Selection)	→	
	←	P04 (Response Frame)
P16 (Identify MIFARE Card Type)	→	
	←	P16 (Response Frame) '2': MIFARE Classic 1K. '3': MIFARE Classic 4K.
P07 (Classic Card Authentication)	→	
	←	P07 (Response Frame)
Now, user can access memory operation for the MIFARE card (like P10, P11, P12, P13, P14)	→	
	←	The response of memory operation command.
P06 (Card HALT)	→	
	←	P06 (Response Frame)
P01 (Disable Mifare)	→	
	←	P01 (Response Frame)

Ref. 2 The quick method for activating the classic MIFARE cards.

HOST	Direction	A10-P
P01 (Enable Mifare)	→	
	←	P01 (Response Frame)
P05 (Classic Card Activation)	→	
	←	P05 (Response Frame)
P16 (Identify MIFARE Card Type)	→	
	←	P16 (Response Frame) ‘2’: MIFARE Classic 1K. ‘3’: MIFARE Classic 4K.
P07 (Classic Card Authentication)	→	
	←	P07 (Response Frame)
Now, user can access memory operation for the MIFRE card (like P10, P11, P12, P13, P14)	→	
	←	The response of the memory operation command.
P06 (Card HALT)	→	
	←	P06 (Response Frame)
P01 (Disable Mifare)	→	
	←	P01 (Response Frame)

Ref. 3 Activate and authenticate for the MIFARE Ultralight card.

HOST	Direction	A10-P
P01 (Enable Mifare)	→	
	←	P01 (Response Frame)
P02 (Query MIFARE Presence)	→	
	←	P02 (Response Frame)
P03 (Anti-collision)	→	
	←	P03 (Response Frame)
P04 (Selection)	→	
	←	P04 (Response Frame)
P16 (Identify MIFARE Card Type)	→	
	←	P16 (Response Frame) '1': MIFARE Ultralight.
Now, user can access memory operation for MIFRE card(like P08, P09, P10, P11, P12, P13)	→	
	←	The response of the memory operation command.
P06 (Card HALT)	→	
	←	P06 (Response Frame)
P01 (Disable Mifare)	→	
	←	P01 (Response Frame)

Ref. 4 The quick method for activating the Ultralight card.

HOST	Direction	A10-P
P01 (Enable Mifare)	→	
	←	P01 (Response Frame)
P05 (Ultralight Card Activation)	→	
	←	P05 (Response Frame)
P16 (Identify MIFARE Card Type)	→	
	←	P16 (Response Frame) '1': MIFARE Ultralight.
Now, user can access memory operation for the MIFRE card (like P08, P09, P10, P11, P12, P13)	→	
	←	The response of the memory operation command.
P06 (Card HALT)	→	
	←	P06 (Response Frame)
P01 (Disable Mifare)	→	
	←	P01 (Response Frame)

Ref. 5 Activating for the DESFire card

HOST	Direction	A10-P
P01 (Enable Mifare)	→	
	←	P01 (Response Frame)
P05 (Mifare Card Activation)	→	
	←	P05 (Response Frame)
P16 (Identify MIFARE Card Type)	→	
	←	P16 (Response Frame) '4': MIFARE DESFire.
(Reselect to Desfire)		
P06 (MIFARE Card HALT)	→	
	←	P06 (Response Frame)
P17 (Desfire Card Activation)	→	
	←	P17 (Response Frame)
P19 、 P20 (Request frame)	→	
	←	P19 、 P20 (Response Frame)
P18 (DESfire Card Deselect)	→	
	←	P18 (Response Frame)
P01 (Disable Mifare)	→	
	←	P01 (Response Frame)

Section 9 Online Transaction Messages with Master/Session

Keys (MK/SK)

Message 70 PIN entry request (MK/SK)

Format: <STX>70 . [Account] <FS> [session key] [Amount] <FS> [timeout] <ETX> [LRC]
(request frame)

Message length: Variable.

Usage: Display prompt to invite user to input PIN.

The PIN pad polls for keypad inputs until one of following criteria happens:

[ENTER] key pressed; PIN pad validates input and send message 71.

[CAN] key pressed; PIN pad aborts PIN entry and sends <EOT>.

Message 72 received; PIN pad aborts PIN entry and sends <EOT>.

Timed out (session or single key idle for 60 seconds); Acts as [CAN] key.

Notes: **User Interactive Sequence:** The prompts on the LCD of each phase are as following.

After message 70 received (The display language can be selected)	Total Amount \$xxx.xx Enter PIN Push “ENTER”
After user presses first digit (Each * represents one digit)	PIN: ****
If [ENTER] pressed	PIN PAD PROCESSING (Until [CLR] pressed or new command received, can be turned off by message Z7)
If [CAN] pressed	Transaction Cancelled (Return to idle prompt after 3 seconds, can be turned off by message Z7)

PIN length: According to the ANSI X9.8 standard, the length of the PIN should be between 4 to 12 digits. If the user inputs less than 4 digits and then press ENTER, A10-P will beep for error and continue to wait for the user's input. After the user inputs the 13th character, the A10-P will beep for error, conserves the PIN characters 1st to 12th, and wait for the ENTER.

This message has implemented the countermeasure to prevent an exhaustive PIN determination attack. See **Appendix A** for details.

Master key must be selected before starting the transaction: A10-P will warn and refuse message 70 if message 08 was not issued before.

The key and algorithm that used to encrypt the PIN block:

Session Key Active Master Key	8 bytes (16 chars) or 16 bytes (32 chars) Session key	24 bytes (48 chars) Session key
TDES "K0" (24 bytes)	Use MK to decrypt SK Use SK to encrypt PIN	Use MK to decrypt SK Use SK to encrypt PIN
TDES "P0" (24 bytes)	SK must be "0" Use MK to encrypt PIN	SK must be "0" Use MK to encrypt PIN
TDES "K0" (16 bytes)	Use MK to decrypt SK Use SK to encrypt PIN	Error.
TDES "P0" (8 or 16 bytes)	SK must be "0" Use MK to encrypt PIN	Error.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
70	2	Message ID
.	1	<2E>, delimiter
[Account]	8..19	Card account number
<FS>	1	<1C>, field separator
[session key]	16 or 32 or 48	Session key encrypted by the active master key. The operating mode to decrypt the session key will be ECB. Format: hexadecimal string. This field should be all zeros if the active master key is usage "P0"
[Amount]	4 to 12	Amount of goods to be displayed on the A10-P. Accepts digit character ('0' ~ '9'), minus ('-'), comma (',') and decimal point ('.')
<FS>	1	(optional) <1C>, field separator
[timeout]	1	(optional) ASCII characters from '1' to '9' which is the timeout value in units of 30 seconds. Default = $9 \times 30 = 270$ seconds.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message 70	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs) Prompt user to enter PIN.
	←	Message 71 or <EOT> when input times out or [CAN] is pressed
<ACK> (Good echo) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	
		Display “PIN PAD PROCESSING” until [CLEAR] is pressed or another message is received.

Message 71 Encrypted PIN Block Response (MK/SK)

Format: **<STX>71.[fkey][PIN len][fmt][PIN blk][LRC] (response frame)**
<STX>71[error code]<ETX>[LRC] (error frame)

Message length: Fixed 27 bytes for TDES PIN block and 6 bytes for error code frame.

Usage: Send the entered PIN to the host in encrypted format.

Message element:

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
71	2	Message ID
.	1	<2E> delimiter
[fkey]	1	Always '0' (This field is kept to retain the old model compatibility.)
[PIN len]	2	Decimal string representing PIN length in the PIN block. Available value: “04” to “12”, or “00” if Z62 allowing null PIN option is enabled.
[fmt]	2	“01” = TDES PIN block.
[PIN blk]	16	Encrypted PIN block Format: hexadecimal string.
<ETX>	1	<03>

[LRC]	1	Checksum
-------	---	----------

Error frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
71	2	Message ID
[Error code]	1	Code to indicate error (see next page)
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message 70/Z60/Z62	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message 71 or <EOT> when input times out or [CAN] is pressed
<ACK> (Good echo) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	
		Display processing prompt

Error codes:

Code	Meaning
'0'	Null Account input field.
'1'	Key value error. (session key value conflicted with the usage of the active master key, or session key length longer than the active master key)
'2'	Account number shorter than 8 digits.
'3'	Account number longer than 19 digits.
'4'	Account number has character(s) other than '0'-'9'.
'5'	Session key format error.
'6'	Timeout value error
'7'	No more DES operation within 60 minutes (refer to Appendix A)
'8'	Amount string format error originated from Message 70 PIN count, Accept Null PIN flag, and Prompt string format error originated from Message Z62
'9'	Active master key not exists
'A'	Active master key usage incorrect (non K0 or P0 key)
'B'	Key storage access error
'C'	Memory buffer allocation error
'E'	Data length error in a field.
'G'	Specified file not found or authentication error.

Message 72 PIN Entry Cancel

Format: <STX>72<ETX>[LRC] (request frame)

Message length: Fixed 5 bytes.

Usage: Cancel the current transaction that is waiting for user's input and return the A10-P to the IDLE state. The commands that can be canceled by message 72 include 70, Z60, Z62 (for both MK/SK and DUKPT), 60, 62, Z42, Z50, Q1, Q8, Q9, QK.

If the A10-P receives message 72 while processing the user input such as swipe card, enter a PIN or keying in data, It will respond with <EOT> to acknowledge that the operation is canceled.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
72	2	Message ID
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

In Idle mode (normal condition)

HOST	Direction	A10-P
Message 72	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)

OR:

In PIN/data entry mode (cancel/abort the session)

HOST	Direction	A10-P
Message 72	→	
	←	<EOT> If the A10-P is in the PIN/Data entry mode or is waiting for the tapping/swiping card.

Message 7G PIN Change Request (MK/SK)

Format: <STX>7G. [Account] <FS>[session key] [Amount]
<FS>[timeout] <ETX> [LRC]

Message length: Variable

Usage: Display prompt and accept customer PIN input for PIN change process. A10 will then wait till the PIN entered and [ENTER] key is pressed twice to make sure the both entered PIN are matched. After ENTER key is pressed, the string "PROCESSING" will be displayed until the CLEAR key is pressed. During this period, A10 will not process any message other than the CANCEL message (message 72).

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
7G	2	Message ID
.	1	<2E>, delimiter
Account	8..19	Card account number
<FS>	1	<1C>, field separator
[session key]	16 or 32 or 48	Session key encrypted by the active master key. The operating mode to decrypt the session key will be ECB. Format: hexadecimal string. This field should be all zeros if the active master key is usage "P0"
[Amount]	4 to 12	Amount of goods to be displayed on the A10-P. Accepts digit character ('0' ~ '9'), minus ('-'), comma (',') and decimal point ('.')
<FS>	1	(optional) <1C>, field separator
[timeout]	1	(optional) ASCII characters from '1' to '9' which is the timeout value in units of 30 seconds. Default = 9x30 = 270 seconds.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

In Idle mode (normal condition)

HOST	Direction	A10-P
Message 7G	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs) Prompt user to enter PIN
	←	Message 71 or <EOT> when input times out or [CAN] is pressed
<ACK> (Good echo) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	
		Display "PIN PAD PROCESSING" until [CLEAR] is pressed or another message is received.

Message Z60 PIN entry request with external prompt (MK/SK)

Format: `<STX>Z60.[account]<FS>[session key]<FS>[timeout]<ETX>[LRC]`
(request frame)

Message length: Variable 48 to 77 bytes.

Usage: This message must be preceded by a display command (Z2 or Z3) with PIN entry authenticated prompt. The PIN pad polls for keypad inputs until one of following criteria happens:

- [ENTER] key pressed; PIN pad validates input and sends message 71.
- [CAN] key pressed; PIN pad aborts PIN entry and sends <EOT>.
- Message 72 received; PIN pad aborts PIN entry and sends <EOT>.
- Timed out (session or single key idle for 60 seconds); Acts as [CAN] key.

Note: The relation between session key and active master key is the same with message 70 (MK/SK).

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
Z60	3	Message ID
.	1	<2E>, delimiter
[Account]	8 .. 19	Account number
<FS>	1	<1C>, Field separator
[Session key]	16 or 32 or 48	Session key encrypted by the active master key. The operating mode to decrypt the session key will be ECB. Format: hexadecimal string. This field should be all zeros if the active master key is usage "P0"
<FS>	1	(Optional) <1C>, Field separator
[timeout]	1	(Optional) ASCII characters from '1' to '9' which is the timeout value in units of 30 seconds. Default = 9x30 = 270 seconds.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message Z2 or Z3 (With authenticated prompt for PIN entry)	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
		Showing the prompt message
Message Z60	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message 71 (after the customer PIN is entered), or <EOT> when input times out or user pressed the [CAN]
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message Z62 PIN entry request with customized prompt (MK/SK)

Format: <STX>Z62.[account]<FS>[session key][minPIN][maxPIN]
 [null flag][prompt1]<FS>[prompt2]<FS>[proc prompt]<FS>
 [timeout]<ETX>[LRC] (request frame)

Message length: Variable.

Usage: This message directs PIN pad to display “prompt1” and “prompt2” strings, and polls for keypad inputs until one of following criteria happens:

[ENTER] key pressed; PIN pad validates input and sends message 71.

[CAN] key pressed; PIN pad aborts PIN entry and sends <EOT>.

Message 72 received; PIN pad aborts PIN entry and sends <EOT>.

Timed out (session or single key idle for 60 seconds); Acts as [CAN] key.

NOTE: The relation between session key and active master key is the same with message 70 (MK/SK).

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
Z62	3	Message ID
.	1	<2E>, delimiter
[account]	8 .. 19	Account number
<FS>	1	<1C>, field separator
[session key]	16 or 32 or 48	Session key encrypted by the active master key. The operating mode to decrypt the session key will be ECB. Format: hexadecimal string. This field should be all zeros if the active master key is usage “P0”
[minPIN]	2	00, 04 .. 12 minimum PIN length. (Only '00' is available if the [null flag] is set to 'Y').
[maxPIN]	2	00, 04 .. 12 maximum PIN length. (Only '00' is available if the [null flag] is set to 'Y').
[null flag]	1	Y Null PIN allowed N Null PIN not allowed
[prompt1]	Var.	Prompt displayed before any key is pressed It

		alternates with prompt2
<FS>	1	<1C>, field separator
[prompt2]	Var.	(optional) Prompt displayed before any key is pressed. It alternates with prompt1
<FS>	1	<1C>, field separator
[proc prompt]	Var.	(optional) Prompt displayed after the PIN is entered
<FS>	1	(optional) <1C>, field separator
[timeout]	1	(optional) ASCII characters from '1' to '9' which is the timeout value in units of 30 seconds. Default = 9x30 = 270 seconds.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message Z62	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
		(Display [prompt1] and [prompt2] and waiting for the user to enter PIN)
	←	Message 71 (after customer PIN is entered)
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	(Display [proc prompt])

Message Z64 Query Key Check Value (KCV)

Format: <STX>Z64 [KeyID] <ETX> [LRC] (request frame)

Message length: Fixed 7 bytes.

Usage: This message will export the KCV of the specified master key.

KCV is calculated as following:

1. Use [KeyID] specified key as encryption key.
2. Use "0000000000000000" (8 bytes zero) as data.
3. If the encrypt key is single length (8 bytes), use TDES algorithm to encrypt the data.
4. Take leftmost 3 bytes as KCV, output KCV as message Z65.

Example: TDES key "0123456789ABCDEF FEDCBA9876543210" will have KCV as "08D7B4".

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
Z64	3	Message ID
[KeyID]	1	'0' ~ '9', 'A' ~ 'G', The ID of master keys.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message Z64	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) (<EOT> after 3 NAKs)
	←	Message Z65
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message Z65 Key Check Value Response

Format: <STX>Z65 [KeyID] [KCV] <ETX>[LRC] (response frame)

Message length: Variable. 13 bytes for KCV, or 8 bytes for error code.

Usage: This message is the response of Z64.

If [KeyID] specified in Z64 is pointing to a valid master key, the KCV will be sent.

Otherwise a question mark '?' will be sent.

Message element:

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
Z64	3	Message ID
[KeyID]	1	'0' ~ '9', 'A' ~ 'G', The ID of master keys.
[KCV]	6 or 1	Success: 6 characters of KCV. Fail: '?'.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

Refer to message Z64.

Message Z66 Message Authentication Code (MAC) Request

Format: <STX>Z66 [PktType] [SeqNo] [KeyId] <FS> [SessionKey] <FS>
 [SecKeyId] <FS> [Message] <ETX> [LRC] (request frame)

Message length: Variable 14 to 270 bytes.

Usage: This message is used to generate the MAC codes according to the algorithm specified in ANSI X9.19 (ISO 9797-1). User can send ASCII strings or hexadecimal strings to the A10-P by message Z66 to generate its MAC. User can also breakdown a long message into multiple Z66 messages with ascending sequence numbers to generate a MAC.

Notes: **Message Length:** A onetime message that can be up to 224 characters (equal to 112 bytes when send as a hexadecimal string because 2 characters form 1 byte). Multiple messages can have sequence numbers from 00 to 99. Thus the maximum capacity of message Z66 is 22,400 characters (or 11,200 bytes in binary mode).
Multiple messages: When using multiple messages, [KeyId], [SessionKey] and [SecKeyId] must be the same. [Message] must be a multiple of 8 characters (or 16 characters in binary mode), or else A10-P will generate a wrong MAC.
MAC algorithm: A10-P generates the TDES MAC according to ISO9797-1 algorithm 3. (Padding with 0s. Initial vector = 0. Refer to **Appendix A** point 10 for the algorithm details.)

Session Key: The value of the session key determines the usage of the specified master keys.

Usage of 1 st Key ID	Usage of 2 nd Key ID	Value of session key	MAC Key
"K0"	N/A	Non-zero	Session key (decrypted by master key specified by [KeyId] using TDES ECB mode.)
"M3" (mode G)	N/A	Zero	Master key specified by [KeyId]. If the specified key is mode 'V', this is for the MAC verification and cannot be used to generate the MAC for message Z66 .
"M1" (mode G)	"M1"	Zero	Master key specified by [KeyId] as left key, and master key specified by [SecKeyId] as right key.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
Z66	3	Message ID
[PktType]	1	'4' = ASCII last or only packet. '5' = ASCII first or middle of multiple packets. '6' = Binary last or only packet. '7' = Binary first or middle of multiple packets.
[SeqNo]	2	'00' to '99'; for onetime only packet, set to 00.
[KeyId]	1	(Optional) Master key to use, range = 'B' to 'E'. If this field is blank, the MAC master key will be the selected key 0 ~ 9.
<FS>	1	<1C>, field separator
[SessionKey]	32	Session key to be decrypted by the Master key pointed by [KeyId]. Format: hexadecimal string. This field should be all zeros if the selected key is usage "M1" or "M3"
<FS>	1	<1C>, field separator
[SecKeyId]	1	(Optional) If first [KeyId] points to key with "K0" or "M3" usage, this field should be omitted.
<FS>	1	<1C>, field separator
[Message]	1-224	ASCII string or Hexadecimal string to be MACed.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow: (Onetime only packets)

HOST	Direction	A10-P
Message Z66 (type 4,6)	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message Z67 (with MAC)
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message flow: (Multiple packets)

HOST	Direction	A10-P
Message Z66 (Seq'00' and types 5,7)	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message Z67 (with status code '1')
<ACK> / <NAK> / <EOT>	→	
Message Z66 (Seq'01'--'98', types 5,7)	→	
	←	<ACK> / <NAK> / <EOT>
	←	Message Z67 (with status code '1')
.....		
Message Z66 (Sequence# larger than last packet, types 4,6)	→	
	←	<ACK> / <NAK> / <EOT>
	←	Message Z67 (with MAC)
<ACK> / <NAK> / <EOT>	→	

Message Z67 Message Authentication Code (MAC) Response

Format: <STX>Z67[status][MAC]<ETX>[LRC] (response frame)

Message length: Fixed 7 (status only) or 23 (with MAC) bytes.

Usage: The A10-P generated MAC response. It contains the status codes or MAC.

Message element:

Response frame (A10-P to HOST)

HOST	Direction	A10-P
<STX>	1	<02>
Z67	3	Message ID
[status]	1	'0'=Success, MAC follows '1'=Ready for the next Z66 packet, used in multiple messages. '2'=Sequence numbers out of order '3'=Master key specified in [KeyId] not exists, or range unacceptable (id 0 to A), or usage not “K0”, “M1”, “M3”. '4'=Master key specified in [SecKeyId] is unreasonable. The [SecKeyId] only exists if [KeyId] points to a “M1” master key, and the [SecKeyId] itself should have “M1” usage. '5'=[Message] length has error (too long, zero length, or not an even number in binary mode) '6'=[PkyType] flag has invalid value '7'=[Message] contents error (i.e. characters larger than 'F' in binary mode) '8'=[SessionKey] invaild '9'=MAC master key length should not be 8 'A'=Session key is incompatible to the usage of the specified master key. (If MK's usage is “M1” or “M3”, SK should contains all zeros, if MK's usage is “K0”, SK should not be zero.) 'B' =Fatal error (malloc and DES error)
[MAC]	16	Calculated MAC, optional if status = 0.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

Refer to message Z66

Section 10 Online Transaction Messages with Derived

Unique Key Per Transaction (DUKPT)

The following messages are designed for the Derived Unique Key Per Transaction (DUKPT) key management scheme described in ANSI X9.24: 2009.

Note that some of the messages have the same IDs as those in the MK/SK scheme, but with different message formats.

The following messages fall into this category:

- 60 Pre-Authorization PIN Entry Request
- 62 Pre-Authorization Amount Authorization Request
- 63 Pre-Authorization Amount Authorization Response
- 70 PIN entry request
- 71 Encrypted PIN block response
- 72 PIN entry cancel
- Z60 PIN entry request with external prompt (DUKPT)
- Z62 PIN entry request with customized prompt
- 76 PIN Entry Test Request
- 90 Load First Initial Key Request
- 91 Load Initial Key Response
- 94 Load Second Initial Key Request
- 96 Select Active Key Set

Message 60 Pre-authorization PIN Entry Request

Format: <STX>60[account]<ETX>[LRC] (request frame)

Message length: Variable 13 to 24 bytes.

Usage: Message 60 and 62 forms the sequence of pre-authorization transaction (authorize before purchase amount decided.)

This message must be preceded by a display command (Z2 or Z3) with PIN entry authenticated prompt. The PIN pad polls for keypad inputs until one of following criteria happens:

- [ENTER] key pressed; PIN pad validates input and sends message 71.
- [CAN] key pressed; PIN pad aborts PIN entry and sends <EOT>.
- Message 72 received; PIN pad aborts PIN entry and sends <EOT>.
- Timed out (session or single key idle for 60 seconds); Acts as [CAN] key.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
60	2	Message ID
[Account]	8..19	Primary account number
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message Z2 or Z3	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
		Show Prompt Messages
Message 60	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	(User enters PIN and press ENTER) Message 71
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	
		Display “PIN PAD PROCESSING” until CLEAR is pressed or another message received.
Message 62	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	(User confirms the amount) Message 63
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message 62 Pre-authorization Amount Authorization Request

Format: <STX>62 [DC Ind] [amount] <ETX> [LRC] (request frame)

Message length: Variable 7 to 15 bytes.

Usage: Display prompts and accept the customer PIN input. The following prompts will be displayed:

“Total Amount \$xxx.xx”

“Enter – Confirm”

”Cancel – Decline”

xxx.x is the content of the amount field, with length between 4 to 8 positions. The A10-P will then wait until either the CAN or ENTER key is pressed. If the ENTER key is pressed, the A10-P will respond with a positive confirmation. If the CAN key is pressed, the A10-P will respond with a negative confirmation. During this period, the A10-P will not process any message other than the message 72 (cancel transaction).

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
62	2	Message ID
[DC Ind]	1	D/C: Debit/Credit Indicator
[amount]	4 to 12	Amount of goods to be displayed on the A10-P. Accepts digit character ('0' ~ '9'), minus ('-'), comma ('.') and decimal point ('.')
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

Refer message 60.

Message 63 Pre-authorization Amount Authorization Response

Format: <STX>63[Confirm] <ETX>[LRC] (response frame)

Message length: Fixed 6 bytes.

Usage: Send the Authentication Code and confirms the transaction amount to the host.

Message element:

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
63	2	Message ID
[Confirm]	1	'0' OK, '1' denied.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

Refer message 60.

Message 70 PIN Entry Request (DUKPT)

Format: <STX>70[account]<FS>[DC Ind][amount]<FS>[timeout]<ETX>[LRC]
(request frame)

Message length: Variable 21 to 40 bytes.

Usage: Display prompt to invite user to input PIN.

The PIN pad polls for keypad inputs until one of following criteria happens:

- [ENTER] key pressed; PIN pad validates input and sends message 71.
- [CAN] key pressed; PIN pad aborts PIN entry and sends <EOT>.
- Message 72 received; PIN pad aborts PIN entry and sends <EOT>
- Timed out (session or single key idle for 60 seconds); Acts as [CAN] key.

NOTE: User Interactive Sequence: The prompts on the LCD of each phase are as the following.

After message 70 received (The displayed language can be selected)	Total Amount \$xxx.xx Enter PIN Push “ENTER”
After user press first digit (Each * represents one digit)	PIN: ****
If [ENTER] pressed	PIN PAD PROCESSING (Until [CLR] pressed or new command received, can be turned off by message Z7)
If [CAN] pressed	Transaction Cancelled (Return to idle prompt after 3 seconds, can be turned off by message Z7)

PIN length: According to the ANSI X9.8 standard, the length of the PIN should be between 4 to 12 digits. If the user inputs less than 4 digits and then press ENTER, the A10-P will beep for error and continue to wait for the user's input. When the user inputs the 13th character, the A10-P will beep for error, conserves PIN characters 1st to 12th, and waits for the ENTER.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
70	2	Message ID
[Account]	8..19	Primary account number
<FS>	1	<1C>, field separator
[DC Ind]	1	D/C: Debit/Credit Indicator
[Amount]	4 to 12	Amount of goods to be displayed on the A10-P. Accepts digit character ('0' ~ '9'), minus ('-'), comma ('.') and decimal point ('.') and decimal point ('.')
<FS>	1	(optional) <1C>, field separator
[timeout]	1	(optional) ASCII characters from '1' to '9' which is the timeout value in units of 30 seconds. Default = 9x30 = 270 seconds.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message 70	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message 71 or <EOT> when [CAN] is pressed or input times out.
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	
		Display "PIN PAD PROCESSING" until CLEAR is pressed or another message received.

Message 71 Encrypted PIN Block Response (DUKPT)

Format: <STX>71<fkey flag>[Key Serial#][PIN][LRC] (response frame)
 <STX>71[error code]<ETX>[LRC] (error frame)

Message length: Variable 32 to 42 bytes.

Usage: Send the entered PIN to the host in encrypted format.

Message element:

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
71	2	Message ID
[fkey flag]	1	Always '0' (This field is kept to retain the old model compatibility.)
[Key Serial#]	10..20	Hex string representing key serial number corresponding to the derived key. The leading 'F' character may be stripped (can be controlled by message 7A). For DUKPT Z62 command: if null PIN is entered, this field is empty.
[PIN]	16	Encrypted PIN block Format: hexadecimal string.
<ETX>	1	<03>
[LRC]	1	Checksum

Error codes:

Code	Meaning
'0'	Null Account input field.
'2'	Account number shorter than 8 digits.
'3'	Account number longer than 19 digits.
'4'	Account number have characters other than '0'-'9'.
'5'	[D/C ind] field not exists or format error.
'6'	Timeout value error.
'8'	Amount string format error originated from Message 70 PIN count, Accept Null PIN flag, and Prompt string format error originated from Message Z62
'A'	No DUKPT key injected
'B'	Flash read/write error
'C'	Memory buffer allocation error
'F'	DUKPT operation limit (1 million) reached, program stops.

Message flow:

Refer to message70 (DUKPT).

Message 72 PIN Entry Cancel

Format: <STX>72<ETX>[LRC] (request frame)

Message length: Fixed 5 bytes.

Usage: Cancel current transaction and return the A10-P to IDLE state. This message is used to interrupt the command in process. If A10-P receives message 72 while processing the user input such as swipe card, enter PIN or key-in data, it will respond with <EOT> to acknowledge that the operation is canceled.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
72	2	Message ID
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

In Idle mode (normal condition)

HOST	Direction	A10-P
Message 72	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)

OR

In PIN/data entry mode (cancel/abort the session)

HOST	Direction	A10-P
Message 72	→	
	←	<EOT> if the A10-P is in PIN/Data entry mode or waiting for tapping/swiping card.

Message Z60 PIN entry request with external prompt (DUKPT)

Format: <STX>Z60.[account]<FS>[timeout]<ETX>[LRC] (request frame)

Message length: Variable 15 to 28 bytes.

Usage: This message must be preceded by a display command (Z2 or Z3) with PIN entry authenticated prompt. The PIN pad polls for keypad inputs until one of following criteria happens:

- [ENTER] key pressed; PIN pad validates input and sends message 71.
- [CAN] key pressed; PIN pad aborts PIN entry and sends <EOT>.
- Message 72 received; PIN pad aborts PIN entry and sends <EOT>.
- Timed out (session or single key idle for 60 seconds); Acts as [CAN] key.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
Z60	3	Message ID
.	1	<2E>, delimiter
[Account]	8 .. 19	Account number
<FS>	1	(Optional) <1C>, Field separator
[timeout]	1	(Optional) ASCII characters from '1' to '9' which is the timeout value in units of 30 seconds. Default = 9x30 = 270 seconds.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message Z2 or Z3	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
		Show prompt messages
Message Z60	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message 71 (after customer PIN is entered), or <EOT> when input times out or user presses [CAN]
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message Z62 PIN entry request with customized prompt (DUKPT)

Format: `<STX>Z62.[account]<FS>[minPIN][maxPIN][null flag]
[prompt1]<FS>[prompt2]<FS>[proc prompt]<FS>[timeout]<ETX>[LRC]
(request frame)`

Message length: Variable.

Usage: This message directs PIN pad to display “prompt1” and “prompt2” strings, and polls for keypad inputs until one of following criteria happens:

- [ENTER] key pressed; PIN pad validates input and sends message 71, and display [procPrompt].
- [CAN] key pressed; PIN pad aborts PIN entry and sends <EOT>.
- Message 72 received; PIN pad aborts PIN entry and sends <EOT>.
- Timed out (session or single key idle for 60 seconds); Acts as [CAN] key..

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
Z62	3	Message ID
.	1	<2E>, delimiter
[account]	8 .. 19	Account number
<FS>	1	<1C>, field separator
[minPIN]	2	00, 04 .. 12 minimum PIN length. (‘00’ only available when [null flag] is set to ‘Y’).
[maxPIN]	2	00, 04 .. 12 maximum PIN length. (‘00’ only available when [null flag] is set to ‘Y’).
[null flag]	1	Y Null PIN allowed N Null PIN not allowed
[prompt1]	Var.	Prompt displayed before any key is pressed, alternates with prompt2
<FS>	1	<1C>, field separator
[prompt2]	Var.	(optional) Prompt displayed before any key is pressed, alternates with prompt1
<FS>	1	<1C>, field separator
[proc prompt]	Var.	(optional) Prompt displayed after PIN is entered
<FS>	1	(optional) <1C>, field separator

[timeout]	1	(optional) ASCII characters from '1' to '9' which is the timeout value in units of 30 seconds. Default = 9x30 = 270 seconds.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message Z2 or Z3	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
		Show prompt messages
Message Z62	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
		(Display [prompt1] and [prompt2], and wait for user to enter PIN)
	←	Message 71 (after customer PIN is entered)
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	(Display [proc prompt])

Message 76 PIN Entry Test Request

Format: <STX>76[account]<FS>[DC Ind][amount]<ETX>[LRC] (request frame)

Message length: Variable 19 to 34 bytes.

Usage: This message is designed to perform the continuous DUKPT PIN entry test. A10-P will generate the encrypted PIN block using specified account and default PIN: 1234.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
76	2	Message ID
[Account]	8..19	Primary account number
<FS>	1	<1C>, field separator
[DC Ind]	1	D/C: Debit/Credit Indicator
[Amount]	4 to 12	Amount string, this field is for format consistency with message 70 (DUKPT) but will not be displayed.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

This message is identical to message70 except that a PIN of '1234' is used instead of getting keypad input.

Message 7A KSN output format

Format: <STX>7A[KSN_format] <ETX>[LRC] (request frame)

Message length: Fixed 6 bytes.

Usage: This message sets the key serial number (KSN) format of message 71 (DUKPT frame).
Format 0 is the original mode (strips leading 'F' of KSN) which is compatible with PP690, PP790SE and A10-P. Format 1 is full mode (output full 20 characters of KSN).

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
7A	2	Message ID
[KSN_format]	1	'0': message 71 outputs KSN without leading 'F' '1': message 71 outputs KSN with leading 'F'.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message 7A	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)

Message 90 Load DUKPT Initial Key (Key Set 1)

Format: <STX>90 [IPEK] [KSN] <ETX> [LRC] (Clear Text)
 <STX>90 [TR-31 Key Block] <ETX> [LRC] (Encrypted)

Message length: Fixed 57 bytes for clear text format; 109 (format B) or 149 (format D) bytes for the TR-31 format .

Usage: Load the first set of DUKPT initial key and serial number key to the A10-P. Consequent keys will be generated using the provided data.

Clear text mode message 90 is only accepted when key injection mode is enabled by dual password authentication. Use it only in security-controlled environment with HSM.

Refer to **Appendix A** Specification for detailed information.

All existing keys in the A10-P will be erased when the first message 02 or 90 is sent in a new clear text key injection session.

Note: The following is the VISA required key serial number format:
 4'F' characters, a 6-digit keyset identifier, with a 5-digit device ID, followed by a '0',
 i.e. "FF FF kk kk kk dd dd d0 00 00"

Message element:

Clear text frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
90	2	Message ID
[IPEK]	32	Initial PIN encryption key. Format: hexadecimal string.
[KSN]	20	Key serial number used in generating consequent keys. Format: hexadecimal string.
<ETX>	1	<03>
[LRC]	1	Checksum

Encrypted frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
90	2	Message ID
[TR-31 Key Block]	104 or 144	TR-31 key block with an optional header block that contains the KSN. See Appendix A for details.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message 90	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message 91
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Example:

Clear Text

IPEK key to be loaded: ABCDEF0123456789FEDCBA9876543210

KSN: FFFF9876543210E00000

The resulting 90 message :

<STX>90ABCDEF0123456789FEDCBA9876543210FFFF9876543210E00000<ETX> [LRC]

TR-31 Key Block

Key Block Protecting Key: AA55AA55AA55AA55 3434343434343434

IPEK key to be loaded: ABCDEF0123456789 FEDCBA9876543210

KSN: FFFF9876543210E00000

Key Block Header: B0104B1TX00N0100 KS18FFFF9876543210E00000

Padded IPEK: 0080 ABCDEF0123456789 FEDCBA9876543210 30111D18CC4C

Derived KBK: 3C50E1B7962F2171DC8643F1D923ABF7

Derived KBMK: 46FBEEB64EAE26A650952DA4F6DD8325

CMAC of (KBH + Padded key data), using KBMK: 93C3D5EBC6C407E4

Use CMAC as IV to do TDES CBC encryption on padded key data, using KBK:

Encrypted key data: C86E6E3B24544F97C629FB0E0586A0285D35BA78E9B13FB

Result: <02>90B0104B1TX00N0100KS18FFFF9876543210E00000EC86E6E3B24544F9

7C629FB0E0586A0285D35BA78E9B13FB93C3D5EBC6C407E4<03>

Message 91 Load Initial Key Response

Format: <STX>91[Status][Error code]<ETX>[LRC] (response frame)

Message length: Variable (max 7 bytes.)

Usage: Confirmation of the initial key loading. A10-P will show a message "IPEK n loaded" (n = 1 or 2) to confirm the successful loading of the initial key of set 1 or set 2 visually.

Message element:

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
91	2	Message ID
[Status]	1	'0' success '1' fail
[Error code]	1	[Error Code] if status = 1
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

Refer to messages 90, 94.

Error codes:

Code	Meaning
'1'	Processing messages 90, 94 without authentication and process the authentication first
'2'	IPEK and KSN format error: not hexadecimal character.
'3'	Internal memory allocation error.
'4'	Internal failure to read the key structure
'A'	TR31 format error.
'B'	Insecure key injection. (New key is longer than the key used to protect it.)
'C'	Failure to verify the MAC value.
'D'	KLK does not exist.
'F'	Key loading count over the limit.

Message 94 Load DUKPT Initial Key (Key Set 2)

Format: <STX>94 [IPEK] [KSN] <ETX> [LRC] (Clear Text)
 <STX>94 [TR-31 Key Block] <ETX> [LRC] (Encrypted)

Message length: Fixed 57 bytes for clear text format; 109 (format B) or 149 (format D) bytes for the TR-31 format.

Usage: Load the second set of DUKPT initial key and serial number key to the A10-P.
 Consequent keys will be generated using the provided data.

Note: Refer to message 90.

Message element:

Clear Text frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
94	2	Message ID
[IPEK]	32	Initial PIN encryption key. Format: hexadecimal string.
[KSN]	20	Key serial number used in generating the consequent keys. Format: hexadecimal string.
<ETX>	1	<03>
[LRC]	1	Checksum

Encrypted frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
94	2	Message ID
[TR-31 Key Block]	104 or 144	TR-31 key block with optional header block that contains the KSN. See Appendix A for details.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message 94	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message 91
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message 96 Select Active Key Set

Format: <STX>96[keyset]<ETX>[LRC] (request frame)

Message length: Fixed 6 bytes.

Usage: Select the active key set for the following transactions. This parameter will be saved and lasts until the next message 96 is received or the DUKPT life cycle ends.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
96	2	Message ID
[keyset]	1	ASCII character '0' = First key set '1' = Second key set
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message 96 (request frame)	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)

Message 98 Check DUKPT Keyset Status

Format: <STX>98[keyset][Key Info Query]<ETX>[LRC] (request frame)

Message length: Fixed 6 bytes.

Usage: Check the status of the specified DUKPT keyset.

Response message 99.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
98	2	Message ID
[keyset]	1	'0' to '1' as key set identifier.
[Key info query]	1	Optional. If omitted, response frame will contain the [result] field only. If set to '1', info fields will be sent.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message 98	→	
	←	<ACK> if LRC ok. <NAK> if LRC error.
	←	Message 99
<ACK> if LRC ok. <NAK> if LRC error. <EOT> if LRC error after 2 retries.	→	

Message 99 DUKPT Keypad Status Response

Format: <STX>99 [status] [usage] <FS> [mode] <FS> [algorithm] <FS>
 [KCV] <ETX> [LRC] (response frame)

Message length: 6 or 15bytes.

Usage: Response of the DUKPT keypad status.

Message element:

Response frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
99	2	Message ID
[status]	1	'O': Key not exists. 'F': Key exists and operative. 'E': Key inoperative.
[usage]	2	(Optional), TR-31 key usage, for DUKPT, always "B1"
<FS>	1	(Optional), <1C>, field separator
[mode]	1	(Optional), TR-31 mode of use, for DUKPT, always 'X'
<FS>	1	(Optional), <1C>, field separator
[algorithm]	1	(Optional), TR-31 key algorithm, for DUKPT, always 'T'
<FS>	1	(Optional), <1C>, field separator
[KCV]	6	(Optional), Hex string represents KCV
<ETX>	1	<03>
[LRC]	1	Check value

Message flow:

Refer to message 98.

Section 11 Prompt Control and Data Entry Messages

The display prompts of the A10-P are built-in and digitally signed with the A10-P application to ensure its integrity and authenticity.

These commands have fixed display prompts:

- Start IC card transaction (T11, T15, T33)
- Start RFID transaction (T61, Q8)
- Swipe Magnetic Stripe Card (Q1)
- Combined card transaction (Q9, QK)
- Standard PIN entry (70)

These commands have selectable display prompts:

- Non-PIN data entry (Z42, Z50): The prompt can be selected from **Appendix G** (Z2/Z3 authenticated mode)
- PIN entry with external prompts (Z60)
- PIN entry with customized prompts (Z62)

The host application can also command A10-P to display any string on the LCD, but if the prompt string is not MAC authenticated, the PIN entry or data entry function will be disabled.

Z2 / Z3 Authenticated mode with fixed prompt

To enable messages Z42 and Z50, the user has to issue Z2 / Z3 messages with a prompt ID supported by the A10-P (See **Appendix G**). These prompts are verified to make sure users will not expose sensitive information (such as PIN) accidentally.

For security reason, to issue the authenticated frame of Z2 / Z3 at the first time, the <SUB> flag is mandatory. After the Z2 authenticated mode is entered, the A10-P will accept two kinds of Z2 packet:

1. Z2 packet in normal mode, without the <SUB> flag and contains only digits (0~9)
2. Z2 packet in authenticated mode, without the <SUB> flag.

For example, issuing Z2<GS>005<SUB> and Z2<GS>016 will show "PLEASE ENTER DRIVER LICENSE" on the screen.

Z2 / Z3 PIN entry mode with fixed prompt

To enable message Z60, the user has to issue Z2 / Z3 messages with a prompt ID supported by the A10-P dedicated for PIN entry (See **Appendix H**). These prompts are verified to make sure users will not misunderstand the PIN entry request message as other non-sensitive data. Also prompt1 and prompt2 of message Z62 will be checked to see if they are listed in this prompt table. If not, the A10-P will reject message Z62.

Messages other than Z2, Z3, Z42, Z50, and Z60, or any unsuccessful Z2 / Z3 messages (wrong prompt ID, format error, Z2 message including non decimal characters), will make the A10-P leaving the Z2 / Z3 authenticated mode to avoid attack.

Message Z0 Move Display Cursor

Format: <STX>Z0[XX][YY]<ETX>[LRC] (request frame)

Message length: Fixed 7 bytes.

Usage: Move the display cursor to the specified location. Z0 message is sent after the A10-P receives the first Z2 message. Under the Z2-authenticated mode, the Z0 message is also disabled.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
Z0	2	Message ID
[YY]	2	Y-coordinate, 01 ~ Max. line
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message Z2	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
		Display string.
Message Z0	→	
	←	<ACK> / <NAK> / <EOT>
		A10-P moves the cursor
Message Z2 (without clear screen)	→	
	←	<ACK> / <NAK> / <EOT>
		Display the 2 nd string from the coordinate specified by Z0.

Message Z1 Return to Idle State

Format: <STX>Z1<ETX>[LRC] (request frame)

Message length: Fixed 5 bytes.

Usage: When A10-P is displaying prompt messages (set by Z2 and Z3), this message resets the display (display idle prompt) and logical state back to idle.

This message cannot be used to abort PIN entry or data entry session, use message 72 in such situation..

This command also reset the magnetic card swiping commands Q1, Q8, Q9.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
Z1	2	Message ID
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message Z1	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)

Message Z2 Display String

Format: `<STX>Z2<SUB><VT>[LinePosition]<INV><US><DI>[string]<ETX><LRC>`
(Request frame, normal)

Message length: Variable, 6 to 70 bytes.

Usage: This message directs PIN pad to display specified string until following the situations:

- User presses [CAN] key on the keypad.
- PIN pad receives another message that changes the display contents (such as message 70), or returns to the idle prompt (message Z1).

Notes: This message can be used to display authenticated prompts for data entry if following criteria are satisfied:

- If the specified string, after stripping non-alphabet characters (00h~40h, 5Bh~60h, 7Bh to 7Fh), is matching any one of the fixed prompts on the “Fixed prompts for Z2 / Z3 authenticated mode”, the string will be treated as “authenticated”, and then message Z42 and Z50 will be enabled.
- If PIN pad is already displaying unauthenticated message, <SUB> will be needed in order to accept authenticated message.
- If <SUB> does not exist, and [string] contains digits ('0' to '9') only, the authenticated state of previous Z2 will be preserved.

Message element:

Z2 request frame (normal mode) (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
Z2	2	Message ID
<SUB>	1	<1A> (optional) If <SUB> exists, A10-P will clear the screen contents.
<VT>	1	<0B>(Optional) User can add this code to set line position value as below.
[LinePosition]	2	(Optional) reply on <VT> Line position value (ASCII) where show message to display.
<INV>	1	(optional) <11>:Inverse mode enable. <12>:Inverse mode disable.
<US>	1	<1F>(Optional)
<DI>	1	(Optional)Rely on <US> <81>: Left-alignment display, <82> :Center-alignment display, <83>: Right-alignment display. <84>: Left-alignment with line display. <85>: Center-alignment with line display. <86>: Right-alignment with line display.
[string]	Var.	String to be displayed. The encoding can be ASCII or UTF-8.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message Z2	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	If format error, return <EOT>. Otherwise display the given string

Examples:

1. **<STX>Z2<SUB>PLEASE ENTER PIN<ETX>**
2. PIN Pad will display “PLEASE ENTER PIN” on the screen, but since this message is not listed in the prompt table, Z42 will be restricted to function key, and Z50 will be rejected, and return to idle prompt.
3. **<STX>Z2<SUB>ENTER<ETX> → <STX>Z002<ETX> → <STX>Z2ACCOUNT NUMBER<ETX>**
4. PIN Pad will display “ENTER” at the first line, then display “ACCOUNT NUMBER” on the second line of screen. Message Z42 and Z50 will be allowed because both strings are listed on the prompt table (ID 024 and 001).
5. **<STX>Z2<SUB>ENTER <ETX> → <STX>Z002<ETX> → <STX>Z2YOUR PIN<ETX>**
6. The first Z2 is matching prompt ID 024 but the second does not match anything.
7. In this case, PIN Pad will revoke the authentication previously enabled by the first Z2, thus Z42 will be restricted to function key, and Z50 will be rejected and back to the idle prompt.
8. **<STX>Z2<SUB>YOUR PIN <ETX> → <STX>Z002<ETX> → <STX>Z2PLEASE<ETX>**
9. The first Z2 does not match any prompt ID, but the second Z2 matches ID 085.
10. In this case, PIN Pad will refuse the second Z2 (response <EOT>) because it violates usage rule.
11. PIN Pad will return to idle prompt, and Z42 / Z50 will be rejected because no valid prompt is displayed.

Message Z2 Display String (Fixed Prompt ID)

Format: `<STX>Z2[op][prompt_id]<SUB><ETX>[LRC]` (request frame)
`<STX>Z2[status]<ETX>[LRC]` (response frame)

Message length: Variable, 6 to 10 bytes.

Usage: This message directs PIN pad to display authenticated prompt string specified by ID number, for data entry command (Z42, Z50), or PIN entry (Z60, 60) until the following situations:

- User presses [CAN] key on the keypad.
- PIN pad receives another message that changes the display contents (such as message 70), or returns to the idle prompt (message Z1).

Note:

- Z2 (prompt ID) can be followed by Z2 (string) which contains digits only.
- The prompts with different usage should not be mixed. PIN pad will revoke all authentications if user tries to change input mode without <SUB>.

Message element:

Z2 request frame (fixed prompt ID) (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
Z2	2	Message ID
[op]	1	Defines which prompt table to be applied for the search: <1D> (GS): data entry (non-PIN) prompt. <1E> (RS): PIN entry prompt.
Prompt ID	3	Prompt ID number that corresponds to the authenticated prompt table. Decimal string: 001 ~ 999.
<SUB>	1	<1A> If <SUB> exists, the A10-P will clear the screen contents.
<ETX>	1	<03>
[LRC]	1	Checksum

Z2 response frame (authenticated mode) (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
Z2	2	Message ID
[status]	1	'0': OK '1': Prompt ID not supported.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message Z2	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message Z2 (response frame)
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	
	←	Display string <EOT> (if received <ACK>)

Message Z2 Display String with Authentication Code

Format: **<STX>Z2<FS>[KeyId] [MAC] [op] [string]<SUB><ETX>[LRC] (Request)**
<STX>Z2[status]<ETX>[LRC] (response frame)

Message length: Variable, 17 to 79 bytes.

Usage: This message directs PIN pad to display MAC authenticated strings, for data entry commands (Z42, Z50), or PIN entry (Z60, 60) until the following situations:

- User presses [CAN] key on the keypad.
- PIN pad receives another message that changes the display contents (such as message 70), or returns to the idle prompt (message Z1).

The MAC value is calculated from the following data concatenated as byte array:

- [op] field.
- All alphabets ('A' to 'Z', 'a' to 'z') in the [string] field. White spaces, punctuation marks and digits are excluded.
- <SUB> character if exists.
- Padding '0' (0x30) to the multiple of 8 bytes.

The algorithm is ISO 9797-1 algorithm 3.

Message element :

Z2 with MAC, request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
Z2	2	Message ID
<FS>	1	<1C>, field separator.
[KeyID]	1	'B' ~ 'E', key to verify the MAC value. The specified key must have usage 'M3' and mode 'V'.
[MAC]	8	Message authentication code of the following message (including <SUB> if exists).
[op]	1	<GS> (0x1D) for Non-PIN entry. <RS> (0x1E) for PIN entry.
[string]	1 .. 62	ASCII string to be displayed
<SUB>	1	<1A> (optional) If <SUB> exists, A10-P will clear the screen contents and also reset the entry mode.
<ETX>	1	<03>
[LRC]	1	Checksum

Z2 with MAC, response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
Z2	2	Message ID
[status]	1	'0': OK '1': MAC key ID error (out of 'B'~'E'). '2': MAC key attribute error. '3': MAC value error. '4': Packet format error.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message Z2	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message Z2 (response frame)
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	
	←	Display string <EOT> (if received <ACK>)

Message Z3 Display Line Prompts

Format: <STX>Z3 [count] <SUB> [prompt1] <FS> [prompt2 . . 7] <ETX> [LRC]
(request frame)

Message length: Variable.

Usage: This message directs PIN pad to display up to seven messages or prompts in rotation. The display changes prompts in 3-second intervals until the following situations:

- User presses [CAN] key on the keypad.
- PIN pad receives another message that changes the display contents (such as message 70), or returns to the idle prompt (message Z1).

This message can be used to display authenticated prompts for data entry if the following criteria are satisfied:

If all prompts, after stripping non-alphabet characters (00h~40h, 5Bh~60h, 7Bh to 7Fh), are matching the fixed prompts on the “Fixed prompts for Z2 / Z3 authenticated mode”, or contains only non-alphabet characters, the string will be treated as “authenticated”, and then message Z42 and Z50 will be enabled.

If PIN pad is already displaying unauthenticated message, <SUB> will be required in order to accept authenticated message.

Elements of each prompt format

<US>	1	<1F>(Optional)
<DI>	1	(Optional)Rely on <US> <81>: Left-alignment display, <82> :Center-alignment display, <83>: Right-alignment display. <84>: Left-alignment with line display. <85>: Center-alignment with line display. <86>: Right-alignment with line display.
[string]	Var.	String to be displayed. The encoding can be ASCII or UTF-8.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
Z3	2	Message ID
[Count]	1	Number of prompts to be displayed ('1' to '7'.)
<SUB>	1	<1A> (optional) If <SUB> exists, A10-P will clear the screen

		contents.
[Prompt1]	var	First string to be displayed. (ASCII or UTF-8),.
<FS>	1	<1C>, field separator
[Prompt2..7]	var	Remaining strings to be displayed. Note. <FS> is required between messages
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message Z3	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
		Display the specified prompts

Examples:

- <STX>Z32<SUB>AMOUNT 1.99<FS>ENTER YOUR PIN<ETX>**

PIN Pad will clear the screen, then display “AMOUNT 1.99” “ENTER YOUR PIN” on the screen. But this message is not authenticated because it does not match any fixed prompts for non-PIN entry. So message Z42 will be restricted to function key, and Z50 will be rejected. PIN Pad will return to idle prompt after Z42/Z50.
- <STX>Z33<SUB>CUSTOMER REF NO<FS>8287<FS>ENTER CASH BACK<ETX>**

PIN Pad will clear screen then display “CUSTOMER REF NO”, “8287”, “ENTER CASH BACK” in three lines. Since line 1 matches ID 016, line 2 contains all digits, and line 3 matches ID 030, this message will be evaluated as authenticated for non-PIN entry. Message Z42 and Z50 will be allowed.
- <STX>Z32<SUB>ENTER<FS>YOUR PIN<ETX>**

PIN Pad will clear screen and display “ENTER”, “YOUR PIN” in two lines. Line 1 matches ID 024 but line 2 does not match any fixed prompt, so this message will not be evaluated as authenticated, Z42 will be restricted to function key, and Z50 will be rejected.
- <STX>Z33<SUB>1. ENTER CARD #<FS>2. ENTER CUST #<FS>3. ENTER ID #<ETX>**

PIN Pad will clear screen and display “1. ENTER CARD #”, “2. ENTER CUST #”, “3. ENTER ID #” in three lines. Line 1 matches ID 031 after stripping non-alphabet characters, line 2 and line 3 also match other IDs. So this message will be evaluated as authenticated, Z42 / Z50 will be allowed.
- <STX>Z32ENTER SSN<FS>OR PHONE #<ETX>**

PIN Pad will reject this message (response <EOT>) because it tries to enter the authenticated display mode without clearing screen.

Message Z3 Display Line Prompts (Fixed Prompt ID)

Format: `<STX>Z3[op][id1]<FS>[id2]...[id7]<ETX>[LRC]` (request frame)
`<STX>Z3[status]<ETX>[LRC]` (response frame)

Message length: Variable 9 to 33 bytes.

Usage: This message directs PIN pad to display up to seven authenticated prompts in rotation.
 The display changes prompts in 3-second intervals until the following situations:

- User presses [CAN] key on the keypad.
- PIN pad receives another message that changes the display contents (such as message 70), or returns to the idle prompt (message Z1).

This message does not have <SUB> character because it will always clear the screen.

Message element:

Z3 request frame (authenticated mode or PIN entry mode) (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
Z3	2	Message ID
[op]	1	Defines which prompt table to be applied for search: <1D> (GS): data entry (non-PIN) prompt. <1E> (RS): PIN entry prompt. (In these 2 modes, A10-P will clear the screen content.)
[Prompt ID1]	3	Prompt ID that corresponds to the fixed prompt provided by the A10-P. Decimal string: 001 ~ 999.
<FS>	1	<1C>, field separator
[Prompt ID2..7]	3	Prompt ID that corresponds to the fixed prompt provided by the A10-P. Decimal string: 001 ~ 999. Note. <FS> is required between prompt IDs.
<ETX>	1	<03>
[LRC]	1	Checksum

Z3 response frame (authenticated mode) (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
Z3	2	Message ID
[status]	1	'0': OK '1': Prompt ID not supported.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message Z3	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message Z3 (response frame)
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	
	←	Display string <EOT> (if received <ACK>)

Message Z3 Display Line Prompts with Authentication Code

Format: `<STX>Z3<FS>[KeyID] [MAC] [count] [op] [prompt1]<FS>[prompt2..7]`
`<SUB><ETX>[LRC] (request frame)`
`<STX>Z3[status]<ETX>[LRC] (response frame)`

Message length: Variable.

Usage: This message directs PIN pad to display up to seven MAC authenticated prompts in rotation. The display changes prompts in 3-second intervals until the following situations:

- User presses [CAN] key on the keypad.
- PIN pad receives another message that changes the display contents (such as message 70), or returns to the idle prompt (message Z1).

The MAC value is calculated from the following data concatenated as byte array:

- [op] field.
- All alphabets ('A' to 'Z', 'a' to 'z') in the [string] field. White spaces, punctuation marks and digits are excluded.
- <FS> character if exists.
- <SUB> character if exists.

Padding '0' (0x30) to the multiple of 8 bytes.

The algorithm is ISO 9797-1 algorithm 3.

Note:

1. If Z3 (string with MAC) is used in combination with Z3 (fixed prompt), their [op] character (GS / RS) must be the same; otherwise A10-P will reject the secondary Z3.
2. A10-P will temporarily turn off the date/timescreen for the first Z2 message it received. The A10-P will return to the idle mode and show the date/time screen after the following possible scenarios: The key entry message command is finished (Z42, Z50 or Z60), the [CAN] is pressed, or the message 72 is received.

Message element:

Z3 with MAC, request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
Z3	2	Message ID
<FS>	1	<1C>, field separator.
[KeyID]	1	'B' ~ 'E', key to verify the MAC value. The specified key must have usage 'M3' and mode 'V'.
[MAC]	8	Message authentication code of the following message (including <SUB> if exists).
[count]	1	'1' ~ '7', number of following prompts.
[op]	1	<GS> (0x1D) for Non-PIN entry. <RS> (0x1E) for PIN entry.
[prompt1]	Var	First string to be displayed. (ASCII or UTF-8).
<FS>	1	<1C>, field separator
[prompt N]	Var.	Second to last string to be displayed. Each prompt is separated by a <FS>.
<SUB>	1	<1A> (optional) If <SUB> exists, A10-P will clear the screen contents.
<ETX>	1	<03>
[LRC]	1	Checksum

Z3 with MAC, response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
Z3	2	Message ID
[status]	1	'0': OK '1': MAC key ID error (out of 'B'~'E'). '2': MAC key attribute error. '3': MAC value error. '4': Packet format error.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message Z3	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message Z3 (response frame)
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	
	←	Display string <EOT> (if received <ACK>)

Example to use Z2 / Z3 with Authentication Code.

[Example 1]

- a. Use message 02 (ANSI TR31 frame) to load the following key to position 'B':
BCDE90123456789ABCDE90123456789A, Usage = M3, Mode = V.
- b. Assume we want to clear the screen and display the following string for PIN entry: "AMOUNT 123456.78 ENTER YOUR PIN".
- c. The data for MAC generation ('A' to 'Z', 'a' to 'z' and ISO8859-15 high page characters from 0xBC to 0xFF, padded with ASCII 0):
<RS>AMOUNTENTERYOURPIN<SUB>0000
1E414D4F554E54454E544552594F555250494E1A30303030
The white spaces and digits are not counted into the MAC. This feature enables the acquirer to issue PIN entry prompts with different amounts, but keeps the same MAC value.
- d. Use the key specified in step a to calculate ISO9797-1 algorithm 3 MAC.
The result is: C51401D727D761E2.
Take leftmost 4 bytes as MAC value: C51401D7.
- e. Send <02>Z2<1C>BCC51401D7<1E>AMOUNT 123456.78 ENTER YOUR PIN<1A><03> to A10-P, Then message Z60 can be issued to request PIN entry.
- f. Send <02>Z2<1C>BCC51401D7 <1A>AMOUNT 123.45 ENTER YOUR PIN<03> to A10-P to see the same MAC applies to different amounts.

[Example 2]

1. Use message 02 (ANSI TR31 frame) to load the following key to position 'B':
6AC292FAA1315B4D8234B3A3D7D5933A, Usage = M3, Mode = V.
2. Assume we want to clear the screen and display for the non-PIN entry: "MESSAGE ONE 1.0" and "MESSAGE TWO 2.0".
3. The data for the MAC generation (padded with ASCII 0):
<GS>MESSAGEONE<FS>MESSAGETWO<SUB>0
1D4D4553534147454F4E451C4D45535341474554574F1A30
4. Use the key specified in step 1 to calculate the ISO9797-1 algorithm 3 MAC.
Take the leftmost 4 bytes as the MAC value: 22C0BAD9.
5. Send <02>Z3<1C>B22C0BAD92<1D>MESSAGE ONE 1.0<1C>MESSAGE TWO 2.0<1A><03> to the A10-P.

Message Z42 Read Key Code

Format: **<STX>Z42[timeout]<ETX>[LRC] (request frame)**

Message length: Variable 7 to 9 bytes.

Usage: Once A10-P receives this command, it begins polling the functional key array until time out happens as specified in [timeout].

If A10-P receives the Z2 / Z3 authenticated frame before message Z42, it will return any key pressed by the user as ASCII key codes via message Z43. Else it will return only the function key codes (F1, F2, F3, CAN, CLR, ENTER), and reject the numerical keys (0 to 9). Multiple key presses or combined key presses will be discarded.

Abort input: Issue message 72 to abort the operation.

Note: **Z2/Z3 required:** Because Z42 will not show any message to prompt the user operation, Z2 or Z3 should be issued before this command, or A10-P will send out <EOT> and stops.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
Z42	3	Message ID
[timeout]	1 to 3	ASCII character from 1 to 255, for example “10” means 10 seconds of timeout.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message Z2 or Z3	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
		Show prompt message
Message Z42	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message Z43
<ACK>/<NAK>/<EOT>	→	

Message Z43 Read Key Code Response

Format: <STX>Z43 [Keycode] <ETX> [LRC] (response frame)

Message length: Fixed 7 bytes.

Usage: This is the response frame of message Z42.

Message element:

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
Z43	3	Message ID
[keycode]	1	'0' to '9' ASCII 'A' to 'C' denote 3 function keys. 'A' = [F1] 'B' = [F2] 'C' = [F3] '*' = [CAN] '#' = [ENTER] '/' = [CLR] '?' means time out.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

Refer to message Z42

Message Z50 String Entry Request

Format: <STX>Z50[echo flag][timeout][max entry]<US><DI>[<ETX>][LRC]
(request frame)

Message length: Variable 10 to 12 bytes.

Usage: This message requests a string of key inputs to be entered from the PIN pad.
The PIN pad polls for keypad inputs until one of the following criteria happens:

- [ENTER] key pressed; PIN pad validates input and sends message 71.
- [CAN] key pressed; PIN pad aborts PIN entry and sends <EOT>.
- Message 72 received; PIN pad aborts PIN entry and sends <EOT>
- Times out (session or single key idle for 60 seconds).

Note:- To enable Z50 input session, send message Z2 or Z3 authenticated for data entry, (Z2 / Z3 with <GS> operand). Otherwise PIN pad will send <EOT> and return to idle state.

- If input character count reaches [max entry], the PIN pad will stop collecting key input and make error beeps. User can press [ENTER], [CLR], or [CAN]. Request the user to input a string from the keypad.

A10-P will wait for the keypad input and store the alphabet data string into the internal buffer. The maximum length of the internal buffer is up to 49 characters.

User can press [CLR] to clear the input buffer and start over the input, or press [CAN] to cancel the input.

Using [0] and [MENU] for the period symbol and white space entries.

.Abort input: Issue message 72 to abort the operation.

Note: A message Z2 or Z3 with authenticated frame must be issued before sending message Z50. Otherwise the A10-P will refuse to execute.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
Z50	3	Message ID
[echo flag]	1	'0': echo input as '*' '1': echo input as is '2': do not echo
[timeout]	3	ASCII characters from 1 to 255 to set the timeout for each keypress. For example "010" means 10 seconds of timeout after the last keypress.
[max entry]	1 or 2	(optional) Maximum entry counts. Range from 00 to 49 (or 0 to 49)
<US>	1	<1F>(Optional)
<DI>	1	(Optional)Rely on <US> <81>: Left-alignment display, <82> :Center-alignment display, <83>: Right-alignment display. <84>: Left-alignment with line display. <85>: Center-alignment with line display. <86>: Right-alignment with line display.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message Z2 or Z3	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
		Show the prompt message
Message Z50	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message Z51 (or <EOT> when input is cancelled)
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message Z51 String Entry Response

Format: <STX>Z51[string]<ETX>[LRC] (response frame)

Message length: Variable, maximum 55 bytes.

Usage: This is the response frame of message Z50.

Message element:

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
Z51	3	Message ID
[string]	1..49	User input string. '?' means time out.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

Refer to message Z50.

Message Z7 Turn ON/OFF CANCEL Message Display

Format: <STX>Z7[option]<ETX>[LRC] (request frame)

Message length: Fixed 6 bytes.

Usage: When a CANCEL message is received or a CANCEL key is pressed to cancel a current transaction, the A10-P will display the “TRANSACTION CANCELLED” message. This display could be turned ON or OFF using message Z7.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
Z7	2	Message ID
[option]	1	'0' = message displayed '1' = message not displayed
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message Z7	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
		message turned ON/OFF

Message Z8 Set Idle Prompt

Format: **<STX>Z8[prompt]<ETX>[LRC] (request frame)**

Message length: Variable: 5 to 21 bytes.

Usage: The A10-P will display an idle prompt when it is in IDLE state. The host can change this idle prompt via message Z8. If the prompt field is filled with a null string, then the A10-P will use the default prompt after receiving this null string command.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
Z8	2	Message ID
[Prompt]	1 to 16	Idle prompt to be used
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message Z8	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
		Displays the idle prompt

Message Z9 Turn ON/OFF LCD Backlight (optional)

Format: <STX>Z9[option]<ETX>[LRC] (request frame)

Message length: Fixed 6 bytes.

Usage: This message can control the global backlight ON or OFF for the LCD of A10-P with backlight option. By default, A10-P will turn on its LCD backlight when it receives PIN entry or clear text entry message such as 70, and turns it off when those function exit.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
Z9	2	Message ID
[option]	1	0 : Turn off LCD backlight 1: Turn on LCD backlight
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message Z9	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) (<EOT> after 3 NAKs)
		LCD backlight turned ON/OFF

Section 12 ICC, SAM, MSR, and PCD Track Reading

Messages

Messages listed in this section are used to:

- Access IC card, SAM card directly (EMV power-up, power-down, APDU exchange.)
- Read magnetic stripe card.
- Read contactless smart card and output MSR track equivalent data.

NOTE: By default, the MSR function is turned off until the user issues the Q4 command to enable it.

I00 ~ I07: IC card manipulation.

I11 ~ I17: SAM card manipulation.

Q1 ~ Q7: Read magnetic stripe card and output control.

Q8 ~ QJ: Read contactless smart card (MSR emulation) and output control.

QK: Multiple interface (MSR, ICC, Contactless) card detection and reading.

81: MSR data output

83: Contactless MSR emulation data output.

Depend on the firmware configuration (PTS only or PTS + SRED), the clear text data output is not available.

PTS Only Configuration

- I01 ~ I06 (ICC manipulation) is available.
- Q6 (MSR operation control) does NOT support options for data encryption.
- 81 (MSR card data) outputs clear text track data.
- 83 (Contactless card data) outputs clear text track equivalent data.

PTS + SRED Configuration

- I01 ~ I06 (ICC manipulation) is NOT available.
- Q6 (MSR operation control) supports data encryption (MK/SK and DUKPT) option.
- QD (Contactless)
- 81 (MSR card data) outputs encrypted (and optionally obfuscated) track data.
- 83 (Contactless card data) outputs encrypted (and optionally obfuscated) track equivalent data.

Message I00 Query Primary Smart Card Presence

Format: **<STX>I00<ETX>[LRC] (request frame)**
<STX>I00[response]<ETX>[LRC] (response frame)

Message length: Fixed 6 bytes (request frame) / 7 bytes (response frame).

Usage: Upon receiving a query card packet, A10-P will check whether the card is present in the primary card reader or not. If a card exists in the reader, A10-P will reply with the code 'F'. Otherwise, the code '0' is used instead.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
I00	3	Message ID
<ETX>	1	<03>
[LRC]	1	Checksum

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
I00	3	Message ID
[Response]	1	0: Smart card not exists. F: Smart card exists
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message I00 (Request)	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
		Check card existence Message I00 (response)
<ACK> (Good echo) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message I01 Primary Smart Card Cold Reset

Format: <STX>I01<ETX>[LRC] (request frame)

Message length: Fixed 6 bytes.

Usage: A10-P performs a cold reset on the active smart card. If the operation is successful, A10-P will send out message I02 with the “Answer to Reset” (ATR) of the smart card to the host. Otherwise, message I0F bearing the error code is sent back.

NOTE: This message is not available in the PTS + SRED configuration.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
I01	3	Message ID
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message I01	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	<i>Smart card cold reset.</i> <i>Message I02 (ATR)</i>
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message I02 Primary Smart Card ATR Response

Format: <STX>I02[ATR]<ETX>[LRC] (response frame)

Message length: Variable, depending on the specification of the smart card..

Usage: The message contains the “Answer to Reset” (ATR) of the smart card to be sent to the host. In general, the ATR conforms to the ISO 7816-3 or EMV 4.3 Book 1 standard.
See these standards for more detailed information.

NOTE: This message is not available in the PTS + SRED configuration.

Message element:

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
I02	3	Message ID
[ATR]	var.	Answer to reset data. Format: Hexadecimal string.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

Refer to message I01.

Message I04 Primary Smart Card Deactivate

Format: <STX>I04<ETX>[LRC] (request frame)

Message length: Fixed 6 bytes.

Usage: Upon receiving the card deactivate packet, A10-P sets the active smart card into the deactivate state. If the operation is successful, A10-P will reply the same packet to confirm. Otherwise, message I0F with the error code will be sent.

NOTE: This message is not available in the PTS + SRED configuration.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
I04	3	Message ID
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message I04	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	<i>Deactivate smart card.</i> <i>Message I04 (success)</i> <i>Message I0F (fail)</i>
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message I06 Primary Smart Card C-APDU

Format: <STX>I06[C-APDU]<ETX>[LRC] (request frame)

Message length: Variable 14 to 530 bytes.

Usage: The C-APDU (application protocol data unit) is the data to be sent to the active smart card. A10-P's smart card reader will handle with base protocols such as T=0 and T=1, and the programmer needs only sending out the APDU. In general, A10-P will reply message I07 (response APDU from card) to the host in 1 second. For any error, A10-P will send back message I0F with an error code.

NOTE: This message is not available in the PTS + SRED configuration.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
I06	3	Message ID
[C-APDU]	8..524	Smart card command APDU. Format: hexadecimal string.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message I01	← →	(Smart card power on and ATR response)
Message I06	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Communication with card. Message I07 or I0F
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)		

Message I07 Primary Smart Card R-APDU

Format: <STX>I07[R-APDU]<ETX>[LRC] (response frame)

Message length: Variable 10 to 530 bytes.

Usage: After the smart card processes the Command-APDU, it puts the result into the Response-APDU. A10-P packs the Response-APDU into this message packet and send it to the host.

NOTE: This message is not available in the PTS + SRED configuration.

Message element:

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
I07	3	Message ID
[R-APDU]	4.. 524	Smart card response APDU Format: hexadecimal string.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

Refer to message I06.

Message I08 Smart Card Offline PIN Verification (EMV)

Format: `<STX>I08[Mode]<FS>[Prompt & Amount]<ETX>[LRC] (Clear-text mode)`
`<STX>I08[Mode]<FS>[Prompt & Amount]<FS>RSA public Exp<FS>[RSA public Modulus]<ETX>[LRC] (Cipher-text mode)`

Message length: Variable.

Usage: User could apply message I01 ~ I06 to implement his EMV transaction flow. In EMV transaction flow, user could apply this message to implement the offline PIN verification operation defined by EMV Co. User applies this message to make PIN pad ask cardholder enter his PIN on PIN pad and then PIN pad will send the PIN block (in clear -text or cipher-text) to IC card. For secure consideration, user will not get any information about cardholder's PIN, user can only get the PIN verification result from PIN pad. In PIN verification operation, PIN pad will issue "Get Challenge" command to get random number from smart card for PIN encryption.

NOTE: This message is not available in the PTS + SRED configuration.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
I08	3	Message ID
[Mode]	1	0: Clear PIN mode 1: Cipher PIN mode
<FS>	1	<1C>
[Prompt & Amount]	var	ASCII string. The data in this filed is used to present to cardholder. The length of this filed could be 1 ~ 32. Ex. Amount: 12.345 PIN pad will append messages on line 3 and 4 to inform user to make a confirmation.
<FS>	1	<1C> (Optional, if Mode is 1)
[RSA public Exp]	1	(Optional, if Mode is 1) 1: 3 2: $2^{16} + 1$
<FS>	1	(Optional, if Mode is 1) <1C>
[RSA public Modulus]	Var.	Hex string. RSA modulus. (Optional, if Mode is 1)
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message I08	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Communication with card. Message I09 or I0F
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)		

Message I09 Response of Smart Card Offline PIN Verification (EMV)

Format: <STX>I09[Response]<ETX>[LRC] (response frame)

Message length: Fixed 7 bytes.

Usage: PIN pad will return the result of PIN verification operation. For EMV rule, it's recommended that terminal applies next CVM if get response equal to 'C', 'E', and 'S', terminates EMV transaction if get response equal to 'D' and 'F'.

NOTE: This message is not available in the PTS + SRED configuration.

Message element:

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
I09	3	Message ID
[Response]	1	ASCII character. A: PIN is verified successfully. C: Cardholder bypass PIN. D: Communication failed (with smart card). E: Exceed PIN try limit S: No random number from smart card. F: Other errors
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

Refer to message I08.

Message I0F Error Codes

Format: <STX>I0F[error code]<ETX>[LRC] (error frame)

Message length: Fixed 8 bytes.

Usage: This message contains the error code indicating what's going on with the erroneous situation. The error code list is as follows:

"01" - ATR error or NO smart card.

"02" - APDU data length is not enough.

"03" – ICC/SAM card not powered yet, use I01 first.

"04" - The alphanumeric value in the APDU data unit is out of range.

"05" - Current smart card reader doesn't have card in.

"06" - Smart card interface switch failed.

"07" - A10-P Smart card module timed out.

"08" - Smart card APDU command failed.

"09" - This parameter is not supported by A10-P.

"0A" - This unit does not have the corresponding peripheral (ICC reader or SAM).

Message element:

Error frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
I0F	3	Message ID
[Error code]	2	Error code
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

Refer to I00 to I07.

Message I11 SAM slot Cold Reset

Format: <STX>I11<ETX>[LRC] (request frame)

Message length: Fixed 6 bytes.

Usage: A10-P performs a cold reset on the active SAM interface. If the operation is successful, A10-P will send message I12 with the "Answer to Reset" (ATR) of the smart card to the host. Otherwise, message I0F bearing the error code is sent back.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
I11	3	Message ID
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message I15	← →	(Select interface and respond)
Message I11	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	<i>Smart card cold reset.</i> <i>Message I12 (ATR)</i>
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message I12 SAM slot ATR Response

Format: <STX>I12[ATR]<ETX>[LRC] (response frame)

Message length: Variable, depending on the specification of the smart card..

Usage: The message contains the “Answer to Reset” (ATR) of the SAM card to be sent to the host. In general, the ATR conforms to the ISO 7816-3 or EMV 2000 level 1 standard. See these standards for more detailed information..

Message element:

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
I12	3	Message ID
[ATR]	N	“Answer to reset” data. Format: Hexadecimal string.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

Refer to message I11.

Message I14 SAM slot Card Deactivate

Format: <STX>I14<ETX>[LRC] (request frame)

Message length: Fixed 6 bytes.

Usage: Upon receiving the card deactivate packet, A10-P sets the active smart card into the deactivate state. If the operation is successful, A10-P will reply the same packet to confirm. Otherwise, message IOF with the error code will be sent.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
I14	3	Message ID
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message I14	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	<i>Deactivate smart card.</i> <i>Message I14 (success)</i> <i>Message IOF (fail)</i>
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message I15 SAM Select Interface

Format: <STX>I15 [IF_code] <ETX> [LRC] (request frame)

Message length: Fixed 7 bytes.

Usage: A10-P sets the active interface of the smart card reader. If the operation succeeds, A10-P will reply the same packet to confirm. Otherwise, message IOF with error code is sent. All SAM slot commands will be sent to the card via this active interface.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
I15	3	Message ID
[IF_code]	1	Interface code. '1' = SAM 1 '2' = SAM 2
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message I15	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message I15 Message IOF (fail)
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message I16 SAM slot C-APDU

Format: <STX>I16[C-APDU]<ETX>[LRC] (request frame)

Message length: Variable 14 to 530 bytes.

Usage: The C-APDU (application protocol data unit) is the data to be sent to the active smart card. A10-P's SAM reader will handle with base protocols such as T=0 and T=1, and the programmer only needs sending out the APDU. In general, A10-P will reply message I17 (response APDU from card) to the host in 1 second. For any error, A10-P will send back message I0F with an error code.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
I16	3	Message ID
[C-APDU]	8..524	Smart card command APDU. Format: hexadecimal string.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message I15	← →	(Select interface and respond)
Message I11	← →	(Smart card power on and ATR response)
Message I16	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Communication with card. Message I17 or I0F
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)		

Message I17 SAM slot R-APDU

Format: <STX>I17[R-APDU]<ETX>[LRC] (response frame)

Message length: Variable 10 to 530 bytes.

Usage: After SAM card processes the Command-APDU, it puts the result into the Response-APDU. A10-P packs the Response-APDU into this message packet and send it to the host.

Message element:

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
I17	3	Message ID
[R-APDU]	4.. 524	Smart card response APDU Format: hexadecimal string.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

Refer to message I16.

Message Q1 Start MSR Reading

Format: <STX>Q1<ETX>[LRC] (request frame)

Message length: Fixed 5 bytes.

Usage: The A10-P arms the MSR for read and displays the “SWIPE CARD” prompt on its LCD until a card is swiped through the reader slot, or message Z1 (reset state) is received, or message 72 is received, or the CAN key is pressed.

After the magnetic stripe card is swiped, A10-P will collect and decode the card data and do formatting designated by message Q6. The data will be output to the host by message 81 and the “PIN PAD PROCESSING” message is displayed.

NOTE: For first time use of the A10-P, issue message Q4 to enable the MSR tracks before issuing Q1.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
Q1	2	Message ID
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message Q1	→	
		“SWIPE CARD” displayed on the LCD
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	(User swipes the magnetic card) Message 81
<ACK> (Good LRC) <NAK> (Bad LRC)	→	

Message Q2 Transaction Completed

Format: **<STX>Q2<ETX>[LRC] (request frame)**

Message length: Fixed 5 bytes.

Usage: Indicate that a transaction is completed. The A10-P displays a “Thank You” message for three seconds and then goes into the IDLE state.

Note: This message can also be used on PIN transaction commands (70, Z60, Z62)

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
Q2	2	Message ID
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message Q2	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
		Display “Thank You” for 3 seconds, followed by the idle prompt.

Message Q3 Ignore Card Swipe

Format: **<STX>Q3<ETX>[LRC] (request frame)**

Message length: Fixed 5 bytes.

Usage: The host should send this message Q3 after receiving the card data and decided not to accept another card temporarily. Upon receiving this message, the A10-P will display the “PIN PAD PROCESSING” prompt and ignore any card swipe.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
Q3	2	Message ID
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message Q3	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
		“PIN PAD PROCESSING” displayed on the LCD

Message Q4 Enable/Disable Magnetic Card Reader

Format: <STX>Q4[flag]<ETX>[LRC] (request frame)

Message length: Fixed 6 bytes.

Usage: All tracks are disabled by default (after each re-power on the A10-P). The host can use this message to enable or disable the magnetic card reader of the A10-P. Only track 2 is enabled if the [flag] field contains '0'. All tracks will be enabled if the [flag] field contains a '2'.

NOTE: Please refer to message 81 for the returned magnetic card data format.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
Q4	2	Message ID
[flag]	1	'0': enable Track2 only '1': disable '2': enable all tracks of MSR
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message Q4	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)

Message Q5 Set MSR Retry Count

Format: <STX>Q5[RetryCount]<ETX>[LRC] (request frame)

Message length: Fixed 6 bytes.

Usage: This command sets a retry count for the MSR swipe initiated by message Q1.
 If A10-P cannot decode any data from the latest swipe, the retry count is larger than “0”, and the timeout has not reached yet, the A10-P will show “BAD READ” on the LCD and waits for the next swipe, until times out or the retry counter reaches the maximum.
 This setting will be kept in the battery powered memory and is effective all the time.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
Q5	2	Message ID
RetryCnt	1	‘0’ to ‘9’, MSR retry count.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message Q5	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
		Save the setting or display an error message if the RetryCnt is out of range.

Message Q6 Set MSR Data Output Format

Format: <STX>Q6.[flag]<ETX>[LRC] (request frame)

Message length: 7 bytes.

Usage: This command is used to set the MSR track data (message 81) output format.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
Q6	2	Message ID
.	1	<2E>, delimiter
[flag]	1	<i>Output format options:</i> '0': Clear-Text Card Data '2': Clear-Text Card Data with Sentinels.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message Q6	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message Q6 (echo of the request frame when successful) or <EOT> (if format error.)
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message Q7 MSR Mode Query

Format: <STX>Q7<ETX>[LRC] (request frame)
 <STX>Q7[status]<ETX>[LRC] (response frame)

Message length: Fixed 5 bytes for request, 6 bytes for response.

Usage: This command will query the MSR processing option.

Message element:

Request fame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
Q7	2	Message ID
<ETX>	1	<03>
[LRC]	1	Checksum

Response fame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
Q7	2	Message ID
[status]	1	'0': Clear-Text Card Data '2': Clear-Text Card Data with Sentinels. --- For PTS + SRED firmware --- '1': Cipher-Text Card Data (MK/SK). '3': Cipher-Text Card Data (MK/SK) with Sentinels. '4': Cipher-Text Card Data (MK/SK), and obfuscated card data. '5': Cipher-Text Card Data (MK/SK) with Sentinels, and obfuscated card data. '6': Cipher-Text Card Data (DUKPT) '7': Cipher-Text Card Data (DUKPT) with Sentinels. '8': Cipher-Text Card Data (DUKPT), and obfuscated card data. '9': Cipher-Text Card Data (DUKPT) with Sentinels, and obfuscated card data.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message Q7 (request)	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message Q7 (response)
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message Q8 Start PCD Reading for Track Equivalent Data

Format: <STX>Q8<ETX>[LRC] (request frame)

Message length: Fixed 5 bytes.

Usage: The A10-P arms the PCD for contactless reading and display the “TAP CARD” prompt on its LCD until a contactless card is taped, the Z1 message (reset state) is received, or the CAN key is pressed.

After the contactless card is tapped, the A10-P will output the MSR track equivalent data to the host by message 83.

NOTE: After the A10-P is re-powered and before issuing message Q8 (to retrieve the contactless card data) , issue message QA to enable the contactless module's tracks.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
Q8	2	Message ID
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message Q8	→	
		“TAP CARD” displayed on the LCD
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	(User taps the contactless card) Message 83
<ACK> (Good LRC) <NAK> (Bad LRC)	→	

Message Q9 Start Dual Interface Reading (MSR and PCD)

Format: <STX>Q9<ETX>[LRC] (request frame)

Message length: Fixed 5 bytes.

Usage: The A10-P arms both MSR and PCD, and displays the prompt “SWIPE / TAP CARD” on its LCD until a card is tapped or swiped through the reader slot, or message Z1 (reset state) is received, or the CAN key is pressed.

If a magnetic card is swiped, A10-P will collect and decode the card data and do formatting designated by message Q6. The data will be output to the host by message 81 and display the “PIN PAD PROCESSING” message.

If a contactless card is tapped, A10-P will output the MSR track equivalent data to the host by message 83 and display the “PIN PAD PROCESSING” message.

NOTE: After the A10-P is re-powered and before issuing message Q9 (to retrieve the contactless / magnetic stripe card data), issue message Q4 (enable the contactless module's tracks) and message QA (enable the magnetic stripe reader tracks).

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
Q9	2	Message ID
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message Q9	→	
		“SWIPE/TAP CARD” displayed on the LCD
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	(User swipes magnetic card) Message 81 (User taps contactless card) Message 83
<ACK> (Good LRC) <NAK> (Bad LRC)	→	

Message QA Enable/Disable Contactless Card Reader

Format: <STX>QA[flag]<ETX>[LRC] (request frame)

Message length: Fixed 6 bytes.

Usage: All the tracks are disabled by default (after each re-power on A10-P). The host can use this message to enable or disable the contact-less card reader of the A10-P. Only track 2 is enabled if the [flag] field contains '0'. Both tracks will be enabled if the [flag] field contains a '2'.

Note: Please refer to message 83 for the returned magnetic card data format.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
QA	2	Message ID
[flag]	1	'0': enable Track2 only '1': disable '2': enable all tracks of MSR
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message QA	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs) (<EOT>Contactless card reader fails in the initialization)

Message QB Control MSR Auto Arm Mode

Format: <STX>QB[flag]<ETX>[LRC] (request frame)

Message length: Fixed 6 bytes.

Usage: If the 'auto arm' is turned on, the A10-P will return the magnetic card data on each swipe under the idle mode (The terminal does not have to issue message Q1 for each card swipe). If this functionality is turned off, the terminal has to issue message Q1 to get the magnetic card data.

Note. If the card tracks are all disabled (refer to message Q4), the terminal will always get the null magnetic card data.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
QB	2	Message ID
[flag]	1	'0': Auto arm mode disabled. '1': Auto arm mode enabled.
<ETX>	1	<03>
[LRC]	1	Checksum

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
QB	2	Message ID
[status]	1	'0' = Setup successful '1' = Invalid flag
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message QB	→	
	←	<ACK> (Good LRC) / <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message QB(response)

Message QC Control PCD Auto Arm Mode for Track Equivalent Data

Format: <STX>QC[flag]<ETX>[LRC] (request frame)

Message length: Fixed 6 bytes.

Usage: If the 'auto arm' is turned on, the A10-P will return the contactless track equivalent data on each tap under the idle mode (the terminal does not have to issue message Q8/Q9 for each card tap). If this functionality is turned off, the terminal has to issue message Q8/Q9 to get the contactless card data.

Note. If the card tracks are all disabled (refer to message QA), the terminal will always get the null contactless card data

Message element:

Request fame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
QC	2	Message ID
[flag]	1	'0': Auto arm mode disabled. '1': Auto arm mode enabled..
<ETX>	1	<03>
[LRC]	1	Checksum

Response fame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
QC	2	Message ID
[status]	1	'0' = Setup successful '1' = Invalid flag
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message QC	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message QB(response)

Message QD Set PCD Track Equivalent Data Output Format

Format: <STX>QD.[flag]<ETX>[LRC] (request frame)

Message length: Fixed 7 bytes.

Usage: This command sets the PCD track equivalent data (message 83) output format.

Message element:

Request fame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
QD	2	Message ID
.	1	<2E>
[flag]	1	'0': Without Start/End sentinels '2': With Start/End sentinels
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message QD	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message QD (echo of the request frame when successful) or <EOT> (if format error.)
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message QF Enable card swipe with JPEG

Format: **<STX>QF<FS>[JPEG filename]<ETX>[LRC] (request frame)**
<STX>QF[status]<ETX>[LRC] (response frame)

Message length: Fixed 7 to 22 bytes for request frame, and 6 bytes for response frame.

Usage: This command message function is the same as message Q1, but the PIN pad LCD can display specified JPEG file of message QF.

Message element:

Request fame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
QF	2	Message ID
<FS>	1	<1C>
[JPEG filename]	1 to 16	JPEG file name
<ETX>	1	<03>
[LRC]	1	Checksum

Response fame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
QF	2	Message ID
[status]	1	'0': User presses [Enter Key]. 'A': File not found. 'B': Format error.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message QF	→	
		JPEG file displayed on the LCD
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	<EOT>(User presses [CANCEL Key].) Message QF(response) Message 81 (User swipes the magnetic card)
<ACK> (Good LRC) <NAK> (Bad LRC)	→	

Message QG Enable card tap with JPEG

Format: <STX>QG<FS>[JPEG filename]<ETX>[LRC] (request frame)
 <STX>QG[status]<ETX>[LRC] (response frame)

Message length: Fixed 7 to 22 bytes for request frame, and 6 bytes for response frame.

Usage: This command message function is the same as message Q8, but the PIN pad LCD can display specified JPEG file of message QG.

Message element:

Request fame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
QG	2	Message ID
<FS>	1	<1C>
[JPEG filename]	1 to 16	JPEG file name
<ETX>	1	<03>
[LRC]	1	Checksum

Response fame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
QG	2	Message ID
[status]	1	'A': File not found. 'B': Format error.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message QG	→	
		JPEG file displayed on the LCD
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message QG(response) Message 83 (User taps the magnetic card)
<ACK> (Good LRC) <NAK> (Bad LRC)	→	

Message QH Enable card swipe/tap with JPEG

Format: <STX>QH<FS>[JPEG filename]<ETX>[LRC] (request frame)
 <STX>QH[status]<ETX>[LRC] (response frame)

Message length: Fixed 7 to 22 bytes for request frame, and 6 bytes for response frame.

Usage: This command message function is the same as message Q9, but the PIN pad LCD can display specified JPEG file of message QH.

Message element:

Request fame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
QH	2	Message ID
<FS>	1	<1C>
[JPEG filename]	1 to 16	JPEG file name
<ETX>	1	<03>
[LRC]	1	Checksum

Response fame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
QH	2	Message ID
[status]	1	'A': File not found. 'B': Format error.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message QH	→	
		JPEG file displayed on the LCD
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message QH(response) Message 81 (User swipes the magnetic card) Message 83 (User taps the magnetic card)
<ACK> (Good LRC) <NAK> (Bad LRC)	→	

Message QI Get Contactless Tag Data

Format: <STX>QI[control][DOL]<ETX>[LRC] (request frame)

Message length: Variable.

Usage: A10-P will retrieve the data that listed on the DOL after the MSD data read. A10-P will send message QJ (Get Contactless Tag Data Response) to the host as a response.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
QI	2	Message ID
[control]	1	'0': Ready for next packet, [DOL] is ignored. '1':Get contactless tag data. (optional)
[DOL]	Var	(optional), Data Object List, each object is expressed by a tag number, and <SUB> is used to delimit each object. For example, 9F12<SUB>9A<SUB>9F02<SUB>....
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message QI1	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Get Contactless Tag Data Response. <i>Message QJ</i> (if pkt No. is not total pkt)
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)		
Message QI0	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Get Contactless Tag Data Response. <i>Message QJ</i> (if pkt No. is not total pkt)
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)		
.....		
		Get Contactless Tag Data Response. <i>Message QJ</i> (if pkt No. is total pkt)
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message QJ Get Contactless Tag Data Response

Format: <STX>QJ[Pkt NO.][Total Pkt]<FS>[DO]<ETX>[LRC] (response frame)
<STX>QJ[Status]<ETX>[LRC] (response frame).

Message length: Variable.

Usage: The message contains the contactless tag data to be sent to the host as a response.

Message element:

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
QJ	2	Message ID
[Pkt NO.]	1	Decimal. Packet sequence number (1 ~ 9).
[Total Pkt]	1	Decimal. Total packets (1~9).
<FS>	1	<1C>
[DO]	Var.	Data Object, each data object is expressed in TLV format with an <FS> delimit in each field, and <SUB> is used to delimit each object. For example: 9F12<FS>0F<FS>CREDITO DE VISA<SUB>9A<FS>06<FS>050806<SUB>
<ETX>	1	<03>
[LRC]	1	Checksum

OR

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
QJ	3	Message ID
[Status]	1	F: Query failed or data empty.
<ETX>	1	<03>
[LRC]	1	Checksum

Data Object:

A10-P will return series of data objects that list on the [DOL] field in the message QJ with TLV (tag || length|| value) format as below:

[EMV Tag Number (2 ~ 8 byte)] <FS> [Length (2byte)] <FS> [Value] <SUB>

[EMV Tag Number (2 ~ 8 byte)] <FS> [Length (2byte)] <FS> [Value] <SUB>

.....

[EMV Tag Number (2 ~ 8 byte)] <FS> [Length (2byte)] <FS> [Value].

Note: When A10-P responds these data objects, it will convert these values from binary value to hex decimal string if the data object format is “b” or “cn” or “n”.

Message flow:

Refer to message QI.

Message QK Start Multiple Interface Detection (MSR, ICC, PCD)

Format: <STX>QK<ETX>[LRC] (request frame)
<STX>QK[Res][Err Msg]<ETX>[LRC] (response frame)

Message length: Fixed 5 bytes for request, 6 or 7 bytes for response.

Usage: This command will poll MSR/ICC/PCD, until detected, 72 message or Z1 message (reset state) received or CAN key pressed.

If A10-P detects MSR/ICC/PCD, it will send the message to the host as a response.

If Res is '1', then send message Q1 to read the track data.

If Res is '2', then send message T11 to start an ICC EMV transaction.

If Res is '3', then send message Q8 to read the track equivalent data or send message T61 to start a PCD EMV transaction.

Message element:

Request fame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
QK	2	Message ID
<ETX>	1	<03>
[LRC]	1	Checksum

Response fame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
QK	2	Message ID
[Res]	1	'0' : Fatal Error. '1' : Detect MSR. '2' : Detect ICC. '3' : Detect PCD.
[Err Msg]	1	(optional), If Res is '0'. '0' : Enable MSR fail. '1' : Get ICC status fail. '2' : Enable antenna fail. '3' : Cancel polling.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message QK (request)	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message QK (response)
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message 81 MSR Card Data

Format: <STX>81.[trk1 data]<FS>[trk2 data]<FS>[trk3 data]<FS>
 [BIN record]<ETX>[LRC] (Clear text / encrypted data)
 <STX>81\$[IV]<ETX>[LRC]
 (MK/SK Initial Vector)
 <STX>81#[KSN] <ETX>[LRC]
 (DUKPT Key serial number)
 <STX>81@[trk1 data]<FS>[trk2 data]<FS>[trk3 data]<FS>
 [BIN record]<ETX>[LRC]
 (Obfuscated track data)
 <STX>81&[Check PAN result 1]<FS>[Check PAN result 2]<ETX>[LRC]
 (PAN Validation result)

Message length: Variable, depending on the card image in buffer.

Usage: A10-P uses this message to send back the information read from the last card swipe. The card data is transmitted in ASCII string as defined by the ISO 7811 format specification. Under different types, the A10-P returns the card data in different formats. If message Q4 is issued and the A10-P is set as track 2 only, the returned message 81 will only contain track 2 data without a field separator.

For PTS only firmware, the track data is always in clear text. For PTS + SRED firmware, the track data will be encrypted, except those PAN number listed in the BIN exception table. Refer to “account data security” for the obfuscation and encryption algorithms. If encryption fails, A10-P will return NULL track data.

If BIN record search is successful, the BIN will be attached in the end of the magnetic strip card data.

Notes:

Packets for different operation mode:

Please refer to message Q6 for operation type information.

Mode 0 and 2 (Clear text)	1 Packet	PIN pad will only return one "81." message., if swiped card is gift card, [trk2 data] would be replaced by gift card number which is Prepaid gift card standard defined by First Data Corporation.
Mode 1, and 3 (MK/SK encrypted)	2 Packet	PIN pad returns "81\$" for MK/SK initial vector, then "81." for encrypted card data.
Mode 4 and 5: (MK/SK encrypted + obfuscated)	4 Packets	PIN pad returns "81\$" for MK/SK initial vector then returns "81." for encrypted card data, then returns "81@" for obfuscated data, and finally returns "81&" to indicate if the PAN in track 1 / track 2 is valid.
Mode 6 and 7: (DUKPT encrypted)	2 Packets	PIN pad returns "81#" for DUKPT key serial number, then "81." for encrypted card data.
Mode 8 and 9: (DUKPT encrypted + obfuscated)	4 Packets	PIN pad first returns "81#" for DUKPT key serial number, then returns "81." for encrypted card data, then returns "81@" for obfuscated data, and finally returns "81&" to indicate if the PAN in track 1 / track 2 is valid.

Message element:

(Clear text / encrypted data)

Field	Length	Value and description
<STX>	1	<02>
81	2	Message ID
.	1	<2E>
[trk1]	N	Variable length, data of track 1. (*)
<FS>	1	<1C>, field separator. (*)
[trk2]	N	Variable length, data of track 2.
<FS>	1	<1C>, field separator. (*)
[trk3]	N	Variable length, data of track 3. (*)
<FS>	1	Optional <1C>, field separator. (*)
[BIN record]	16	Optional Hex string. If BIN processing is enabled and a match record is found.
<ETX>	1	<03>
[LRC]	1	Checksum

(*): If MSR is set to Tk2 output only, this field is omitted

(MK/SK Initial Vector)

Field	Length	Value and description
<STX>	1	<02>
81	2	Message ID
\$	1	<24>
[IV]	16	Hex string. MK/SK initial vector.
<ETX>	1	<03>
[LRC]	1	Checksum

(DUKPT Key Serial Number)

Field	Length	Value and description
<STX>	1	<02>
81	2	Message ID
#	1	<23>
[KSN]	20	Hex string. DUKPT key serial number.
<ETX>	1	<03>
[LRC]	1	Checksum

(Obfuscated data)

Field	Length	Value and description
<STX>	1	<02>
81	2	Message ID
@	1	<40>
[trk1]	N	Variable length, data of track 1. PAN is partially masked. (*)
<FS>	1	<1C>, field separator. (*)
[trk2]	N	Variable length, data of track 2. PAN is partially masked.
<FS>	1	<1C>, field separator. (*)
[trk3]	N	Variable length, data of track 3. (*)
<FS>	1	Optional <1C>, field separator. (*)
[BIN record]	16	Optional Hex string. If BIN processing is enabled and a match record is found.
<ETX>	1	<03>
[LRC]	1	Checksum

(*): If MSR is set to Tk2 output only, this field is omitted

(PAN Validation Result – using mod10 method)

Field	Length	Value and description
<STX>	1	<02>
81	2	Message ID
&	1	<26>
[RES1]	1	PAN checking result of track 1 (*) '0': Invalid PAN '1': Valid PAN '2': No PAN exists
<FS>	1	<1C>, field separator. (*)
[RES2]	1	PAN checking result of track 2. '0': Invalid PAN '1': Valid PAN '2': No PAN exists
<ETX>	1	<03>
[LRC]	1	Checksum

(*): If MSR is set to Tk2 output only, this field is omitted

Message flow:

Under Operation Type 0, 1, 2 and 3

HOST	Direction	A10-P
Message Q1 or Q9	→	
		“SWIPE CARD” displayed on the LCD
	←	<ACK> (Good LRC) / <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	(User swipes the magnetic card) Message 81 (81.)
<ACK> (Good LRC) <NAK> (Bad LRC)	→	

Clear Text of MK/SK encrypted data (Mode 0, 1, 2, 3)

HOST	Direction	A10-P
Message Q1 or Q9	→	
		“SWIPE CARD” displayed on LCD
	←	<ACK> (Good LRC) <NAK> (Bad LRC) (<EOT> after 3 NAKs)
	←	(User swipes magnetic card) Message 81 (81\$) for initial vector.
<ACK> (Good LRC) <NAK> (Bad LRC) (<EOT> after 3 NAKs)	→	
	←	(User swipes magnetic card) Message 81 (81.)
<ACK> (Good LRC) <NAK> (Bad LRC) (<EOT> after 3 NAKs)	→	

MK/SK encrypted data with partially masked PAN (Mode 4, 5)

HOST	Direction	A10-P
Message Q1 or Q9	→	
		“SWIPE CARD” displayed on LCD
	←	<ACK> (Good LRC) <NAK> (Bad LRC) (<EOT> after 3 NAKs)
	←	(User swipes magnetic card) Message 81 (81\$) for initial vector
<ACK> (Good LRC) <NAK> (Bad LRC) (<EOT> after 3 NAKs)	→	
	←	Message 81 (81.) for encrypted data.
<ACK> (Good LRC) <NAK> (Bad LRC) (<EOT> after 3 NAKs)	→	
	←	Message 81 (81@) for partially masked PAN.
<ACK> (Good LRC) <NAK> (Bad LRC) (<EOT> after 3 NAKs)	→	
	←	Message 81 (81&) for PAN validation result.
<ACK> (Good LRC) <NAK> (Bad LRC) (<EOT> after 3 NAKs)	→	

DUKPT encrypted data (Mode 6, 7)

HOST	Direction	A10-P
Message Q1 or Q9	→	
		“SWIPE CARD” displayed on LCD
	←	<ACK> (Good LRC) <NAK> (Bad LRC) (<EOT> after 3 NAKs)
	←	(User swipes magnetic card) Message 81 (81#) for KSN
<ACK> (Good LRC) <NAK> (Bad LRC) (<EOT> after 3 NAKs)	→	
	←	Message 81 (81.) for encrypted data.
<ACK> (Good LRC) <NAK> (Bad LRC) (<EOT> after 3 NAKs)	→	

DUKPT encrypted data (Mode 8, 9)

HOST	Direction	A10-P
Message Q1 or Q9	→	
		“SWIPE CARD” displayed on LCD
	←	<ACK> (Good LRC) <NAK> (Bad LRC) (<EOT> after 3 NAKs)
	←	(User swipes magnetic card) Message 81 (81#) for KSN
<ACK> (Good LRC) <NAK> (Bad LRC) (<EOT> after 3 NAKs)	→	
	←	Message 81 (81.) for encrypted data.
<ACK> (Good LRC) <NAK> (Bad LRC) (<EOT> after 3 NAKs)	→	
	←	Message 81 (81@) for partially masked PAN.
<ACK> (Good LRC) <NAK> (Bad LRC) (<EOT> after 3 NAKs)	→	
	←	Message 81 (81&) for PAN validation result.
<ACK> (Good LRC) <NAK> (Bad LRC) (<EOT> after 3 NAKs)	→	

Message 83 PCD Track Equivalence Data

Format: <STX>83.[trk1 data]<FS>[trk2 data]<FS>[trk3 data]<ETX>[LRC]
 (Clear text / encrypted data)
 <STX>83\$[IV]<ETX>[LRC]
 (MK/SK Initail Vector)
 <STX>83#[KSN] <ETX>[LRC]
 (DUKPT Key serial number)
 <STX>83@[trk1 data]<FS>[trk2 data]<FS>[trk3 data]<ETX>[LRC]
 (Obfuscated track data)
 <STX>83&[Check PAN result 1]<FS>[Check PAN result 2]<ETX>[LRC]
 (PAN Validation result)

Message length: Variable, depending on the card image in buffer.

Usage: Refer to message 81

Message element: Refer to message 81, except message ID becomes "83"

Message flow:

Refer to message Q8 or Q9.

Section 13 Display Functionality Messages

A10-P provides a graphic display with resolution 320x240, 6 character heights in text mode (12, 16, 24, 30, 48, and 60 pixels), and supports many display functions to be used by the user.

Font size selection:

A10-P supports 6-character heights (12, 16, 24, 30, 48, and 60 pixels) for selection.

Foreground/Background color selection:

User can alternate the foreground and background colors.

Screen save setting:

Set the parameters for the screen save functionality.

Enable / disable screen save:

Enable or disable the screen save functionality.

Screen save preview:

The screen save will be launched once.

Message B1 Font Size Selection

Format: <STX>B1[Font Size]<ETX>[LRC] (request frame)
 <STX>B2[Response]<ETX>[LRC] (response frame)

Message length: Fixed 6.

Usage: A10-P supports 4 font sizes: 12, 16, 24, 30, 48, and 60 pixel height. Use this message can change the displayed font size. Please note that only low-page characters (<0x80) can be displayed, high-page characters should be encoded using UTF-8 code.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
B1	2	Message ID
[Font Size]	1	'0': 12 pixel height font '1': 16 pixel height font '2': 24 pixel height font '3': 30 pixel height font '4': 48 pixel height font '5': 60 pixel height font
<ETX>	1	<03>
[LRC]	1	Checksum

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
B2	2	Message ID
[Response]	1	'0': Setup successful. '1': Format error '2': Not support
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message B1 (Request)	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
		Message B2 (response)
<ACK> (Good echo) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message B3 Font Color (Foreground / Background) Selection

Format: **<STX>B3 [Foreground] [Background]<ETX>[LRC] (request frame)**
<STX>B4 [Response]<ETX>[LRC] (response frame)

Message length: Fixed 17 bytes.

Usage: Use this message to set the new colors of the font and background. The format is RR/GG/BB.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
B3	2	Message ID
[Foreground]	6	Hex String. Format: RRGGBB
[Background]	6	Hex String. Format: RRGGBB
<ETX>	1	<03>
[LRC]	1	Checksum

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
B4	2	Message ID
[Response]	1	'0': Setup successful '1': Format error
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message B3 (Request)	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
		Message B4 (response)
<ACK> (Good echo) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message BB Screen Saver Setting

Format: <STX>BB[Waiting Time][Screen Saver Type]<ETX>[LRC] (request frame)
<STX>BC[Response]<ETX>[LRC] (response frame)

Message length: Fixed 9 bytes.

Usage: A10-P has the screen saver functionality. This message is used to select one of screen savers and set the waiting time. Message BD is called to enable the screen saver. While A10-P is in idle (without receiving messages, keystrokes, or screen touch) passing the period of time specified in [Waiting Time], the screen saver will be activated. The minimum waiting time should be greater than or equal to 5 seconds.

Screen saver - Jpeg Show: display all jpeg pictures in rotation (at least one JPEGImage should be loaded into the A10-P).Screen saver will be launched each time in rotation.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
BB	2	Message ID
[Waiting Time]	3	005~999
[Screen Saver Type]	1	'1': Jpeg show
<ETX>	1	<03>
[LRC]	1	Checksum

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
BC	2	Message ID
[Response]	1	'0': Valid parameter. '1': Invalid parameter.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message BB (Request)	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
		Message BC (response)
<ACK> (Good echo) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message BD Enable / Disable Screen Saver

Format: <STX>BD[Op]<ETX>[LRC] (request frame)
<STX>BE[Response]<ETX>[LRC] (response frame)

Message length: Fixed 6 bytes.

Usage: Use this message to enable / disable the screen saver.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
BD	2	Message ID
[Op]	1	'0': Disable the screen saver functionality. '1': Enable the screen saver functionality.
<ETX>	1	<03>
[LRC]	1	Checksum

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
BE	2	Message ID
[Response]	1	'0': Valid parameter. '1': Invalid parameter.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message BD (Request)	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
		Message BE (response)
<ACK> (Good echo) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message BF Screen Saver Preview / Stop Preview

Format: <STX>BF[Op]<ETX>[LRC] (request frame)

Message length: Fixed 6 bytes.

Usage: Use this message to launch the screen saver directly or turns it off. User can also issue other messages to turn off the screen saver.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
BF	2	Message ID
[Op]	1	'0': Stop the screen saver preview '1': Start the Screen Saver Preview
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message BF (Request)	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)

Section 14 JPEG File Operation Messages

A10-P supports several JPEG file operation messages. Customers can use the UIC provided JPEG Tool to query the A10-P to retrieve the JPEG file table. From that the user can delete the existed JPEG files from the A10-P or select the JPEG files for display. In addition, customers can download a JPEG file into the A10-P or upload a JPEG file from the A10-P. Below are the JPEG file operation messages A10-P supports:

J0	JPEG File Table Initialization
J1	Query JPEG File Table
J2	Select JPEG File
J3	Delete JPEG File
J4	Download JPEG File
J5	Upload JPEG File
J6	Play JPEG File
J7	Set JPEG File As Idle Prompt
J8	Enable/Disable Idle Logo Functionality
J9	Show JPEG File
JA	Setup Boot Logo

Message J0 JPEG File Table Initialization

Format: <STX>J0<ETX>[LRC] (request frame)
 <STX>J0[Status]<ETX>[LRC] (response frame)

Message length: Fixed 5 bytes for the request frame and 6 bytes for the response frame.

Usage: This message is used to initialize the JPEG file table in the terminal.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
J0	2	Message ID
<ETX>	1	<03>
[LRC]	1	Checksum

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
J0	2	Message ID
[Status]	1	'0': Success. '1': Failure.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message J0	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message J0
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message J1 Query JPEG File Table

Format: <STX>J1<ETX>[LRC] (request frame)

 <STX>J1[PktType][Status_1][FileName_1]<FS>[Status_2][FileName_2]

 <FS>...[Status_N-1][FileName_N-1]<FS>[Status_N][FileName_N]

 <ETX>[LRC] (where the value of 'N' is the total number of the JPEG files) (response frame)

Message length: Fixed 5 bytes for the request frame and a variable of 5 to 540 bytes for the response frame.

Usage: This message is used to query the JPEG file table from the terminal.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
J1	2	Message ID
<ETX>	1	<03>
[LRC]	1	Checksum

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
J1	2	Message ID
[PktType]	1	'0' = first or middle of multiple packets. '1' = last or only packet.
[Status_1]	1	'0': Unselect. (optional) '1': Select.
[FileName_1]	1-15	File name 1 (optional)
<FS>	1	<1C>, field separator (optional)
[Status_2]	1	'0': Unselect. (optional) '1': Select.
[FileName_2]	1-15	File name 2 (optional)
<FS>	1	<1C>, field separator (optional)
.....		
[Status_N-1]	1	'0': Unselect. (optional) '1': Select.
[FileName_N-1]	1-15	File name N-1 (optional)
<FS>	1	<1C>, field separator (optional)

[Status_N]	1	'0': Unselect. (optional) '1': Select.
[FileName_N]	1-15	File name N (optional)
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message J1	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message J1 ([PktType] = '0')
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	
Message J1	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message J1 ([PktType] = '1')
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message J2 Select JPEG File

Format:

```

<STX>J2[control][FileName_1]<FS>[FileName_2]<FS>...
[FileName_N-1]<FS>[FileName_N]<ETX>[LRC]
(where the value of 'N' is the total number of the JPEG files)
(request frame)

<STX>J2[Status_1]<FS>[Status_2]<FS>...
[Status_N-1]<FS>[Status_N]<ETX>[LRC]
(where the value of 'N' is the total number of the JPEG files)
(response frame)

```

Message length: Variable 7 to 540 bytes for the request frame, and variable 5 to 44 bytes for the response frame.

Usage: This message is used to select or unselect the JPEG files from the terminal's display list.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
J2	2	Message ID
[control]	1	'0': Unselect. '1': Select.
[FileName_1]	1-15	File name 1
<FS>	1	<1C>, field separator (optional)
[FileName_2]	1-15	File name 2 (optional)
<FS>	1	<1C>, field separator (optional)
..... ...		
[FileName_N-1]	1-15	File name N-1 (optional)
<FS>	1	<1C>, field separator (optional)
[FileName_N]	1-15	File name N (optional)
<ETX>	1	<03>
[LRC]	1	Checksum

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
J2	2	Message ID
[Status_1]	1	'0': Success, '1': Failure, '2': Format error.
<FS>	1	<1C>, field separator (optional)
[Status_2]	1	'0': Success, '1': Failure. (optional)
<FS>	1	<1C>, field separator (optional)
..... ...		
[Status_N-1]	1	'0': Success, '1': Failure. (optional)
<FS>	1	<1C>, field separator (optional)
[Status_N]	1	'0': Success, '1': Failure. (optional)
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message J2	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message J2
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message J3 Delete JPEG File

Format:

```
<STX>J3[FileName_1]<FS>[FileName_2]<FS>...
[FileName_N-1]<FS>[FileName_N]<ETX>[LRC]
(where the value of 'N' is the total number of the JPEG files)
(request frame)

<STX>J3[Status_1]<FS>[Status_2]<FS>...
[Status_N-1]<FS>[Status_N]<ETX>[LRC]
(where the value of 'N' is the total number of the JPEG files)
(response frame)
```

Message length: Variable 6 to 540 bytes for the request frame, and variable 5 to 44 bytes for the response frame.

Usage: This message is used to delete the JPEG files from the terminal's display list.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
J3	2	Message ID
[FileName_1]	1-15	File name 1
<FS>	1	<1C>, field separator (optional)
[FileName_2]	1-15	File name 2 (optional)
<FS>	1	<1C>, field separator (optional)
..... ...		
[FileName_N-1]	1-15	File name N-1 (optional)
<FS>	1	<1C>, field separator (optional)
[FileName_N]	1-15	File name N (optional)
<ETX>	1	<03>
[LRC]	1	Checksum

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
J3	2	Message ID
[Status_1]	1	'0': Success, '1': Failure, '2': Format error.
<FS>	1	<1C>, field separator (optional)
[Status_2]	1	'0': Success, '1': Failure. (optional)
<FS>	1	<1C>, field separator (optional)
..... ...		
[Status_N-1]	1	'0': Success, '1': Failure. (optional)
<FS>	1	<1C>, field separator (optional)
[Status_N]	1	'0': Success, '1': Failure. (optional)
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message J3	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message J3
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message J4 Download JPEG File

Format: <STX>J4 [PktType] [SeqNo] [Force] <FS> [FileName] <FS> [Size]
 [Data] <ETX> [LRC] (request frame)
 <STX>J4 [Status] <ETX> [LRC] (response frame)

Message length: Variable 22 to 538 bytes for the request frame, and fixed 6 bytes for the response frame.

Usage: This message is used to download a JPEG file from the terminal.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
J4	2	Message ID
[PktType]	1	'0' = first or middle of multiple packets. '1' = last or only packet.
[SeqNo]	3	'000' to '999'; for onetime only packet, set to '000'.
[Force]	1	'0' = ask before overwrite, '1' = forced overwrite
<FS>	1	<1C>, field separator
[FileName]	0-15	JPEG file name (optional)
<FS>	1	<1C>, field separator
[Size]	3	'000' to '523', size of [ImgData] in bytes
[ImgData]	0-523	Base 64 format data (optional)
<ETX>	1	<03>
[LRC]	1	Checksum

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
J4	2	Message ID
[Status]	1	'0': Ready for the next J4 packet. '1': [PktType] has invalid value. '2': [SeqNo] has invalid value. '3': [Force] has invalid value. '4': [FileName] has invalid value. '5': [Size] has invalid value. '6': [ImgData] has invalid value. '7': Number of <FS> is wrong. 'A': File size is zero or too large. 'B': Null file name. 'C': File name is too long or user abort. 'D': No more space. 'E': File is too large to store 'F': Download Success.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message J4 ([PktType] = '0' and [SeqNo] = '000')	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message J4 ([Status] = '0')
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	
Message J4 ([PktType] = '0' and [SeqNo] = '001'-'998')	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message J4 ([Status] = '0')
.....		
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	
Message J4 ([PktType] = '1')	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message J4 ([Status] = 'F')
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message J5 Upload JPEG File

Format: <STX>J5[control]<FS>[FileName]<ETX>[LRC] (request frame)
<STX>J5[PktType][SeqNo][Size][Data]<ETX>[LRC] (response frame)

Message length: Variable 6 to 21 bytes for the request frame, and variable 12 to 536 bytes for the response frame.

Usage: This message is used to upload a JPEG file from the terminal.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
J5	2	Message ID
[control]	1	'0': First command for upload, [FileName] is required. '1': Ready for the next J5 packet, [FileName] is ignored.
[FileName]	1-15	File name (Option)
<ETX>	1	<03>
[LRC]	1	Checksum

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
J5	2	Message ID
[PktType]	1	'0' = First or middle of multiple packets. '1' = Last or only packet. '2': Null file name. '3': File name is not found. '4': Buffer is not enough. '5': Format error.
[SeqNo]	3	'000' to '999', for onetime only packet, set to '000'.
[Size]	3	'000' to '524', size of [Data] in bytes
[Data]	0-524	Base 64 format data
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message J5 ([control] = '0')	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message J5 ([PktType] = '0' and [SeqNo] = '000')
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	
Message J5 ([control] = '1')	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message J5 ([PktType] = '0' and [SeqNo] = '001'-'998')
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)		
.....		
Message J5 ([control] = '1')	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message J5 ([PktType] = '1')
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message J6 Play JPEG File

Format: <STX>J6<ETX>[LRC] (request frame)

Message length: Fixed 5 bytes.

Usage: This message is used to play the JPEG files that have been selected from the terminal. Those selected JPEG files will be displayed sequentially until an operation (pressing any key, or receiving message 72 or Z1) that changes the display.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
J6	2	Message ID
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message J6	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)

Message J7 Set JPEG File As Idle Prompt

Format: <STX>J7[Jpeg Name]<ETX>[LRC] (request frame)
 <STX>J7[Result]<ETX>[LRC] (response frame)

Message length: Variable 6 ~ 20 bytes.

Usage: This message is used to set a JPEG file as the idle logo.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
J7	2	Message ID
[Jpeg Name]	1~15	ASCII string, Jpeg file name.
<ETX>	1	<03>
[LRC]	1	Checksum

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
J7	2	Message ID
[Result]	1	'0' = Setup successful '1' = Invalid length of file name. '2' = Fail to setup.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message J7	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
		Message J7 (response)
<ACK> (Good echo) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message J8 Enable/Disable Customized Idle Logo

Format: <STX>J8[Op]<ETX>[LRC] (request frame)
 <STX>J8[Result]<ETX>[LRC] (response frame)

Message length: Fixed 6 bytes.

Usage: This message is used to enable/disable the customized idle logo. When it is enabled, the default idle logo and the date/time will not be displayed. In order to successfully enable the customized idle logo, A JPEG file has to be loaded as the customized idle logo using message J4 and set by J7.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
J8	2	Message ID
[Op]	1	0 : Disable 1 : Enable
<ETX>	1	<03>
[LRC]	1	Checksum

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
J8	2	Message ID
[Result]	1	'0' = Setup successful '1' = Invalid Op parameter. '2' = Fail to setup (No Jpeg file is assigned)
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message J8	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
		Message J8 (response)
<ACK> (Good echo) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Message J9 Show JPEG File

Format: <STX>J9[Jpeg Name]<ETX>[LRC] (request frame)
 <STX>J9[Result]<ETX>[LRC] (response frame)

Message length: Variable 6 ~ 20 bytes.

Usage: This message is used to make the A10-P to show the assigned Jpeg file. The assigned Jpeg file will be displayed until any operation (pressing the Cancel key or receiving a command) that changes the display.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
J9	2	Message ID
[Jpeg Name]	1~15	ASCII string, Jpeg file name.
<ETX>	1	<03>
[LRC]	1	Checksum

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
J9	2	Message ID
[Result]	1	'0' = Success. '1' = Invalid length of file name. '2' = No assigned Jpeg file.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message J9	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
		Message J9 (response)
<ACK> (Good echo) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	
		The screen will not be changed until pressing Cancel or receiving any command.

Message JA Set Boot Logo

Format: <STX>JA[PktType] [SeqNo]<FS>[Size] [Data]<ETX>[LRC] (request frame)
 <STX>JA[Status]<ETX>[LRC] (response frame)

Message length: Variable 22 to 538 bytes for the request frame, and fixed 6 bytes for the response frame.

Usage: This message is used to download a bitmap file for boot logo from the terminal to the A10-P.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
JA	2	Message ID
[PktType]	1	'0' = first or middle of multiple packets. '1' = last or only packet.
[SeqNo]	3	'000' to '999', for onetime only packet, set to '000'.
<FS>	1	<1C>, field separator
[Size]	3	'000' to '525', size of [ImgData] in bytes
[ImgData]	0-525	Base 64 format data (optional)
<ETX>	1	<03>
[LRC]	1	Checksum

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
JA	2	Message ID
[Status]	1	'0': Ready for next JA packet. '1': [PktType] has invalid value. '2': [SeqNo] has invalid value. '3': Number of <FS> is wrong. '4': [Size] has invalid value. '5': [ImgData] has invalid value. '6': No more space. '7': Bitmap file format error. 'F': Download Success.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message JA ([PktType] = '0' and [SeqNo] = '000')	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message JA ([Status] = '0')
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	
Message JA ([PktType] = '0' and [SeqNo] = '001'-'998')	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message JA ([Status] = '0')
.....		
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	
Message JA ([PktType] = '1')	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)
	←	Message JA ([Status] = 'F')
<ACK> (Good LRC) <NAK> (Bad LRC) <EOT> (after 3 NAKs)	→	

Section 15 White List Configuration Messages

Message ED Load BIN Exception Table

Format: <STX>ED.[Signature data]<FS>[BIN Exception Table]<ETX>[LRC]
(request frame)

Message length: Variable.

Usage: This message is used to load the BIN exception table (white list that excludes from encryption) to the reader.

The reader will create a temporary buffer to collect all data, and then use the application public key to verify the signature. (Since the application public key is issued by the UIC, the acquirer needs to request UIC to sign the BIN exception table.)

If verification succeeds, reader will replace the original BIN Exclusion Table; if verify failed, reader will abandon this data and return error code in the EE packet.

Note: To clear the whole white list, just send an empty entry to the reader: <STX>ED. <ETX>

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
ED	2	Message ID
.	1	<2E>, delimiter.
[Signature data]	Var.	Base64 encoded signature of BIN exception table entries.
<FS>	1	<1C> as field separator.
[BIN exception table]	Var.	ASCII string, refer to BIN Exception Rules for detailed data format. Maximum 50 entries are allowed.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message ED	→	
	←	<ACK> if LRC ok. <NAK> if LRC error. <EOT> if LRC error after 2 retries.
	←	Message EE
<ACK> if LRC ok. <NAK> if LRC error. <EOT> if LRC error after 2 retries.	→	

Message EE BIN Exception Table Loading Response

Format: <STX>EE[status]<ETX>[LRC] (response frame)

Message length: Fixed 6 bytes.

Usage: This message is the response of message ED.

Message element:

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
EE	2	Message ID
[status]	1	'0': Signature Verification fail. '1': Data successfully loaded.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

Refer to message ED.

Message EF Get Check Digits of BIN Exception Table

Format: **<STX>EF<ETX>[LRC] (request frame)**
<STX>EF[status].[check digits]<ETX>[LRC] (response frame)

Message length: Fixed 5 bytes for request / fixed 13 bytes for response.

Usage: This message is used to get the “check digits” of the current white list. Check digits are the leftmost 3 bytes of BIN Exception Table’s SHA 1 hash, represented in hexadecimal string.

Message element :

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
EF	2	Message ID
<ETX>	1	<03>
[LRC]	1	Checksum

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
EF	2	Message ID
[status]	1	'0': Cannot generate check digits '1': Check digits generated.
.	1	Optional, only exists with valid check digits. <2E> as delimiter.
[check digits]	6	Optional, only exists with valid check digits. Part of SHA 1 hash of BIN exception table.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message EF (request)	→	
	←	<ACK> if LRC ok. <NAK> if LRC error. <EOT> if LRC error after 2 retries.
	←	Message EF (response)
<ACK> if LRC ok. <NAK> if LRC error. <EOT> if LRC error after 2 retries.	→	

Section 16 Remote Key Loading Messages

Remote Key Loading module allows the PIN Pad device to download and replace the key remotely from the HOST. The PIN Pad and HOST shall verify each other base on PKI (Public Key Infrastructure). The PIN Pad shall communicate with the Payment Terminal Application through USB or COM port and all the communications between the PIN Pad and HOST shall be passive and controlled by the POS.

The details of the RKL module are listed in **Appendix I**.

Terminology

The following table lists the definitions of certificates, keys, and data to be exchanged in the RKL functions.

1. Certificates and keys at HOST side:

Certificates	Related Keys	Description
	PK_{Root, CA}	The CA root public key. This will be used to verify the new CER_{HOST, intermediate} . This root key shall be sent to UIC and then hard coded into PIN Pad firmware.
CER_{HOST, intermediate}	PK_{HOST, intermediate}	The host's CA public key certificate, signed by the host's CA root Private Key. This will be pre-loaded into PIN Pad and used to verify the CER_{HOST, RKL} .
	SK_{HOST, intermediate}	The host CA private key, used to sign CER_{HOST, RKL} . It's stored at the host side and is never exposed.
CER_{HOST, RKL}	PK_{HOST, RKL}	The host public key's Certificate, signed by the host's CA Private Key. This will be used to verify the acknowledgement from the host in RKL process.
	SK_{HOST, RKL}	The host private key. This key is used to sign the TDES key during the remote key loading. And it is stored at the host side and is never exposed.

2. Certificates and keys in the PIN pad side:

Certificates	Related Keys	Description
	PK_{UIC}	UIC CA public key. This will be used to verify the acknowledgement of CER_{PINPad, RKL} when the host receives the certificate from PIN Pad in RKL process. The host shall have an interface to pre-load it.
	SK_{UIC}	The UIC CA private key. This is only used inside of UIC to sign the PK_{PINPad} and generate CER_{UIC,CA} . This private key will never be exposed.
CER_{PINPad, RKL}	PK_{PINPad, RKL}	PIN Pad public key certificate, signed by UIC CA Private Key, SK_{UIC} . The expiration date of the certificate should be beyond the life cycle of the PIN Pad.
	SK_{PINPad, RKL}	PIN Pad private Key. This key is preloaded to PIN Pad during manufacturing and will be used to decrypt the enciphered TDES key.

3. Key to be injected:

Element	Description
CI	TDES cryptogram data (16 bytes binary data), generated by HOST and will be injected into PIN Pad in Remote Key Loading process. The key is used to do the PIN encryption or in other encryption algorithm like MAC. Each key only has its unique purpose for secure consideration.
B	Key block of TDES Key encrypted by PK_{PIN Pad} . $B = (SN_{HOST} // CI)PK_{PINPad}$

4. Data for verification:

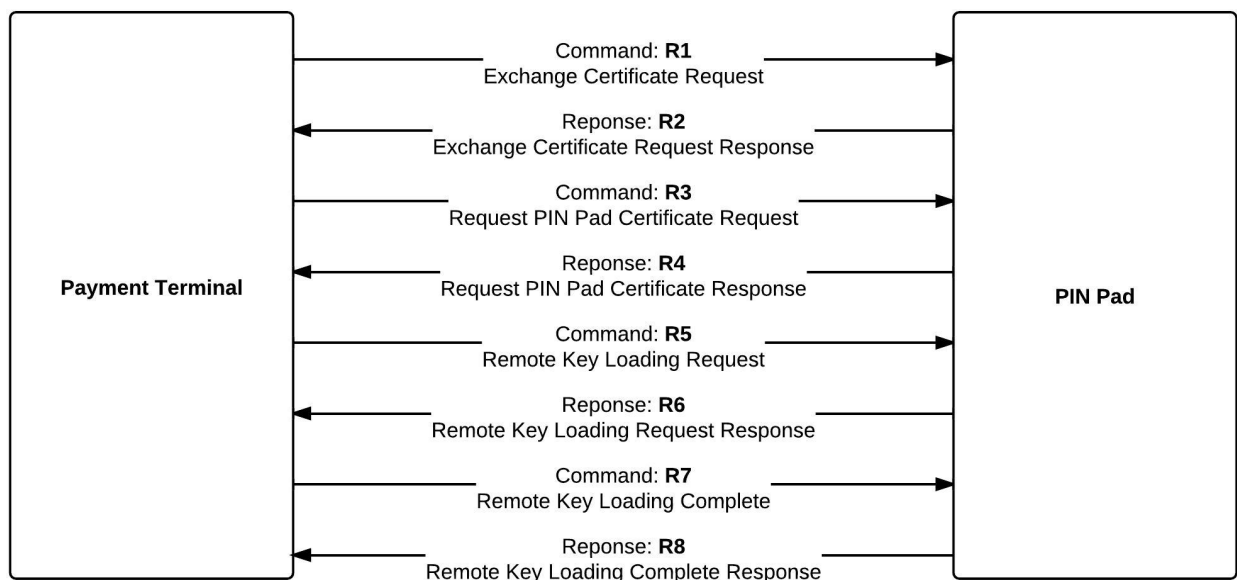
Element	Description
SN_{PINPad}	The serial number of the PIN Pad.
SN_{HOST}	The serial number of the host RKL certificate, CER_{HOST, RKL} .
RA & RB	8-byte Random numbers which are generated by PIN Pad (RA) and HOST (RB).

5. Signed message and thumb print.

Element	Description
TP(CER_{HOST})	Thumb print of the HOST's Certificate in PKCS#7 encoded in base64.
(RA//RB//SN_{PINPad}//B)SK_{HOST}	RA, RB, Pin Pad Serial Number and B which are signed by private key of the host, SK _{HOST, RKL} . It is a PKCS#7 format encoded in base64. The literal sequence in PKCS#7 is RA, RB, SN _{PINPad} , and B and SHA-1 for the integrity.
(RA//RB//SN_{HOST})SK_{PINPad}	The signature of RA//RB//SN _{HOST} which is signed by private key of the PIN Pad, SK _{PINPad, RKL} . It is a PKCS#7 format encoded in base64.

Command Flow

The payment terminal needs to initiate the Remote Key Loading process. The following is the command flow:



Message R1 Exchange Certificate Request

Format: <STX>R1[CER]<ETX>[LRC] (request frame)

Message length: Variable.

Usage: Initiate the certificate exchange by sending **CER**_{HOST, RKL} to PIN pad. The certificate is RSA PKCS#7 enveloped PEM (data is base 64 encoded with the “-----BEGIN PKCS7-----” and “-----END PKCS7-----” marks).

This command will initialize an “RKL session”. During RKL session, only R series command and “72” command can be accepted by PIN pad.

Message element:

Request fame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
R1	2	Message ID
[CER]	Var.	RSA PKCS#7 eneveloped CER _{HOST,RKL} in PEM format.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message R1	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) (<EOT> after 3 NAKs)
	←	Message R2
<ACK> (Good LRC) <NAK> (Bad LRC) (<EOT> after 3 NAKs)	→	

Message R2 Exchange Certificate Request Response

Format: <STX>R2[TP][LRC] (response frame)
<STX>R2[error code]<ETX>[LRC] (error frame)

Message length: 125 bytes.

Usage: Response of message R1

Message element:

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
R2	2	Message ID
[TP]	120	RSA PKCS#7 enveloped TP(CER _{HOST}) in PEM format. (See details below)
<ETX>	1	<03>
[LRC]	1	Checksum

Error frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
R2	2	Message ID
[Error]	1	Error code if R1 command failed. '1' PEM data format error (i.e. non-base64 character found). '2' Stored intermediate public key hash error. '3' Stored intermediate public key length error. '4' Unreasonable public key contents. '5' Certificate verification failed. '6' Cannot allocate enough memory to parse certificate. '7' Certificate parsing failed. '8' Unknown element found when parsing certificate. '9' Certificate data element error (i.e. wrong version or signature) 'Z' Any other internal error.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

Refer to message R1.

PKCS#7 Data structure of [TP] (ASN.1 annotation):

```
SEQUENCE {  
    version                INTEGER,  
    digestAlgorithms  DigestAlgorithmIdentifier,  
    contentInfo        ContentInfo  
}
```

```
DigestAlgorithmIdentifier ::= SEQUENCE {  
    algorithm  OBJECT IDENTIFIER,  
    parameters ANY DEFINED BY algorithm OPTIONAL  
}
```

```
ContentInfo ::= SEQUENCE {  
    contentType OBJECT IDENTIFIER,  
    content      [0] EXPLICIT ANY DEFINED BY contentType OPTIONAL  
}
```

Where version = 0, digestAlgorithm = sha1 (parameters=NULL), contentType = {pkcs7-1}, content = OCTET STRING of thumb print (20 bytes sha-1 hash).

Message R3 PIN Pad Certificate Request

Format: <STX>R3<ETX>[LRC] (request frame)

Message length: Fixed 5 bytes.

Usage: Get the **CER**_{PINPad, RKL} from PIN pad.

This command must be issued in RKL session.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
R3	2	Message ID
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message R3	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) (<EOT> after 3 NAKs)
	←	Message R4
<ACK> (Good LRC) <NAK> (Bad LRC) (<EOT> after 3 NAKs)	→	

Message R4 PIN Pad Certificate Response

Format: <STX>R4[RA][CERPIN Pad][LRC] (response frame)
 <STX>R4[error code]<ETX>[LRC] (error frame)

Message length: Variable.

Usage: Response of message R3

Message element:

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
R4	2	Message ID
[RA]	16	Random Number A generated by PIN Pad
[CER _{PIN Pad}]	Var.	RSA PKCS#7 enveloped CER _{PINpad,RKL} in PEM format.
<ETX>	1	<03>
[LRC]	1	Checksum

Error frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
R4	2	Message ID
[Error]	1	Error code if R3 command failed. 'A' Not in RKL session.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

Refer to message R3.

Message R5 Remote Key Loading Request

Format: <STX>R5[Key ID][Signed data][LRC] (request frame)

Message length: Variable.

Usage: Send Remote Key Loading request with validation information.

This command must be issued in RKL session..

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
R5	2	Message ID
[Key ID]	1	'0' to '9', 'B' to 'F'. ('A' is not available.)
[Signed data]	Var.	RSA PKCS#7 "signed data" structure in PEM format. (RA//RB//SN_{PINpad}//B//key ID//usage//mode) signed by SKHOST RA: Random number generated by PIN Pad. RB: Random number generated by Host. SN_{PINpad}: Serial number of PIN Pad. B: (SN_{HOST}//CI) encrypted by PK_{PINpad} - SN_{HOST}: Serial number of the Host certificate - CI: TDES key data. Key ID: the same with [Key ID] field Usage: ANSI TR-31 usage of the key Mode: ANSI TR-31 mode of the key
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message R5	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) (<EOT> after 3 NAKs)
	←	Message R6
<ACK> (Good LRC) <NAK> (Bad LRC) (<EOT> after 3 NAKs)	→	

PKCS#7 Data structure of [Signed data] (ASN.1 annotation):

```

ContentInfo ::= SEQUENCE {
    contentType      ContentType,
    content           [0] EXPLICIT ANY DEFINED BY contentType
OPTIONAL
}
ContentType ::= OBJECT IDENTIFIER -- value={pkcs-7 1}

```

The content of PKCS#7-1 data defined as following

```

Data ::= SEQUENCE {
    keyAndAuthData  KeyAndAuthData, -- See definition below
    signature        BIT STRING -- PKCS#1 signature of keyAndAuthData
}
KeyAndAuthData ::= SEQUENCE {
    rA               OCTET STRING -- 8 bytes random A
    rB               OCTET STRING -- 8 bytes random B
    sn_PinPad        Printable String -- 12 bytes serial # of PIN pad
    b                OCTET STRING -- encrypted TDES master key and
SNHOST
    key_id           INTEGER -- 1 byte key id from 0~9, 11~15
    usage            Printable String -- 2 bytes key usage
    mode             Printable String -- 1 byte key mode
}

```

Encryption detail of TDES master key data "b":

Clear text data (according to RSA PKCS#1 v1.5) = 0x00 | 0x02 | PS | 0x00 | DATA

Where PS = non-zero random data of 256 - 3 - (length of DATA) bytes,

DATA = ASN.1 DER encoding of following structure

```
{sn_Host      OCTET STRING,  
key_value     OCTET STRING}
```

Message R6 Remote Key Loading Response

Format: <STX>R6[Signed data][LRC] (response frame)
 <STX>R6[error code]<ETX>[LRC] (error frame)

Message length: Variable.

Usage: Response of message R5

Message element:

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
R6	2	Message ID
[Signed data]	Var.	RSA PKCS#7 “signed data” structure in PEM format which contains a digital signature of following data. Message M = RA//RB//SN_{HOST} RA: Random number generated by PIN Pad. RB: Random number generated by Host. SN_{HOST}: Serial number of the Host certificate. Signature = RSA-PKCS1_V1_5-SIGN (SK_{PINPad}, M).
<ETX>	1	<03>
[LRC]	1	Checksum

Error frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
R6	2	Message ID
[Error]	1	Error code if R6 command fails '1' PEM data format error (i.e. non-base64 character found). '5' RKL request verify failed. '8' Unknown element found when parsing RKL request. 'A' Not in RKL session. 'B' Flash read failed (when injecting master key). 'C' Master key value duplicated. (Do not use the same value for different key) 'D' Flash write error (when injecting master key). 'E' Key id range error 'Z' Any other internal error.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

Refer to message R5.

PKCS#7 Data structure of [Signed Data]:

```
SEQUENCE {  
    version                INTEGER,  
    digestAlgorithms DigestAlgorithmIdentifier,  
    contentInfo            ContentInfo  
}
```

```
DigestAlgorithmIdentifier ::= SEQUENCE {  
    algorithm OBJECT IDENTIFIER,  
    parameters ANY DEFINED BY algorithm OPTIONAL  
}
```

```
ContentInfo ::= SEQUENCE {  
    contentType OBJECT IDENTIFIER,  
    content      [0] EXPLICIT ANY DEFINED BY contentType OPTIONAL}
```

Where version = 0, digestAlgorithm = sha1WithRSAEncryption (parameters=NULL),
contentType = {pkcs7-1}, content = OCTET STRING as signature (256 bytes)

Message R7 Remote Key Loading Complete

Format: <STX>R7<ETX>[LRC] (request frame)

Message length: Fixed 5 bytes.

Usage: Indicate the remote key loading process is complete.

This command ends the RKL session.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
R7	2	Message ID
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message R7	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) (<EOT> after 3 NAKs)
	←	Message R8
<ACK> (Good LRC) <NAK> (Bad LRC) (<EOT> after 3 NAKs)	→	

Message R8 Remote Key Loading Complete Response

Format: <STX>R8[status][LRC] (response frame)

Message length: Fixed 5 bytes.

Usage: Response of message R7

Message element:

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
R8	2	Message ID
[status]	1	Execution status of R7. '0': Success.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

Refer to message R7.

Message R9 Load CA Certificate Request for RKL Verification

Format: <STX>R9.[CER data]<ETX>[LRC] (request frame)

Message length: Variable.

Usage: This command loads **CER**_{HOST, intermediate} to PIN pad for later RKL request usage. The certificate is RSA PKCS#7 enveloped PEM (data is base 64 encoded with the “-----BEGIN PKCS7-----” and “-----END PKCS7-----” marks).

This command does not need to be in RKL session.

Message element:

Request frame (HOST to A10-P)

Field	Length	Value and description
<STX>	1	<02>
R9	2	Message ID
.	1	Field separator, 0x2E.
[CER]	Var.	RSA PKCS#7 enveloped CER _{HOST,intermediate} in PEM format.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Direction	A10-P
Message R9	→	
	←	<ACK> (Good LRC) <NAK> (Bad LRC) (<EOT> after 3 NAKs)
	←	Message RA
<ACK> (Good LRC) <NAK> (Bad LRC) (<EOT> after 3 NAKs)	→	

Message RA Load Certificate Request for RKL Verification Response

Format: <STX>RA[status][LRC] (response frame)

Message length: Fixed 5 bytes.

Usage: Response of message R9

Message element:

Response frame (A10-P to HOST)

Field	Length	Value and description
<STX>	1	<02>
RA	2	Message ID
[status]	1	Status Code. '0' Success. Others: Refer to message R2.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

Refer to message R9.

Appendix A Key Management References

Clear Text Key Injection

The initial key transport key (KTK) of the A10-P should be injected in a physically secured environment by the FIPS 140-2 validated hardware security module (HSM).

Enter Clear Text Key Injection Mode

1. Press [ENTER] + '1' from the A10-P keypad and then press [ENTER] to enter System Manager.
2. Select "Key Injection" item, the A10-P will prompt the user to enter two passwords for key injection.
 - ENTER PASSWORD 1:
The 1st authorized operator enters his key injection password.
(For the initial key injection, the default password should be updated. The new password should be different from the previous password and longer than or equal to 7 digits.)
 - ENTER PASSWORD 2:
The 2nd authorized operator enters his key injection password.
(For the initial key injection, the default password should be updated. The new password should be different from the previous password and longer than or equal to 7 digits.)
3. After passwords are verified, there will be three options shown on the LCD, use function key to navigate and press [ENTER] to select:
 - INJECT MKEY/IPEK:
Enable packet handler for clear text key injection packet. Refer to message 02 (clear text) and message 90 (clear text). Messages 08 and 96 are also allowed in this mode.
 - SET MK ALGORITHM:
Choose the key management scheme (TDES).
 - UPDATE PASSWORDS:
Change the key injection passwords on demand.

Switching Between TDES Key Management Scheme

TDES is the default key management scheme of the A10-P. A10-P will treat all keys injected by clear text mode as TDES keys, and the encrypted key bundle should be ANSI TR-31 format "B" (TDES bundle).

The default attributes of the clear text injected key will be:

usage = "K0" ("K1" for KTK), algorithm = "T", mode = "D", version = "00" and export = "N"

The default attributes for clear text injected MK (id '0' to '9')

usage = "K0", algorithm = "A", mode = "D", version = "00" and export = "N"

The default attributes for clear text injected MK (id 'B' to 'E' and 'G')

usage = "K0", algorithm = "T", mode = "D", version = "00" and export = "N"

The default attributes for clear text injected KTK (id 'F')

usage = "K1", algorithm = "A", mode = "D", version = "00" and export = "N"

The life cycle of the key serial number of a DUKPT key:

When a transaction uses data encryption functions (such as MSR track encryption and EMV data encryption), a 10-second countdown timer will be triggered. Only when the POS-A AP is in the IDLE state, the countdown timer is decreased. When it reaches 0, the firmware will delete the DUKPT key and its KSN in the cache so that the KSN will advance for the next transaction. However, if the POS-A AP detects the second PIN input within 10 seconds, it will force KSN to roll.

ANSI TR-31 Key Bundle

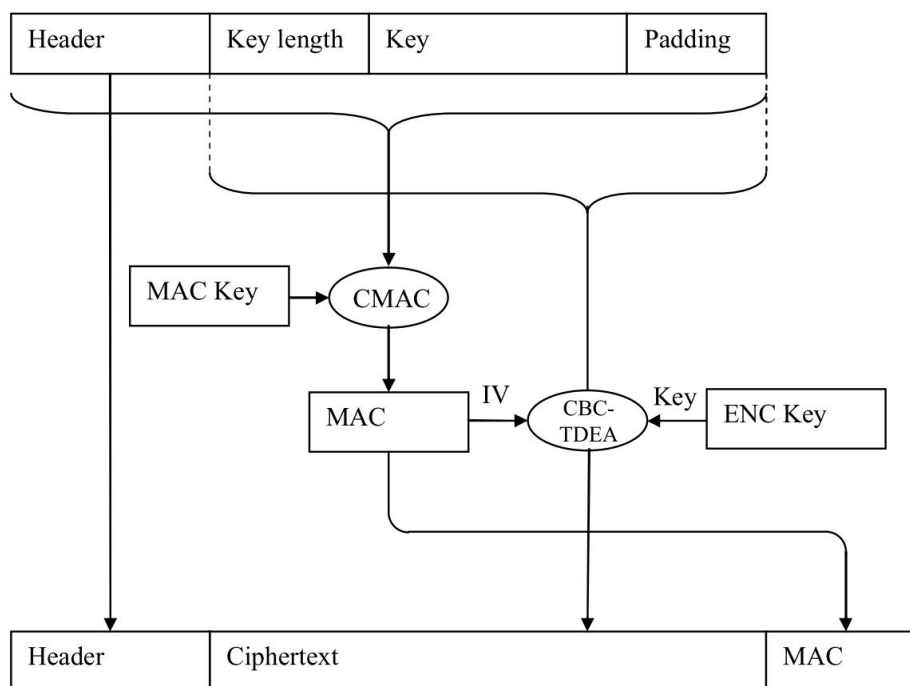
Overview of the TR-31 Bundle

ANSI TR-31 key bundle contains three components:

Component	Description
Key block header (KBH)	Contains the attribute information about the key and the key block.
Confidential data	Key length data, key value, and random padding to be encrypted.
Message authenticate code (MAC)	The authenticate code covering the whole clear text data (KBH + confidential data) with the NIST SP 800-38B CMAC algorithm.

The following graph shows the general flow of the data binding and encryption:

For TDES key bundle the CMAC is using TDES algorithm and generates a 8-bytes CMAC.



Message Elements of the Key Block Header

Field	Length	Value and description
Version ID	1	'B': TR31:2010 key derivation method. (for TDES KTK)
Key block length	4	ASCII numeric digits providing the key block length after encoding. i.e. "0112" means the whole key block contains 112 bytes.
Key usage	2	Provides information about the intended function of the protected key. "K1" = TR-31 key block protecting key. "K0" = Key encryption key. "P0" = PIN encryption key "D0" = Data encryption or decryption key "M3" = MAC generation or verification key "B1" = DUKPT initial key
Algorithm	1	The algorithm for which the protected key may be used. 'T' = TDES
Mode of Use	1	Defines the operation of the protected key that can be performed. 'D' = Decryption only 'E' = Encryption only 'B' = Both for encryption and decryption 'G' = MAC generation 'X' = Key derivation (DUKPT)
Key version number	2	Represents version of the key bundle. A10-P interprets this string as a hexadecimal number and the allowed value is "00" to "FF". "00" means don't care about the version number.
Exportability	1	Defines whether the protected key may be transferred outside the cryptographic domain. Always 'N' on A10-P.
Number of Optional Blocks	2	ASCII numeric digits defining the number of Optional Blocks included in the key block. "00": No optional block (for master keys). "01": One optional block (for DUKPT initial key serial number). "02": Two optional blocks (for version D bundle with key serial number and padding block)
Reserved	2	Always "00"
Opt. block ID [1]	2	(Optional) Identifier of optional block. "KS": for the DUKPT key serial number.

Opt. block length [1]	2	(Optional) Hexadecimal string defining the length of the optional block. “18”: 24 (0x18) bytes for the DUKPT key serial number. (Including Opt. block ID and Opt. block length field.)
Opt. block data [1]	20	(Optional) Hexadecimal string of DUKPT key serial number, always 20 bytes. i.e. “FFFF9876543210E00000”.
Opt. block ID [2]	2	(Optional) Identifier of optional block. “PB”: for version D key bundle, padding optional block data to the multiple of 16 bytes.
Opt. block length [2]	2	(Optional) Hexadecimal string defining the length of the optional block.
Opt. block data [2]	n	(Optional) The padding block can be filled by any printable ASCII character.

Message Elements of the Confidential Data

Field	Length	Value and description
Key length	2 bytes	Key length in bits represented in binary, with MSB first. i.e. 24 bytes (192 bits) TDES key represented as 0x00 0xC0
Key	16 or 24 bytes	Byte array of the key value. TDES Master keys can use 16 or 24 bytes DUKPT initial keys are 16 bytes.
Padding	n	Random data to make the whole key data becoming a multiple of 8 bytes (for TDES bundle).

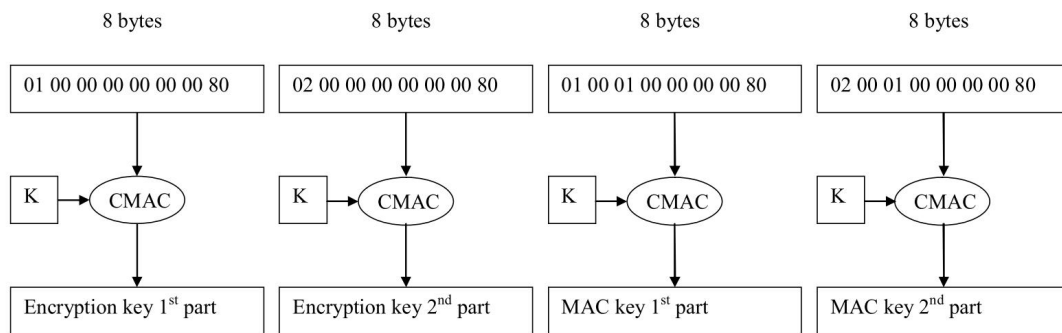
TDES Derivation of the MAC Key and the ENC Key

The TR-31 MAC key and ENC key are derived from the key block protect key (KBPK). A10-P uses the Key Transport Key (KTK) in master key slot #20 (0x14) as the KBPK.

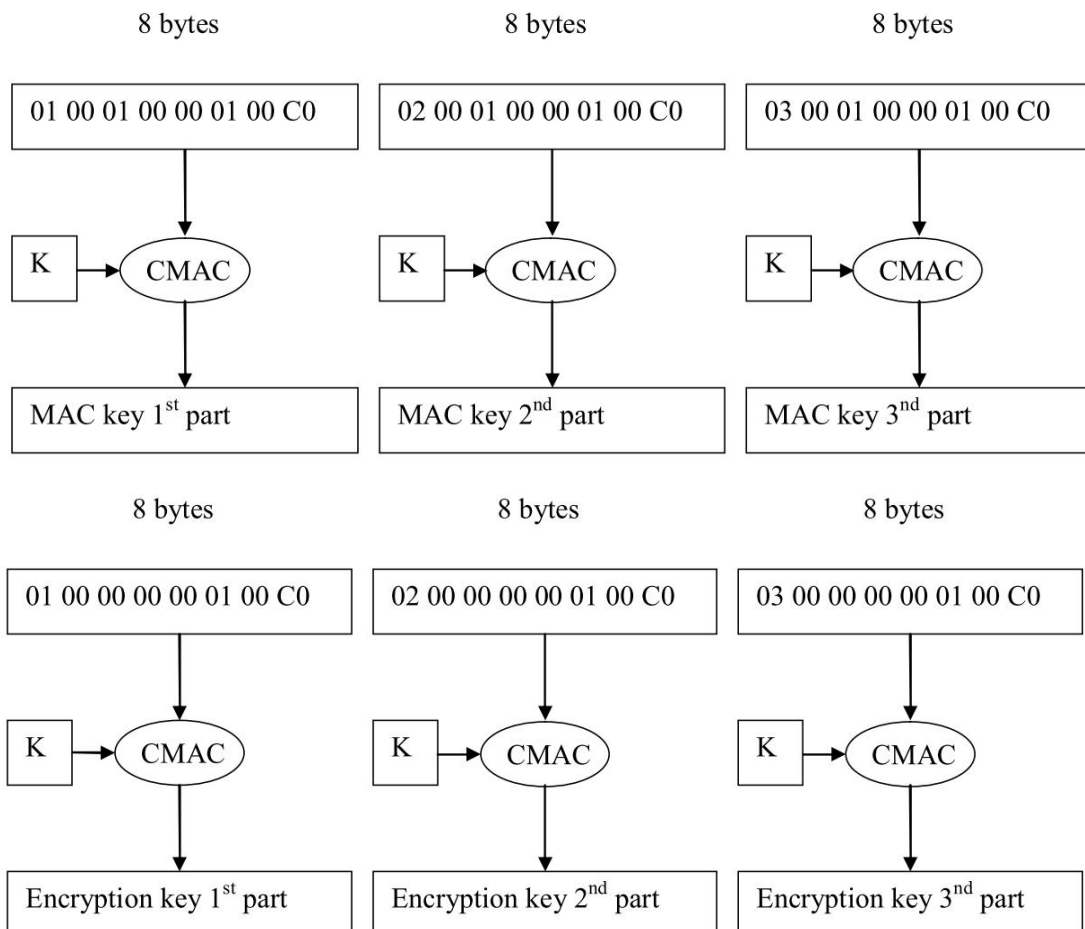
KBPK is used to calculate the CMAC for some pre-defined data source, taking the output as the MAC key or ENC key for TR-31 data.

(CMAC: Refer to NIST Special Publication 800-38B “The CMAC Mode for Authentication”)

The following graph is the data source and CMAC for the double-length KBPK (denoted as 'K').



The following graph is the data source and CMAC for the triple-length KBPK (denoted as 'K').



TDES TR-31 Examples

Example 1:

KLK: 0123456789ABCDEF FEDCBA9876543210

MK to be loaded: 89E88CF7931444F3 34BD7547FC3F380C, usage = "K0", slot = 0

- Generate KBH:
KBH = B | 0080 | K0 | T | D | 00 | N | 0000
- Generate MAC:
Key length = 16 bytes (128 bits = 0x80),
Generate 6 bytes random padding = B43E503C4E6A
Confidential Block = 008089E88CF79314 44F334BD7547FC3F 380CB43E503C4E6A
According to the "TDES Derivation of the MAC Key and the ENC Key" section we get following:
Derived ENC key: 34ADB3718BA916F6 A4EDF76EBA91175B
Derived MAC key: 5E1DAFEC63A36C6A 38EF3AA673758CE1
CMAC of the confidential data (by MAC key) = CB3D4C2890ADB93E
- Generate Encrypted Confidential Block:
Use TDES-CBC to encrypt confidential block, the key is ENC key, with IV = CMAC of the confidential data:
The encrypted key block = D575EF40B8B9B8AE 41A0D235D57E8306 6698212B146AB3A5
- Final output:
KBH | Encrypted Confidential Block | CMAC
Placed in message 02 with Key ID = 0 becomes:
<SI>020B0080K0TD00N0000
D575EF40B8B9B8AE41A0D235D57E83066698212B146AB3A5
CB3D4C2890ADB93E<SO> [LRC]

Example 2:

KLK: 6AC292FAA1315B4D 858AB3A3D7D5933A 5F81E94128C763A1

IPEK to be loaded: E8BC63E5479455E2 6577F715D587FE68

KSN = FFFF9876543210E00000

- Generate KBH:
KBH = B | 0104 | B1 | T | X | 00 | N | 0100
Optional Block = KS | 18 | FFFF9876543210E00000
- Generate MAC:
Key length = 16 bytes (128 bits = 0x80),
Generate 6 bytes random padding = D9CD26E52E18
Confidential Block = 0080E8BC63E54794 55E26577F715D587 FE68D9CD26E52E18
According to the "TDES Derivation of the MAC Key and the ENC Key" section we get following:
Derived ENC key: D6175E41CA9E9216 4060669D7519014C 60D8BD0FE5648F90

Derived MAC key: 776DFA36EC1D51F1 F4BC4D13DE741A8C B393198BE5B86098

CMAC of the confidential data (by MAC key) = 0B2330B55985892B

- Generate Encrypted Confidential Block:

Use TDES-CBC to encrypt confidential block, the key is ENC key, with IV = CMAC of the confidential data:

The encrypted key block = 036EAA0FD4B32C79 04B8CC26A5A8F2DB DD2AA2581F6597C4

- Final output:

KBH | Encrypted Confidential Block | CMAC

Placed in message 90 becomes:

<STX>90B0104B1TX00N0100KS18FFFF9876543210E00000

036EAA0FD4B32C7904B8CC26A5A8F2DBDD2AA2581F6597C4

0B2330B55985892B<ETX> [LRC]

Appendix B PIN Block Format

A10-P supports the following PIN block formats for different usage:

- ISO 9564 format 0 (also known as ANSI X9.8), for TDES encrypted PIN blocks (MK/SK and DUKPT.)
- ISO 9564 format 2, for EMV offline PIN verification.

ISO9564 format 0 (ANSI X9.8)

PIN Block Format

Nibble	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Value	C	N	P	P	P	P	P/F	P/F	P/F	P/F	P/F	P/F	P/F	P/F	F	F

Bit fields explanation:

C - Control field (Format number). Value = 0000 (Does not support Format 1 or Format 3)

N - PIN length entered field. Value = 0100 to 1100 (4-12) (0x4 – 0xC)

P - PIN digit. Value = 0000 to 1001 (0-9)

F - Fill digit. Value = 1111 (F)

P/F - Pin digit or fill digit, as determined by the PIN Length N. PIN Length is 4 to 12

Primary Account Number Block (PANB) Format

Nibble	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Value	0	0	0	0	A1	A2	A3	A4	A5	A6	A7	A8	A9	A10	A11	A12

Bit field explanation:

A - The twelve rightmost digits of the primary account number (PAN), excluding the check digit. A1 is the most significant digit and A12 is the digit immediately preceding the PAN's check digit. If the primary account number excluding the check digit is less than twelve digits, the digits are right justified and are padded on the left with zeroes. Permissible values are 0000 to 1001.

0 - Pad digit = 0000. The first four digits of the account number block are always padded with this value.

Formatted Clear-Text PIN Block

The PIN and account number blocks are Exclusive ORed before being assembled in the DES (Data Encryption Standard) input register. When the account number is not available, only the PIN block is assembled in the DES input register. A10-P will output the DES/TDES encrypted PIN block with message 71 and delete the clear-text PIN block immediately after transaction completed.

Example:

Account Number: 1234567890-6 (6 is the check number and will be ignored)

PIN: 8780

The PIN block = 04 87 80 FF FF FF FF FF

The PANB = 00 00 12 34 56 78 90 00

Formatted PIN block = 04 87 92 CB A9 87 6F FF (Data to be encrypted)

ISO9564 format 2 (EMV Offline PIN)

PIN Block Format

The PIN block of the ISO9564 format 2 has the same structure with the format 0, but the 1st nibble (control field) is 0x2.

In the EMV transaction flow, the plain text offline PIN verification is placing the format 2 PIN block into the EMV VERIFY command APDU. (Refer to EMV v4.3 Book 3 chapter 6.5.12.)

The enciphered offline PIN verification has additional procedure to use random number provided by the IC card and A10-P to become the data block, and then encrypt the data with the RSA public key provided by the IC card:

Field Name	Length	Description
Data Header	1	Hex value 0x7F
PIN block	8	Format 2 PIN block
ICC Unpredictable Number	8	Unpredictable number obtained from the IC card.
Random Padding	$N_{IC} - 17$	Random paddin generated by A10-P to pad the data to the length of the RSA key (obtained from the IC card.)

Appendix C ICC Tag Definition on EMV Data

Some data items not provided with tag definitions by EMV are needed for EMV transactions. The following are the defined tags used by the A10-P.

Terminal Configuration Data:

Name	Description	Format	Tag	Length
Terminal UI Capability	0: Reader will select the highest priority application directly. 1: Reader will provide a list of candidate applications.	b	FFFF8101	1
Language Selection	0: Disable customer language selection. 1: Enable customer language selection.	b	FFFF8102	1
MSR Processing Batch Tag List	A10-P will store the transaction data according to the tag list while the transaction is performed as MSR processing. Ex, for tag list (829F365F3 09F399F029F03), A10-P will store transaction data of tag number 0x82, 0x9F36, 0x5F30, 0x9F39, 0x9F02 and 0x9F03.	b	FFFF8103	var
Transaction Log	User can enable/disable log function Transaction Log function is to prevent split sales. If Transaction Log is enable,during terminal risk management floor limit checking if there are several log entries with the same PAN, the terminal selects the most recent entry. The temrinal adds the Amount,Authorised for current transaction to the amount store in the log for that PAN to determine if the sum exceeds the Terminal Floor Limit. If the sum is greater than or equal to the Terminal Floor Limit, the terminall shall set the Transaction exceeds floor limit bit in the TVR to 1.	b	FFFF8105	1
Config ID	An indicator of the configuration for the EMV contact certification.	b	FFFF8106	1
Bypass PIN support	0: Do not allow bypass PIN processing. 1: Allow bypass PIN processing.	b	FFFF8107	1
POS Entry Mode	User can define a POS Entry mode code here	b	FFFF810A	1~4
US Common Debit	To enable US Common Debit flow 0x00:Disable	b	FFFF8112	1

	0x01:Enable Please refer to Appendix J.			
CAPK Revocation List	CA Public Key revocation list. The format is <RID><len of PKI+S/N><PKI><len of S/N><S/N>	b	FFFF8211	var

EMV Application Configuration Data:

Name	Description	Format	Tag	Length
Application Selection Indicator	0: Full AID matching. 1: Partial AID matching	n	FFFF8214	1
Threshold Value for Biased Random Selection	Format is same as the “Amount Authorized (Numeric)” defined by EMV. It must be zero or a positive number less than the floor limit (Refer to EMV.).	n	FFFF8215	6
Target Percentage to be used for Biased Random Selection	0x00 ~ 0x63	b	FFFF8216	1
Maximum Target Percentage to be used for Biased Random Selection	0x00 ~ 0x63	b	FFFF8217	1
US Common Debit	When Debit flow is enabled, the AID that contain TagFFFF8218 will be selected to card. After AID is matched, value of priority will be replace with value of TagFFFF8218 in Application config if value is 0x01 to 0x0f. If TagFFFF8218 exist but value is 0x00 this AID will be selected but priority won't replace with value of this Tag Please refer to Appendix J.	b	FFFF8218	1
Terminal Action Code - Default	Refer to EMV 4.3 BOOK 3, section 10.7	b	FFFF8205	5
Terminal Action Code - Denial	Refer to EMV 4.3 BOOK 3,	b	FFFF8206	5

	section 10.7			
Terminal Action Code - Online	Refer to EMV 4.3 BOOK 3, section 10.7	b	FFFF8207	5
Data tags required in Online message (ARQC)	See below	b	FFFF8208	var.
Data tags required in reversal message	See below	b	FFFF8209	var.
Data tags for batch data capture	See below	b	FFFF820A	var.
ARC Approve	See below	b	FFFF820B	var.
ARC Decline	See below	b	FFFF820C	var.
ARC Referral	See below	b	FFFF820D	var.
Data tags required in Referral	See below	b	FFFF820E	var.
Data tags required in APRC	See below	b	FFFF820F	var.
Data tags required in Financial Request	See below	b	FFFF8212	var.
Data tags required in Advice	See below	b	FFFF8213	var.

Note. The contents in the data object with tag values 0xFFFF8208 ~ 0xFFFF820A, 0xFFFF820E ~ 0xFFFF820F and 0xFFFF8212 ~ 0xFFFF8213 are a list of tags with asterisks as separators. Ex. The contents 95*9F02*82* in the data object with tag value (0xFFFF8208) means that there are three data objects: Terminal Verification Results (0x95), Amount Authorized (0x9F02) and Application Interchange Profile (0x82) will be necessary for online transaction.

Note. The data objects with tag values 0xFFFF8208 ~ 0xFFFF820A, 0xFFFF820E ~ 0xFFFF820F and 0xFFFF8212 ~ 0xFFFF8213 are used for customer-defined application on the A10-P. If the customer installs this information into the A10-P previously, the customer-defined application could apply the APIs provided by the A10-P to implement an EMV transaction and without modifying on the application if the data required for online message is changed.

Note. ARC Approve / Decline / Referral is used by the customer to define their definition of ARC code. Customer can concatenate a list of ARC values for definition of Approve, Decline or Referral.

Ex. If ARC Approve is 0x303031313536, A10-P will treat ARC with values 0x3030, 0x3131 and 0x3536 when the host approves this transaction online.

Transaction Result Data:

Name	Description	Format	Tag	Length
Issuer Script Result	The result of terminal script processing.	b	9F5B	var
Enciphered PIN block	Transaction PIN enciphered at the A10-P for online verification.	b	99	8
DUKPT KSN	To indicator the KSN if DIKPT PIN method is processed.	b	FFFF810B	var

These data objects are defined in EMV 4.3 BOOK3, Annex A without tag values. The tag values are defined by the A10-P.

Appendix D ICC Minimum Set of EMV Configuration Data

For a complete transaction, the user has to load a minimum set of terminal configuration data and EMV application data into the A10-P. In A10-P, the terminal configuration is applied to contact and contactless transactions. That is, there is only one terminal configuration.

Minimum set of Terminal configuration Data.

Mandatory:

Merchant Category (9F15), Merchant ID (9F16), Terminal Country Code (9F1A), Terminal ID (9F1C), IFD Serial NO. (9F1E), Terminal Capabilities (9F33), Terminal Type (9F35), Additional Terminal Capabilities (9F40), User Application Confirmation (FFFF8101), MSR Batch data tag list (FFFF8103).

Optional:

Merchant Name and Location (9F4E), Terminal Risk Management Data (9F1D)

Terminal Configuration Example:

```
FFFF8101 b 01
FFFF8102 b 01
FFFF8103 b 829F365F309F399F029F03
FFFF8105 b 00
FFFF8106 b 11
FFFF810A n 80
9F15 n 0100
9F16 ans 123456789012345
9F1A n 0840
9F1B b 00002710
9F1C an SmartPOS
9F1E an SmartPOS
9F33 b E0F0C8
9F35 n 22
9F39 n 05
9F40 b F000B0A001
9F4E ans UIC
9F01 n 000000000010
FFFF8211 b A00000000405F803000010B01234567805F803000010
```

Note: About the format of tags such as 9F15, 9F16, 9F1A, 9F1C, 9F1E, 9F33, 9F35, 9F40, please refer to the specification of EMV Book 3 (Annex A Data Elements Dictionary).

Note. User Application Confirmation is used to enable / disable the A10-P to build an AID list for the cardholder to select an EMV application.

Minimum set of EMV application configuration Data.

Application Selection Indicator (FFFF8214), Threshold Value for Biased Random Selection (FFFF8215), Target Percentage for Random Selection (FFFF8216), Maximum Target Percentage for Biased Random Selection (FFFF8217), ARQC Tag list (FFFF8208), Reversal Tag list (FFFF8209), Batch Tag list (FFFF820A), Default TDOL (97), Acquirer Identifier (9F01), Application Identifier (9F06), Application Version Number (9F09), Default Dynamic Data Authentication Data Object List (9F49), Terminal Floor Limit (9F1B),

Optional:

Terminal Action Code (Default) (FFFF8205), Terminal Action Code (Denial) (FFFF8206), Terminal Action Code (Online) (FFFF8207)

Note: Each Tag-Format-Value entry needs to be in the same line. DO NOT separate the entry to be two or more lines.

Application Configuration Example:

```

A0000000031010
FFFF8214 n 01
FFFF8215 n 000000000500
FFFF8216 b 00
FFFF8217 b 00
FFFF8205 b 0000000000
FFFF8206 b 0000000000
FFFF8207 b 0000000000
FFFF8208 b
829F369F278E9F1E9F109F339F35959F379F019F029F035F255F245A5F349F159F169F399F1
A9F1C575F2A9A9F219C9F25
FFFF8209 b
829F369F1E9F109F5B9F339F35959F015F245A5F348A9F159F169F399F1A9F1C579F029F035
F2A9A9F219C
FFFF820A b
829F369F079F278E9F349F1EDF8120DF8121DF81229F109F5B9F339F35959B9F379F109F029
F035F255F245A5F34898A579F159F169F399F1A9F1C5F2A9A9F219C9F249F199F269F25
FFFF820B b 3030
FFFF820C b 3531
FFFF820D b 30313032
97 b 9F0802
9F01 n 000000000001
9F06 b A0000000031010
9F09 b 0096
9F1B b 00002710
9F49 b 9F3704

```

Appendix E PCD Tag Definition on EMV Data

Some data items not given tag definitions by the EMV Contactless specification v2.4 are needed when performing the EMV transactions. A10-P defines its unique tags with the following data.

Table C-1 describes the self-define tag for PCD EMV Level2

Table C-2 describes the self-define tag for Visa contactless (payWave).

Table C-3 describes the self-define tag for Mastercard contactless (MCL).

Table C-4 describes the self-define tag for American Express contactless (Expresspay).

Table C-5 describes the self-define tag for Discover contactless (D-PAS).

EMV Application Configuration Data:

Table C-1: The self-define tag for PCD EMV Level2

Name	Description	Format	Tag	Length
Kernel ID	Used to build the matching kernel identification during the Combination Selection process 020000: Mastercard contactless (MCL) 030000: Visa contactless (payWave) 040000: American Express contactless (Expresspay) 050000: JCB contactless (J/Speedy) 060000: Discover contactless (D-PAS) 070000: UnionPay contactless (QuickPass)	b	DF810C	3
Status Check	Option within the terminal related to the checking of a single unit of currency. A single unit of currency has the value of 1 of the (major) unit of currency as defined in ISO 4217. As an example a single unit of currency for Euro is 1.00. 0 : Disable Status Check. 1 : Enable Status Check. Please refer to EMV Contactless Specifications v2.4 BOOK B, section 3.1	b	FFFF8007	1
Zero Amount Allowed	Indicates the zero amount transaction is accepted or not. 0x00: Not allowed 0x01: Allowed Please refer to EMV Contactless Specifications v2.4 BOOK B, section 3.1	b	FFFF8005	1
Contactless Floor Limit	Indicates the contactless floor limit relating to the Combination.	n	DF8123	6

	Please refer to EMV Contactless Specifications v2.4 BOOK B, section 3.1			
Contactless Floor Limit Check	Determine whether the contactless floor limit need to be checked 0x00: Disable 0x01: Enable	b	FFFF800A	1
Contactless Transaction Limit	Indicates the limit for which contactless transactions can be conducted relating to the Combination. Please refer to EMV Contactless Specifications v2.4 BOOK B, section 3.1 Please refer to EMV Contactless Specifications v2.4 BOOK B, section 3.1	n	DF8124	6
Disable Contactless Transaction Limit Check	Determine whether the contactless limit needs to be checked 0x00: check the transaction limit 0x01: not check the transaction limit	b	FFFF8004	1
CVM Required Limit	Indicates the limit for which a CVM is necessary relating to the Combination. Please refer to EMV Contactless Specifications v2.4 BOOK B, section 3.1	n	DF8126	6
CVM Required Limit Check	Determine whether the contactless CVM limit needs to be checked 0x00: Disable 0x01: Enable	b	FFFF8009	1
Extended Selection Support	An option in which the terminal appends the value indicated by the Extended Selection data element (Tag '9F29') to the ADF name in the SELECT command. 0x00: Not supported 0x01: Supported	b	FFFF8010	1
Display Message Hold Time	If provided, it indicates that the reader is to delay the processing of the Message Identifier in the next User Interface Request until the Hold Time has elapsed. The Hold Time is an integer in units of 100ms; the default value is zero. Only messages that must be displayed for a minimum period of time should be given a Hold Time. Note: An alert status with a timed operation (e.g. LED sequence or audio tone) will not be truncated by a subsequent User Interface Request.	b	FFFF8104	1
US Common Debit	When Deibt flow is enabled, the AID that contain	b	FFFF8218	1

	TagFFFF8218 will be selected to card. After AID is matched, value of priority will be replace with value of TagFFFF8218 in Application config if value is 0x01 to 0x0f. If TagFFFF8218 exist but value is 0x00 this AID will be selected but priority won't replace with value of this Tag Please refer to Appendix J.			
List of AID Support	The method of setting up the application selection priority List of AIDs has been used. Tag not present: Not supported 0x00: Not supported > 0: Supported with selection priority (1: first priority)	b	FFFF8219	1
Exception File Check	Check the exception file 0x00: check 0x01: not check	b	FFFF800C	1

Table C-2: The self-define tag for payWave

Name	Description	Format	Tag	Length
CVN 17 switch	0 : Disable CVN 17 1 : Enable CVN 17	b	FFFF8006	1
Track 1&2 configure	configure the data output with track1/track2 bit 1: 0x2: track1 enable bit 2: 0x4: track2 enable 0x6: enable both tracks	b	FFFF8012	1
Amount, Authorized of Zero Check Option	0: do not check zero amount 1: Option 1 2: Option 2	b	FFFF8008	1
Terminal Entry Capability	5: reader supports VSDC contact chip 8: reader does not support VSDC contact chip	b	FFFF8011	1
Dynamic Reader Limit (DRL) Support	0: DRL not support 1: DRL support Default (Tag not present) is DRL support.	b	FFFF800F	1
Application Usage Control(AUC) support	Cash and Cashback transaction default support AUC function, using below values to change setting. 0 : AUC not support 1 : AUC support Default (Tag not present) is AUC support.	b	FFFF800D	1
Process Issuer Script	This tag will be automatically generated during issuer script processing, get this tag to determine	b	FFFF800E	1

	AOSA(Available Offline Spending Amount) is present or not.			
--	--	--	--	--

Table C-3: The self-define tag for MCL

Name	Description	Format	Tag	Length
None	None	None	None	None

Table C-4: The self-define tag for Expresspay

Name	Description	Format	Tag	Length
Terminal Action Code - Default	Please refer to EP3.0 section 12, [EMV4.2 iii] section 10.7 and [EMV4.2 iv] section 6.3.6.	b	DF8120	5
Terminal Action Code - Denial	Please refer to EP3.0 section 12, [EMV4.2 iii] section 10.7 and [EMV4.2 iv] section 6.3.6.	b	DF8121	5
Terminal Action Code - Online	Please refer to EP3.0 section 12, [EMV4.2 iii] section 10.7 and [EMV4.2 iv] section 6.3.6.	b	DF8122	5
Magstripe Unpredictable Number Range	A10-P uses the random number of month (RNM) for the unpredictable number with '0000YYMM' format. The formula is $RNM = UNemv \bmod (UNR + 1)$	n	FFFF8250	2
Dynamic Reader Limit (DRL) Support	0: DRL not support 1: DRL support Default (Tag not present) is DRL support.	b	FFFF800F	1

Table C-5: The self-define tag for Discover D-PAS

Name	Description	Format	Tag	Length
None	None	None	None	None

Table C-6: Transaction Result Data

Name	Description	Format	Tag	Length
Issuer Script Result	The result of terminal script processing.	b	9F5B	var
Enciphered PIN block	Transaction PIN enciphered at the A10-P for online verification.	b	99	8
DUKPT KSN	To indicate the KSN if DIKPT PIN method is processed.	b	FFFF810B	var

These data objects are defined in EMV 4.3 BOOK3, Annex A without tag values. The tag values are defined by the A10-P.

Appendix F PCD Minimum Set of EMV Configuration Data

To perform an EMV transaction, the user has to setup a minimum set of terminal configuration data and EMV application data into the A10-P. In A10-P, the terminal configuration is applied to contact and contactless transactions. That is, there is only one terminal configuration.

Minimum set of Terminal configuration Data.

Mandatory:

Terminal Country Code (9F1A).

Optional:

Merchant Category (9F15), Merchant ID (9F16), Merchant Name and Location (9F4E),

Terminal ID (9F1C), IFD Serial NO. (9F1E), Terminal type (9F35).

Terminal Configuration Example:

Please refer to **Appendix D**.

Minimum set of EMV application configuration Data.

PCD EMV Level2:

Application Identifier(9F06),

Transaction Type(9C)

Kernel ID(DF810C),

Contactless Floor Limit(DF8123),

Contactless Transaction Limit (DF8124),

CVM Required Limit (DF8126)

For payWave:

Terminal Transaction Qualifiers(9F66),

qVSDC and payWave CVN 17 switch(FFFF8006),

payWave Application Configuration Example:

9C n 00

DF810C b 030000

FFFF8007 b 00

FFFF8005 b 01

DF8124 n 000000010000

DF8126 n 000000006000

DF8123 n 000000008000

FFFF8010 b 00

FFFF8006 b 01

FFFF8012 b 07

FFFF8008 b 01

95 b 0000000000

9F35 n 22

9F66 b B600C000

9F06 b A000000003
 9F7A b 01
 9F1B b
 FFFF8011 b 05
 FFFF800F b 00

For MCL:

None.

Note:

Please refer to EMV Contactless Specifications for Payment Systems Book C-2 to find out the detailed tag definitions. And refer the Table 4.3 of Book C-2 to find out the default values of the Configuration Data. Mastercard contactless configuration data allow empty tag input. To represent the empty tag in A10-P, we need to reserve a blank space after the format character.

MCL Application Configuration Example:

FFFF8005 b 01
 9C n 00
 9F06 b A0000000041010
 9F01 n
 9F40 b 0000000000
 9F09 b 0002
 DF8117 b 00
 DF8118 b 60
 DF8119 b 08
 DF811A b 9F6A04
 DF8108 b
 DF60 b
 DF8109 b
 DF62 b
 DF810A b
 DF63 b
 DF810D b
 DF811B b 20
 DF810C b 02
 9F6D b 0001
 DF811E b 10
 DF812C b 00
 DF811C b 0000
 DF811D b 00
 9F7E b
 DF8123 n 000000010000
 DF8124 n 000000030000
 DF8125 n 000000050000
 DF8126 n 000000001000
 DF811F b 08
 DF8120 b 0000000000
 DF8121 b 0000000000
 DF8122 b 0000000000
 9F33 b
 9F1D b 6CFF000000000000
 9F35 n 22

For Expresspay:

Expresspay Terminal Capabilities (9F6D),

Terminal Transaction Capabilities (9F6E),
 Application Version Number (9F09),
 Terminal Floor Limit (9F1B),
 Terminal Capabilities (9F33),
 Terminal Type (9F35)
 Terminal Action Code – Default (DF8120),
 Terminal Action Code – Denial (DF8121),
 Terminal Action Code – Online (DF8122),
 List of AID Support (FFFF8219),
 Expresspay Magstripe Unpredictable Number Range (FFFF8250),

Expresspay Application Configuration Example:

9F06 b A00000002501
 9C n 00
 DF810C b 04
 9F6D b C0
 9F6E b 48000000
 FFFF8004 b 00
 FFFF8005 b 01
 FFFF8007 b 00
 FFFF8009 b 01
 FFFF8010 b 00
 9F09 b 0001
 9F33 b 000888
 9F35 b 22
 DF8123 n 000000010000
 DF8124 n 000000015000
 DF8126 n 000000005000
 9F1B b 00010000
 DF8120 b 0000000000
 DF8121 b 0000000000
 DF8122 b 0000000000
 5F2A n 0978
 FFFF800F b 00
 FFFF8219 b 01

For D-PAS:

Terminal Transaction Qualifiers(9F66),
 Application Version Number(9F09),
 Terminal Floor Limit(9F1B),
 List of AID Support (FFFF8219),

D-PAS Application Configuration Example:

9F06 b A0000001523010
 9C n 00
 DF810C b 06
 FFFF8005 b 01
 FFFF8007 b 01
 FFFF8009 b 01
 FFFF8010 b 01
 9F40 b 6000000001

9F09 b 0001
9F33 b E0E808
9F35 b 22
FFFF800A b 01
9F1B b 00003A98
DF8123 n 0000000015000
DF8124 n 0000000030000
DF8126 n 0000000002000
9F66 b B6204000
DF8120 b 00000000000
DF8121 b 00000000000
DF8122 b 00000000000

Appendix G Fixed Prompts for Z2/Z3 Authenticated Mode

Prompt ID	Display	Prompt ID	Display
001	ENTER VALUE	112	CUSTOMER ID
002	ENTER PHONE	113	CUSTOMER NUMBER
003	ENTER CUST ID	114	CUSTOMER REF
004	ENTER AMOUNT	115	CUSTOMER REF NO.
005	PLEASE ENTER	116	DATE OF BIRTH
006	ENTER CARD ID	117	DEPARTMENT NO.
007	ENTER POINTS	118	DRIVER ID
008	ENTER ACCOUNT	119	DRIVER LICENSE
009	ENTER LOYALTY	120	DRIVER NUMBER
010	ENTER ZIP CODE	121	EMPLOYEE ID
011	ENTER VEHICLE	122	EMPLOYEE NUMBER
012	ENTER EMPLOYEE	123	ENTER
013	ENTER ID NUMBER	124	ENTER ACCOUNT #
014	ENTER DRIVER ID	125	ENTER AIR TAIL #
015	ENTER CASH BACK	126	ENTER BADGE #
016	DRIVER LICENSE	127	ENTER BIRTH DATE
017	PLEASE RE-ENTER	128	ENTER CARD #
018	REENTER DRIVER ID	129	ENTER CASH BACK
019	ENTER DRIVER LICENSE	130	ENTER CUST #
020	ENTER BIRTH DATE	131	ENTER CUST CODE
021	REENTER ZIP CODE	132	ENTER CUST DATA
022	ENTER MEMBERSHIP	133	ENTER CUST ID
100	ACCOUNT NUMBER	134	ENTER CUST REF
101	AIRCRAFT TAIL NO	135	ENTER CUST REF #
102	BADGE NUMBER	136	ENTER CID CODE
103	CARD NUMBER	137	ENTER CVC CODE
104	CARD SEC CODE	138	ENTER CVN CODE
105	CASH BACK AMOUNT	139	ENTER CVV CODE
106	CID CODE	140	ENTER DEPT #
107	CVC CODE	141	ENTER DOB
108	CVN CODE	142	ENTER DRIVER #
109	CVV CODE	143	ENTER DRIVER ID
110	CUSTOMER CODE	144	ENTER DRIVER LIC
111	CUSTOMER DATA	145	ENTER EMP ID

Prompt ID	Display	Prompt ID	Display
146	ENTER EMPLOYEE #	183	PHONE NUMBER
147	ENTER EXP DATE	184	PLEASE
148	ENTER FLEET #	185	PLEASE ENTER
149	ENTER FLEET DATA	186	PLEASE RE-ENTER
150	ENTER HOME PHONE	187	PO NUMBER
151	ENTER ID #	188	RE-ENTER
152	ENTER JOB #	189	REFERENCE NUMBER
153	ENTER ODOMETER	190	RESTRICTION CODE
154	ENTER PHONE #	191	ROUTE NUMBER
155	ENTER PO #	192	SECURITY CODE
156	ENTER REF #	193	SERIAL NUMBER
157	ENTER ROUTE #	194	SOCIAL SEC NO.
158	ENTER SEC CODE	195	STREET NUMBER
159	ENTER SERIAL #	196	SWIPE CARD
160	ENTER SOC SEC #	197	SWIPE CARD OR
161	ENTER SSN	198	TRAILER NUMBER
162	ENTER STREET #	199	USER ID
163	ENTER TRAILER #	200	V-CODE
164	ENTER USER ID	201	VEHICLE CARD NO.
165	ENTER V-CODE	202	VEHICLE ID
166	ENTER VEH CARD #	203	VEHICLE NUMBER
167	ENTER VEHICLE #	204	WORK PHONE NO.
168	ENTER VEHICLE ID	205	ZIP CODE
169	ENTER WORK PHONE	206	Enter Value \$
170	ENTER ZIP CODE	207	Enter Term
171	EXPIRATION DATE	208	Enter last 4 digits
172	FLEET DATA		
173	FLEET NUMBER		
174	HOME PHONE NO.		
175	ID NUMBER		
176	JOB NUMBER		
177	MMDDYY		
178	MMDDYYYY		
179	MMYY		
180	ODOMETER READING		
181	OR PHONE #		
182	OR PHONE NUMBER		

Appendix H Fixed Prompts for Z2/Z3 PIN Entry Mode

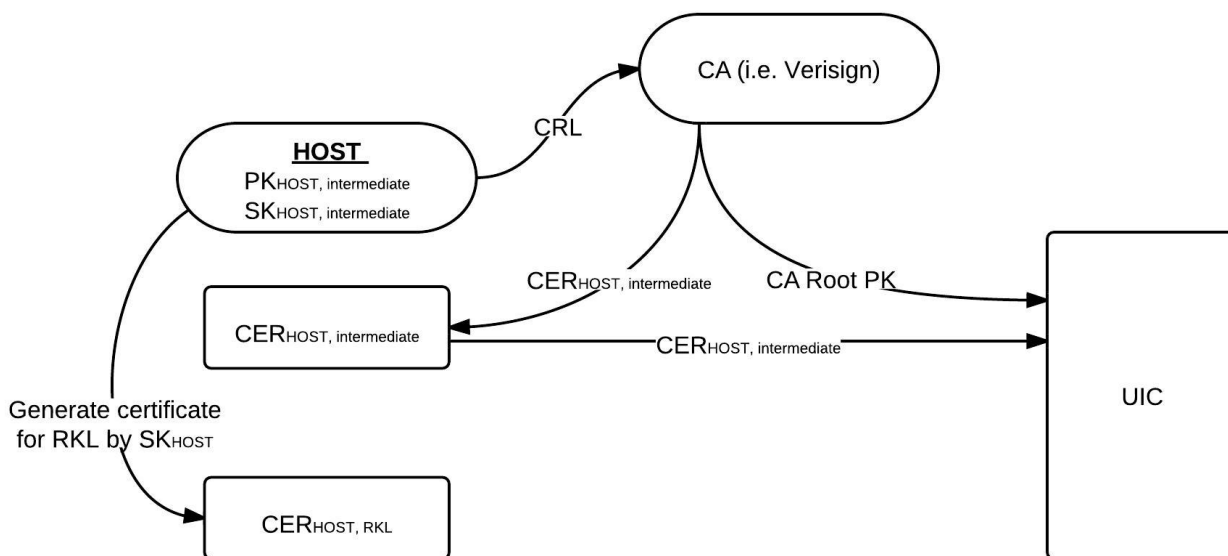
Prompt ID	Display
001	Enter Your PIN
002	Then Push ENTER
003	Thank You
004	PIN, Please
005	Thank You Very Much
006	Enter PIN
007	Type Your Password
008	Press [Enter]
009	Verifying PIN
00A	Type PIN\nthen Push ENTER
00B	Enter\nNew Password
00C	Confirm\nNew Password

Appendix I Remote Key Loading Details

Certificate Hierarchy

Host Certificate Hierarchy

The bank shall generate the RSA key pairs ($PK_{\text{HOST, intermediate}}$ and $SK_{\text{HOST, intermediate}}$) and use the private key to generate the certificate ($CER_{\text{HOST, RKL}}$) for the certificate exchange. After being certified by CA, the CA root public key and the host intermediate certificate need to be pre-loaded or hardcoded into PIN Pad.

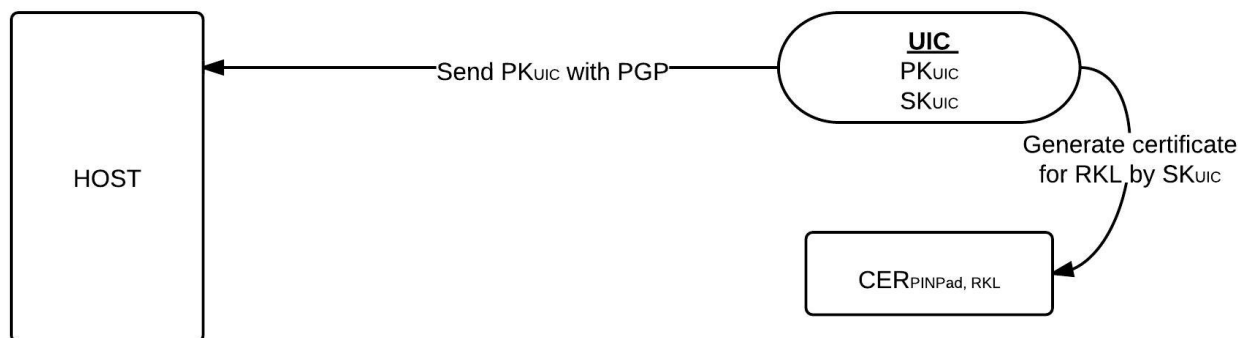


Host Certificate Hierarchy (three-layer):

CA Root Key $\rightarrow CER_{\text{HOST, intermediate}}$ ($PK_{\text{HOST, intermediate}}$) $\rightarrow CER_{\text{HOST, RKL}}$

PIN Pad Certificate Hierarchy

UIC shall generate the RSA key pairs (PK_{UIC} and SK_{UIC}) and use the private key to generate the certificate ($CER_{\text{PINPad, RKL}}$) for the certificate exchange. The UIC public key will be sent to host in a secure way (i.e. PGP) and need to be pre-loaded into host.

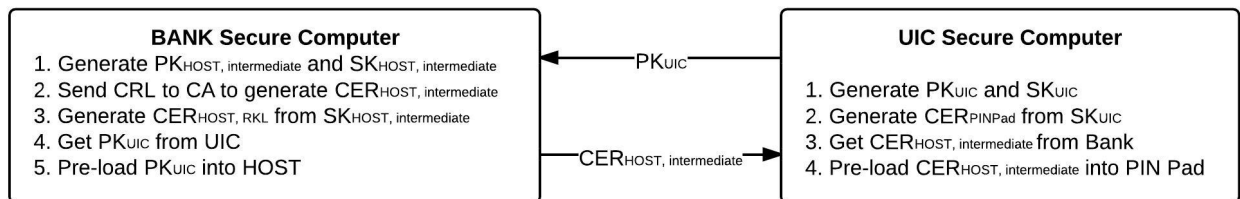


PIN Pad Certificate Hierarchy (Two-layer):

$PK_{\text{UIC}} \rightarrow CER_{\text{PINPad, RKL}}$

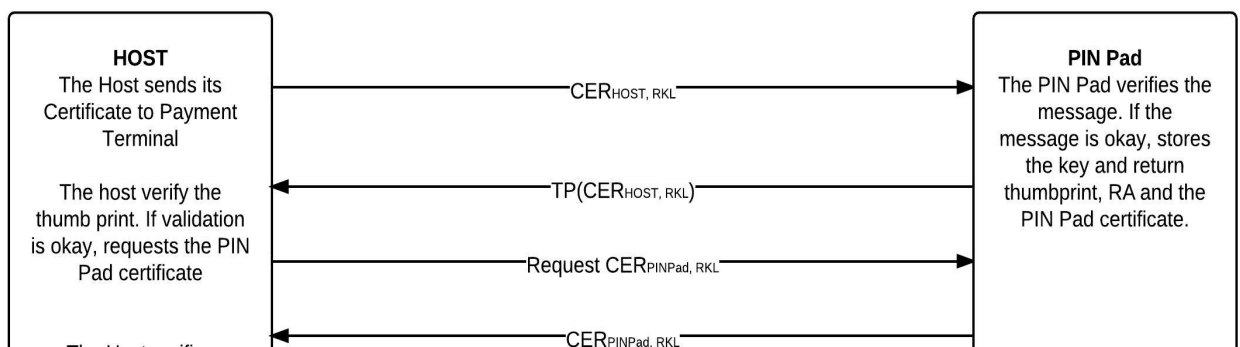
Preload Keys and Certificate

Both bank and UIC need to generate their RSA key pairs. The public key of bank ($PK_{HOST, intermediate}$) shall be sent to CA and generate the intermediate certificate, $CER_{HOST, intermediate}$, to verify the host certificate, $CER_{HOST, RKL}$, which is generated from the private key of host ($SK_{HOST, intermediate}$). Also, UIC needs to send its public key, PK_{UIC} , to bank for the PIN Pad certificate ($CER_{PINPad, RKL}$) verification in the remote key loading process.



RKL Step 1: Certificate Exchange

In the first step of the remote key loading, RKL server and PIN Pad will exchange their certificates and verify each other. The $CER_{HOST, RKL}$ can be verified by $CER_{HOST, intermediate}$ which is pre-loaded into PIN Pad in the previous step. The $CER_{PINPad, RKL}$ can be verified by PK_{UIC} which is pre-loaded into Host in the previous step.



The host sends the host certificate to PIN Pad. (Message R1)

The PIN Pad verifies the host certificate by the host intermediate public key. If the validation is okay, PIN Pad returns the thumb print of the host certificate. If the validation is not okay, PIN Pad returns the error code.

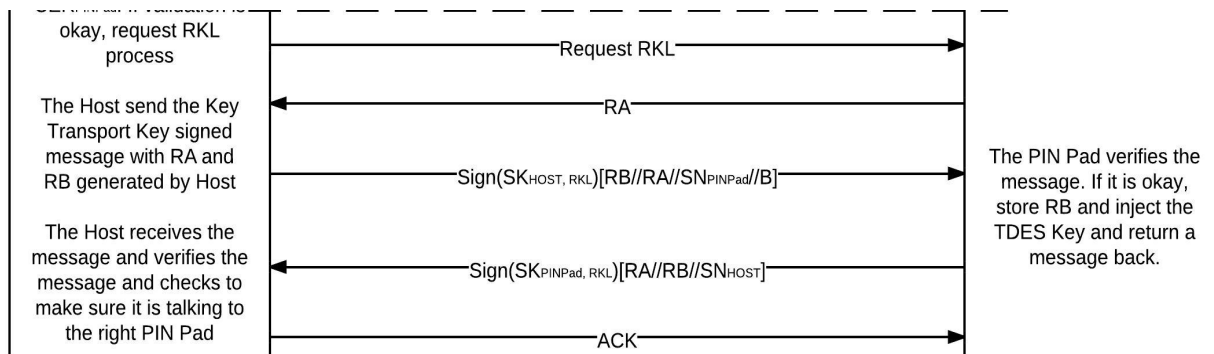
The host verifies the thumb print which is sent from PIN Pad in the previous step. If it is okay, the host issues the request to get PIN Pad certificate and Random number A. (Message R3 / R4)

The host verifies the PIN Pad certificate by using the UIC Public Key. If the validation is okay, the PIN Pad is ready for the remote key loading.

Once the exchange of certificate is done, the remote key loading is ready to start.

RKL Step 2: Send secure data and verify response

In the Second step of the remote key loading, RKL server and PIN Pad will exchange their random numbers RA and RB. The injection key information from RKL server can be verified by $CER_{HOST, RKL}$ which is saved into PIN Pad in the previous step. The message sent from PIN Pad can be verified by $CER_{PINpad, RKL}$ which is saved into Host in the previous step.



The host issues the request to starting the RKL with key data (B) encrypted by public key in the PIN pad certificate ($PK_{PINpad, RKL}$), and other data for verification (RA, RB, SN_{PINpad}). The data is signed by private key of host ($SK_{HOST, RKL}$).

PIN Pad will verify the RKL data by $CER_{HOST, RKL}$. If verification is OK, PIN pad will decrypt the key data by $SK_{PINpad, RKL}$, and then inject the key.

After key is injected, PIN pad will generate a digital signature by RA, RB, SN_{HOST} and send back to host for final verification.

Message example: Load intermediate certificate of the Host

PC --> DTE : 1202

Text :<02>R9.-----BEGIN<20>PKCS7-----<0A>MIIDTgYJKoZIhvcNAQcCoIIDPzCCAzsCAQExADALBgkqhkiG
9w0BBWgGggMhMIID<0A>HTCCAgUCEAoBQT4AAAEdkVE3hwAAWDEwDQYJKoZIhvcNAQEFBQAwwZDELMAkGA1UE<0A>
BhMCMVFcxDzANBgNVBAgTBIRhaxdhbjEMMAoGA1UEBxMDVFBFMQwwCgYDVQQKEwNV<0A>SUMxDTALBgNVBAstBFRT
UkQxGTAXBgNVBAMTEFVJQyBjZXJ0IGZvcibSS0wwHhcN<0A>MTIwNTMwMTAyOTMxWhcNMjMwNTEzMTEyOTMxWjA2
MQswCQYDVQQGEwJVUzEQMA4G<0A>A1UECgwHRGllYm9sZDEVMBMGA1UEAwMUmtsVGVzdEludGVyMIIBIjANBgkq
hkiG<0A>9w0BAQEFAAOCAQ8AMIIBCgKCAQEAp1eT/JPCHAHSgkLzoQ1gxRBgnMuCi1BopsuC<0A>17cSxGuwfnngp
XI5GbLkzo667ch8zRCXcjYbpxVoq0z7u8NItI0OSNaNT0nXivPZ+<0A>0NsoCwwIuHW3XM8diSLUBTHufMX7LWaz
0Scf+XlQLDl5zZoaiFYcwIu/VmZ40VWZ<0A>pxYFavq8jepdRXVSynBM+Xx5At/+rQOE6HxbqzEdGKhrkSiejrgS
Lpaype9D1FZI<0A>1HF5mTjyA64YxgoUAoUR29ybdNPpqC008pCUkfDeYhF9dhwL0U7ohFXgljppIVGF<0A>kNOP
WcIboxw+ggDJtdPG7XNp6ki2XVv1zRlss7qJ4KicxAQrKwIDAQABMA0GCSqG<0A>Sib3DQEBBQUAA4IBAQBPCGdZ
8EkmFv6oqCkVCY9/HSzrNrsqAtHj/iTj/P9y+c1Z<0A>Zx000OA+nAmqm6QLdQxDX46fxddznj5YwNMgolAj6Yye
M2Ga/LzZu5HDx+LzfsDY<0A>yfiGUIG3pFoRclBuq5j5ztHI2gvsL9vQfBPo0Ia7NnzPYskej6Ndp0gOQEehgmKG
<0A>01/5ndoOeWP0zGf5G68NycTWd7oqQuinLsyRt4W0tweb7SW7vGqaBDhNWK7z8wHh<0A>u3DxTsU7h4urDcg5
1KDN2IVKDo4bxcBrHPYwh875nmYGKQdcOIyJXpEffP9x+fZ<0A>dhBl2PC9ffuXMC/rvEzckZXecC8ibh4P52J+
DQoZoQAxA==<0A>-----END<20>PKCS7-----<0A><03><1B>

PC <-- DTE : 1

Text :<06>

PC <-- DTE : 7

Text :<02>RA0<03>a

PC --> DTE : 1

Text :<06>

Message example: Begin RKL session by exchange certificate request

PC --> DTE : 1140

Text :<02>R1-----BEGIN<20>PKCS7-----<0A>MIIDIQYJKoZIhvcNAQcCoIIDEjCCAw4CAQEExADALBgkqhkiG9w0BBwGgggLOMIIC<0A>8DCCAdgCEAoBQT4AAAEdkVE3hwAAWDEwDQYJKoZIhvcNAQEFBQAwnjELMAkGA1UE<0A>BhMCMVVMxEDAOBgNVBAoMB0RpbZWJvbGQxFTATBgNVBAMMDfJrbFRlc3RjbhRlcjAe<0A>Fw0xMjAlMzAxMDM1MDNaFw0yMzAlMTMxMDM1MDNaMDcxZzAJBgNVBAYTA1VTRAw<0A>DgYDVQQKDAdEaWVib2xkMRYwFAYDVQQDDA1Sa2xUZgXN0VmVuZG9yMIIBIjANBgkq<0A>hkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEAAzkttvx7stRA75AaJIKWsgSjD2mmJJZyn<0A>Wph2OlbEvaG1iKTFvyHJM0KAeq9+aRj1D65Z0pGkEOwffWTwzJ+XHRXCIfxfrA8<0A>AvRXXs0z1Wlbe ytd1zQTCcKro1U6MqaK/cxze/WWhtMPBE2BapsrfXPliKMU45GC<0A>NavZ8zHPa8agFvW8nyVFWnaNV4CR6pdj8OADst3m7c9byI3MsLEHps7J6z0VeF3V<0A>hfvHE6okvGLQGEe3aTVkXdrPdmbjN+gcIlzRvDj2/ZOC2D+w0lq9tcwDBzFHHq/8<0A>/+5o84Fp5nFPGeWhnnu8faSFQjAurYlAzHQQ5J3DuRRNE8xf7khq5QIDAQABMA0G<0A>CSqGS Ib3DQEBBQUAA4IBAQBKlncHZPUgzCtK8hjGQ1ZIJRj/Q/pWXIBCh4meVnD9<0A>yIqVqVoJeGS2v5TIJGBbKyOfSOXQcxOPGKVZIVEMmmYhyJIUq1CC13UCocveieHg<0A>w6VLCexbCAkPwxB2vrlZajcF6sE5cQjMGyrXqcKgYRRLUvint/o00op0ztpwFtye<0A>ux/ie+uKJlbKDNZt7YCmi6n7awmX5lhza/MGvDCcWHTZtGTORQsqoxtn7WY2j1TW<0A>rAjCOC3tFaBr1Q2sunzoqWommOm5vw0nc1CeyXvN4tv6pXYxN2v0w1f/WvwJzYMc<0A>N+ZO4uh1wd5cpJrcyRGYefIuLUwxo9Ztu2XTlyX9LPucoQAxA==<0A>-----END<20>PKCS7-----<0A><03><14>

PC <-- DTE : 1

Text :<06>

PC <-- DTE : 125

Text :<02>R2-----BEGIN<20>PKCS7-----<0A>MDECAQAwCQYFKw4DAhoFADAlAAAGCSqGS Ib3DQEHAQQUAABrarMgpOnk2EU7yS4pWSLWynvfCQ==<0A>-----END<20>PKCS7-----<0A><03>\$

PC --> DTE : 1

Text :<06>

Message example: Get PIN pad certificate and Random A

PC --> DTE : 5

Text :<02>R3<03>b

PC <-- DTE : 1

Text :<06>

PC <-- DTE : 1254

Text :<02>R4732DA7057E454B32-----BEGIN<20>PKCS7-----<0A>MIIDaQYJKoZIhvcNAQcCoIIDWjCCAlYCAQExADALBgkqhkiG9w0BBwGgggM8MIID<0A>ODCCAIAACAQEwDQYJKoZIhvcNAQEFBQAQAwZDELMAkGA1UEBhMCVFcxDzANBgNVBAgt<0A>BlRhaXdhbjEMMAoGA1UEBxMDVFBFMQwwCgYDVQQKEwNVSUMxDTALBgNVBAsTBFBRT<0A>UkQxGTAXBgNVBAMTEFVJQyBjZXJ0IGZvcjBSS0wwHhcNMTEwNTE3MDY1NDU1WhcN<0A>MjMwNDMwMDY1NDU1WjBgMQswCQYDVQQGEwJUVzEPMA0GA1UECBMGVGFpd2FuMQww<0A>CgYDVQQHEwNUUEUxDDAKBgNVBAoTA1VJQzENMAsgA1UECixMEVFNSRDEVMBMGA1UE<0A>AxMMmjA2MTFSRDAwMDAzMIIBIjANBgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEAA<0A>zWv/I9jLO2buZBE+q6Rz+qsM7y7ceWA2RFzeGzulyWTFfOLla87z6uYpXrocbHsM<0A>IogqbLYFgU4XOJ9q0vKN2BxzvAlvYn8h/E+QGQeYuPsJePzWq1TSrg9/nkuNsSka<0A>XNTD0vVeso7HbrU2LYBLt7DATElnUdTdbOoBw8SaO+6gzhyBKxHmBBSJkkaIZMo<0A>9Ru1Zkjz2FAGfRUM4ymzhfUGIcdec1YQShkwwIu4pZMRfKnpCs+oCHb9m2dnbtZa<0A>aQLHc793FUaN5XjCbSDeTH97217QCK9d+YmJv+su5v+fNgJ3+1zwi8lBBfkBg0di<0A>aW9FWeeNeDhNLUFgQxSiuQIDAQABMA0GCSqGSIb3DQEBAQUAA4IBAQBB0kTJiUu3g<0A>/5vhW0XvwzKr5KSl7Rdj7vfvyiq1vBf4xJTMDOvVqNiGTi3IW0/L5wM58fGhQupm<0A>uH+vVqO30WV1Nfd6AqP1Vgt2EpijRmjvDlgui9dQa1/9LMNyKQx1KdxeHKzdzaSq<0A>MHKfCRVz1pCFRAlpba951TT11wmFrKXHiNgfdTX5N9nI2Tq2CwGgxx7a2jIoKRHD<0A>Gg8TiQyRmTMrXtPt2tOSr6dCU8Y86OqF67bHjs00BQMon5B91a1AwggBFZUlej2M<0A>tL6GrcQdK+KuUL3PIamkL1wlh4UcYd17g5UQ+SrAnPtFXw+NuD9X/waQdDjfl5b0<0A>Q384sDzGX57ooQAxA==<0A>-----END<20>PKCS7-----<0A><03>\$

PC --> DTE : 1

Text :<06>

Message example: Generation of signed data for RKL request

Assume we want to transmit the following data to PIN pad:

Serial number of host certificate: 0a:01:41:3e:00:00:01:1d:91:51:37:87:00:00:58:31.
TDES key value: 01:23:45:67:89:ab:cd:ef:fe:dc:ba:98:76:54:32:10.
Random number B: f1:21:ca:3c:b5:c4:48:ed
Key id: 0x0
Key usage: "K0"
Key mode: "D"

Also we got serial number, public key and random number of PIN pad from R3 command:

Serial number of PIN pad: "20611RD00003"
Public key of PIN pad: <PK_{PINpad,RKL}>
Random number A: 73:2d:a7:05:7e:45:4b:32

1. Combine serial number of host certificate and TDES key to build the clear text data:

04:0f: --Tag of 16 bytes octet string
0a:01:41:3e:00:00:01:1d:91:51:37:87:00:00:58:31: -- SN of host certificate
04:0f: -- Tag of 16 bytes octet string
01:23:45:67:89:ab:cd:ef:fe:dc:ba:98:76:54:32:10 -- TDES key value

2. Set up RSA PKCS#1 encryption block (EB):

00:02:<217 bytes non-zero random>:00:<Data we generated in step 1>

3. Encrypt EB by the public key of PIN pad to generate the key block [B].

B = RSAEP (EB, PK_{PINpad,RKL})

4. Build the keyAndAuthData structure

04:08: -- 8 bytes OCTET STRING
73:2d:a7:05:7e:45:4b:32 -- rA
04:08: -- 8 bytes OCTET STRING
f1:21:ca:3c:b5:c4:48:ed -- rB
13:0c: -- 12 bytes PrintableString
32:30:36:31:31:52:44:30:30:30:30:33 -- S/N of PIN pad
04:82:01:00 -- 256 bytes OCTET STRING
.....[B]..... -- Encrypted S/N of host certificate and key.
02:01: -- 1 byte INTEGER as key id
00 -- id = 0.
13:02: -- 2 bytes PrintableString as key usage
4b:30 -- "K0"
13:01: -- 1 byte PrintableString as key mode
44 -- "D"

5. Generate digital signature of keyAndAuthData according to RSASSA-PKCS1-v1_5

hash = sha1(keyAndAuthData)

T = SEQUENCE {

digestAlgorithm AlgorithmIdentifier, -- identifier of SHA-1

digest OCTET STRING -- sha-1 hash value of keyAndAuthData

}

→ T = 30:21:30:09:06:05:2b:0e:03:02:1a:05:00:04:14:<hash>

EM = 0x00 | 0x01 | 0xFF.....0xFF | 0x00 | T

signature = RSASP1(SK_{HOST,RKL}, EM)

6. Build PKCS#7 data structure for DER encoding

30:82:xx:xx: -- SEQUENCE with length =xxxx

06:09: -- ContentType as OBJECT IDENTIFIER

2A:86:48:86:F7:0D:01:07:01 -- RSA PKCS#7-1 data

A0:82:yy:yy: -- [0] EXPLICIT ANY DEFINED BY contentType (len=yyyy)

30:82:01:26: -- SEQUENCE as keyAndAuthData (301 bytes)

.....(contents of keyAndAuthData, built in step 4)...

03 82 01 01 -- BIT STRING as signatureValue (257 bytes)

00 -- unused bits (none)

.... (contents of signature, built in step 5)...

7. Base64 encodes the result of 6 into PEM format with -----BEGIN PKCS7----- and -----END PKCS7----- marks.

Message example: Generation of RKL response data inside PIN pad

Assume following data:

Random number A: 73:2d:a7:05:7e:45:4b:32

Random number B: f1:21:ca:3c:b5:c4:48:ed

Serial number of host certificate: 0a:01:41:3e:00:00:01:1d:91:51:37:87:00:00:58:31.

1. Composite message M:

M = RA | RB | SN_{host} =

73:2d:a7:05:7e:45:4b:32:f1:21:ca:3c:b5:c4:48:ed:

0a:01:41:3e:00:00:01:1d:91:51:37:87:00:00:58:31

2. Digest message M with sha-1

H = sha1(M) = 18:28:81:35:17:ed:9a:26:cb:21:36:38:0a:6e:69:2b:0d:53:89:1c

3. Make PKCS#1 v1.5 DigestInfo

T = DigestInfo = 30:21:30:09:06:05:2b:0e:03:02:1a:05:00:

04:14:18:28:81:35:17:ed:9a:26:cb:21:36:38:0a:6e:69:2b:0d:53:89:1c

EM = 0x00 | 0x01 | PS | 0x00 | T

Where PS = 0xFF with length = 256-3-35 = 218 bytes.

4. Sign (private encrypt) EM by PIN pad private key

S = RSASP1(SK_{PINpad}, EM)

5. Build PKCS#7 data structure DER encoding

30:82:01:25: -- SEQUENCE with length = 293 bytes

02:01: -- INTEGER as version

00 -- value = 0

30:0d -- SEQUENCE as DigestAlgorithmIdentifier

06 09 -- OBJECT IDENTIFIER as algorithm

2a:86:48:86:f7:0d:01:01:05 -- sha1WithRSAEncryption

05 00 -- NULL as parameter

30:82:01:0f -- SEQUENCE as contentInfo

06 09 -- OBJECT IDENTIFIER as contentType

2A:86:48:86:F7:0D:01:07:01 -- RSA PKCS#7-1 data

04:82:01:00 -- OCTET STRING as content

[S] -- 256 bytes data S created in step 3.

6. Base64 encodes the result of 5 into PEM format with -----BEGIN PKCS7----- and -----END PKCS7----- marks.

PC --> DTE : 868

Text :<02>R50-----BEGIN<20>PKCS7-----<0D><0A>MIICTAYJKoZIhvcNAQcBoIICPQSCAjkwggEwBAhzLacFfkVLMgQI8SHKPLXES00T<0D><0A>DDIwNjExUkQwMDAwMwSCAQbkGHdEFMxbgJfot2VnTNzaWWepQGuxA1JjZd4CenN1<0D><0A>aF7935yaxBCWekH/ktykeXnzAaOrzRuZFkFRc+eSrmvLEsEmH6PCQj6E3tObZj2Q<0D><0A>XVBHyHgHBpqZQAD4e81PMFTZmpEvu5gGKqDAK0ilsPquWZcWcKcrMGg/7+QnXGx0<0D><0A>ZfBrOh2wEIjpbEbegiBzHEgqsM1M1MGk9ruX2RI9lq9AIhP64eo84bv+xl2uzU0z<0D><0A>SjWBR6H6dk70+gGF9m4Xh7LOvGzK6FJfDIBN7ub+KF+9WoehGxtaQEq+4UnVqkm<0D><0A>8AEuOjhlcWchW7+TZGHT1uGU1cQwEOJw1NqvQ2OYCT7TAgeAEwJLMBMBRAOCAQEa<0D><0A>JcgTEM56+bdFCb94cTmb6pyzhF8L0HuBLfVDP0GEDSutyS8/wjPcDiFCi7fNLWc9<0D><0A>5UMNWftRfcr/qmddvXx2hZoRoahz10f/HezJhMyeYg8TnfzxmBG8wn1jKtCC5Il1<0D><0A>GJufHJIA8tUh0Uw5RrC93BgjI7bxv/Xa4D5aqQ1SA40NyArrR15xeBsJpQvaqYN6<0D><0A>98VrD80i3OunyLQx+Kieuu3YTptNLG4p6000PVVgkxVWEXgss03E3qo2qGs2Qy8F<0D><0A>4cUJ4UGKFuuOiZzfBibzpiv4R5YKvHLFW9Oh9HmZrNifL2qchw6D6siJP5K3nV0X<0D><0A>5wz6BsyOJ7LLCP++9YXfsg==<0D><0A>-----END<20>PKCS7-----<0D><0A><03>q

PC <-- DTE : 1

Text :<06>

PC <-- DTE : 444

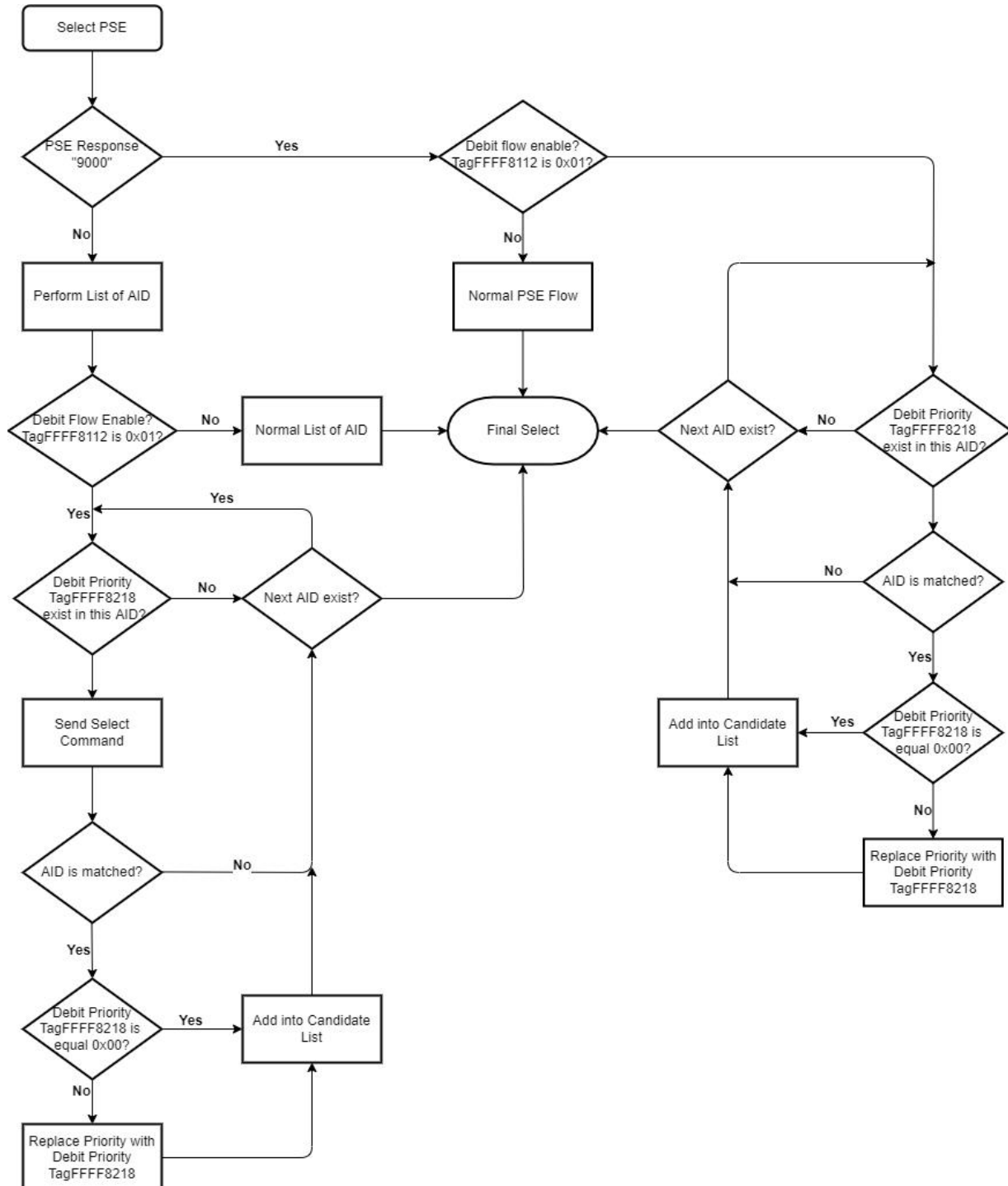
Text :<02>R6-----BEGIN<20>PKCS7-----<0A>MIIBJQIBADANBgkqhkiG9w0BAQUFADCCAQ8GCSqGSIB3DQEHAQSCAQAZ+pbftFwq06aUd5uIsTsTklzqdFRtCURAKCXBqZWPzsUNca42ibdCfq7hlMwGpGUDxBMoLvvvFUgv1q6d4gcf/KISHQE9C/v74SRMxwuvYGUCV0N4WXFMDAo4WGC+RVDiLiahY4pwKkGaf/k9xvLLK0ceMbMb7zLcBNFTbsoH5OW8pN80h72m7OfCtEbqsV8Vilh28KhcvB37H9Pev4KiQHGXGzZninFWMxUwHzxEPrNpdm7XjfyhtOVrFCBdb0LqpWJNZetlu81iFk0XKamf9s+AFvPjRhi8uN4cHP8DvwAlcmoq3Lxp5XSd//OHbbcXeuPjRP9LOSmTg16HTJxED<0A>-----END<20>PKCS7-----<0A><03><20>

PC --> DTE : 1

Text :<06>

Appendix J US Common Debit Flow

Contact Flow



Contactless Flow

